More Advanced Accounting

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Overview & Key Rules of Thumb

This guide contains a mix of accounting-related topics that don't fit into the other technical guides.

These topics are all **more advanced** and, therefore, less likely to come up in interviews.

You're more likely to get these types of questions if you've had significant experience in investment banking or other finance roles, or you're in a region where one of these topics has special importance.

For example, in many continental European countries, bankers like to test your knowledge of **consolidation accounting** – how a parent company records the acquisition of a majority or minority stake in a subsidiary – so the sections on Noncontrolling Interests and Equity Investments are more important there.

On the other hand, the section on LIFO vs. FIFO vs. Average Weighted Cost for Inventory and COGS is less relevant in Europe because IFRS does not allow LIFO.

All these topics are less important than the core subjects: Basic Accounting, Equity Value and Enterprise Value, Valuation and DCF Analysis, and Merger Models and LBO Models.

So, you should *not* spend a ton of time studying this guide unless you're already very confident in your knowledge of everything else.

Key Rule #1: Book Value, Face Value, and Market Value of Debt

Just like there's a difference between the **Book Value** and **Market Value** of Equity, there may also be a difference between the Book Value of Debt (what goes on the Balance Sheet) and the Market Value of Debt (how much another investor would pay for it in the market).

But there's *another* value with Debt as well: Face Value.

Book Value vs. Face Value is easiest to explain, so let's start there:

- **Face Value** = The amount the company initially issues and pays interest on. This one is affected only by Debt Issuances, Principal Repayments, and PIK Interest.

- **Book Value** = The amount shown on the company's Balance Sheet. This one is affected by unamortized Issuance Fees, Debt Discounts and Premiums, Debt Issuances, Principal Repayments, and PIK Interest.

The most important point here is that the number listed for Debt on a company's Balance Sheet is NOT necessarily what it's paying interest on!

It may be close, but it's rarely the same.

The first difference emerges when a company issues Debt.

When this happens, the company must pay "Issuance Fees" to the banks and lenders that arranged the deal, as payment for their services.

These fees are paid upfront in cash, but they're **deducted** from the Debt number that goes on the Balance Sheet (the Book Value), and then they amortize until the Debt matures.

Here's what a $100 Debt issuance with 3% in Issuance Fees looks like:

		Historical	Projected				
Debt Schedule and Assumptions:	*Units:*	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Tax Rate:	%	25.0%					
Debt Issuance Fee %:	%	3.0%					
Initial Debt Issuance:	$ M	$ 100.0					
Debt Maturity:	Year #	5					
Coupon Rate (Fixed Interest Rate):	%	5.0%					
Year # for Full Debt Repayment:	Year #	5					
Beginning Face Value of Debt:	$ M		$ 100.0	$ 100.0	$ 100.0	$ 100.0	$ 100.0
(-) Debt Principal Repayments & Maturities:	$ M		-	-	-	-	(100.0)
Ending Face Value of Debt:	$ M	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**-**
Beginning Book Value of Debt:	$ M		97.0	97.6	98.2	98.8	99.4
(+) Amortization of Debt Issuance Fees:	$ M		0.6	0.6	0.6	0.6	0.6
(+) Loss on Debt Extinguishment:	$ M		-	-	-	-	(0.0)
(-) Debt Principal Repayments:	$ M		-	-	-	-	(100.0)
Ending Book Value of Debt:	$ M	**97.0**	**97.6**	**98.2**	**98.8**	**99.4**	**-**

On the Balance Sheet, the "Debt" line item keeps increasing each year until maturity.

The Amortization is based on the Issuance Fee of $3 divided by the Debt Maturity of 5 years, so $3 / 5 = $0.6 per year.

Then, in the Interest Expense line item on the Income Statement, **both** the cash interest expense of $100 * 5.0% = $5 *and* the $0.6 Amortization of Issuance Fees are included.

If the company repays its Debt *early* (before Year 5), then it records a "Loss on Debt Extinguishment" that's equal to the remaining Unamortized Issuance Fees:

Initial Debt Issuance:	*$ M*	$ 100.0
Debt Maturity:	*Year #*	5
Coupon Rate (Fixed Interest Rate):	*%*	5.0%
Year # for Full Debt Repayment:	*Year #*	3

Beginning Face Value of Debt:	*$ M*		$ 100.0	$ 100.0	$ 100.0	$ -	$ -
(-) Debt Principal Repayments & Maturities:	*$ M*		-	-	(100.0)	-	-
Ending Face Value of Debt:	*$ M*	**100.0**	**100.0**	**100.0**	**-**	**-**	**-**
Beginning Book Value of Debt:	*$ M*		97.0	97.6	98.2	-	-
(+) Amortization of Debt Issuance Fees:	*$ M*		0.6	0.6	0.6	-	-
(+) Loss on Debt Extinguishment:	*$ M*		-	-	1.2	-	-
(-) Debt Principal Repayments:	*$ M*		-	-	(100.0)	-	-
Ending Book Value of Debt:	*$ M*	**97.0**	**97.6**	**98.2**	**-**	**-**	**-**
Unamortized Debt Issuance Fees:	*$ M*	3.0	2.4	1.8	-	-	-
(+) Interest Expense (Cash):	*$ M*		5.0	5.0	5.0	-	-
(+) Amortization of Debt Issuance Fees (Non-Cash):	*$ M*		0.6	0.6	0.6	-	-
Total Interest Expense on Income Statement:	*$ M*		**5.6**	**5.6**	**5.6**	**-**	**-**
Loss on Debt Extinguishment on IS:	*$ M*		-	-	1.2	-	-

		Historical	Projected				
Income Statement:	*Units:*	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Total Revenue:	*$ M*	**$ 200.0**	**$ 200.0**	**$ 200.0**	**$ 200.0**	**$ 200.0**	**$ 200.0**
Cost of Sales:	*$ M*	75.0	75.0	75.0	75.0	75.0	75.0
Gross Profit:	*$ M*	**125.0**	**125.0**	**125.0**	**125.0**	**125.0**	**125.0**
Operating Expenses:							
Selling, General & Administrative:	*$ M*	50.0	50.0	50.0	50.0	50.0	50.0
Depreciation & Amortization:	*$ M*	25.0	25.0	25.0	25.0	25.0	25.0
Total Operating Expenses:	*$ M*	**75.0**	**75.0**	**75.0**	**75.0**	**75.0**	**75.0**
Operating Income:	*$ M*	**50.0**	**50.0**	**50.0**	**50.0**	**50.0**	**50.0**
Operating Margin:	*%*	*25.0%*	*25.0%*	*25.0%*	*25.0%*	*25.0%*	*25.0%*
Other Income / (Expense):							
(-) Interest Expense:	*$ M*	-	(5.6)	(5.6)	(5.6)	-	-
(-) Loss on Debt Extinguishment:	*$ M*	-	-	-	(1.2)	-	-
Total Other Income / (Expense):	*$ M*	**-**	**(5.6)**	**(5.6)**	**(6.8)**	**-**	**-**
Pre-Tax Income:	*$ M*	**50.0**	**44.4**	**44.4**	**43.2**	**50.0**	**50.0**
(-) Provision For Income Taxes:	*$ M*	(12.5)	(11.1)	(11.1)	(10.8)	(12.5)	(12.5)
Net Income:	*$ M*	**$ 37.5**	**$ 33.3**	**$ 33.3**	**$ 32.4**	**$ 37.5**	**$ 37.5**

And on the Cash Flow Statement, both the Amortization of Issuance Fees and the Loss on Debt Extinguishment are non-cash add-backs:

		Historical	Projected				
Cash Flow Statement:	*Units:*	**Year 0**	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
CASH FLOWS FROM OPERATING ACTIVITIES:							
Net Income:	*$ M*	**$ 37.5**	**$ 33.3**	**$ 33.3**	**$ 32.4**	**$ 37.5**	**$ 37.5**
Adjustments for Non-Cash Charges:							
(+) Depreciation & Amortization:	*$ M*	25.0	25.0	25.0	25.0	25.0	25.0
(+) Amortization of Debt Issuance Fees:	*$ M*	-	0.6	0.6	0.6	-	-
(+) Loss on Debt Extinguishment:	*$ M*	-	-	-	1.2	-	-
(+/-) Deferred Taxes:	*$ M*	-	-	-	-	-	-
(+/-) Change in Op. Lease Assets:	*$ M*	-	(2.0)	(2.0)	(2.0)	(2.0)	(2.0)
(+/-) Change in Op. Lease Liabilities:	*$ M*	-	2.0	2.0	2.0	2.0	2.0
Changes in Operating Assets and Liabilities:							
Inventory:	*$ M*	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)
Accounts Receivable:	*$ M*	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)
Accounts Payable:	*$ M*	5.0	5.0	5.0	5.0	5.0	5.0
Accrued Liabilities:	*$ M*	5.0	5.0	5.0	5.0	5.0	5.0
Net Cash Provided by Operating Activities:	*$ M*	**62.5**	**58.9**	**58.9**	**59.2**	**62.5**	**62.5**

The **Market Value**, or "Price," of Debt is based on the Present Value of its future cash flows.

Both the interest payments and the eventual principal repayment are components of those future cash flows.

So, everything that affects the Present Value also affects the Market Value of Debt: the Discount Rate, the expected future cash flows, and the repayment probability upon maturity.

The Discount Rate here is the Yield to Maturity (YTM) on other, similar Debt issuances in the market.

The Market Value of Debt changes when the Discount Rate changes due to changing interest rates in the macro environment; a change in the company's credit quality could also cause it.

For example, if Company A has a $100 Debt issuance with a 5.0% fixed coupon rate, and prevailing yields on similar Debt in the market decrease from 5.0% to 4.0%, the Market Value of its Debt issuance will increase.

Why?

Because it's now more attractive than other, similar options in the market, so buyers are willing to pay more to get that higher coupon rate.

Here's what the calculation looks like in Excel:

Debt Schedule and Assumptions:	Units:	Historical Year 0	Projected Year 1	Year 2	Year 3	Year 4	Year 5
Tax Rate:	%	25.0%					
Debt Issuance Fee %:	%	3.0%					
Initial Debt Issuance:	$ M	$ 100.0					
Debt Maturity:	Year #	5					
Coupon Rate (Fixed Interest Rate):	%	5.0%					
Year # for Full Debt Repayment:	Year #	5					
Prevailing Yields on Similar Debt in the Market:	%	4.0%					
"Redemption Value" (Repayment Probability):	#	100.0					
Market Value of Debt:	$ M	$ 104.5	<-- This is what someone else would pay for the company's Deb if they wanted to buy it in the secondary market.				
(+) Present Value of Future Interest Payments:	$ M	=NPV(E20,F30:J30)					
(+) Present Value of Principal Repayments:	$ M	82.2					
Market Value of Debt:	$ M	**104.5**					
(+) Future Interest Payments:	$ M		5.0	5.0	5.0	5.0	5.0
(+) Future Principal Repayments:	$ M		-	-	-	-	100.0
Total Cash Flow to Investors:	$ M		**5.0**	**5.0**	**5.0**	**5.0**	**105.0**

The Market Value of Debt increases because Prevailing Yield in the market decrease!

You could also use the PRICE function to calculate this in Excel, but it won't match up exactly to the manual approach used above.

The Market Value increases when Prevailing Yields decrease, or when the Repayment Probability increases. It will decrease when Prevailing Yields increase, or when the Repayment Probability decreases.

The Market Value of Debt makes <u>no direct impact</u> on the financial statements. The Book Value and Face Value of Debt are not affected by its Market Value.

The Market Value matters mostly for bond analysis and investing in fixed income.

However, if this difference between the coupon rate on the Debt issuance and Prevailing Yields <u>exists</u> <u>at the time of the initial issuance</u>, then the company may issue the Debt at a **premium or discount** to its Face Value.

When that happens, it's called "Original Issue Discount" (OID) or "Original Issue Premium" (OIP), and, once again, the accounting changes (see the next section).

A company might also issue the Debt at a discount if investors are not confident of its credit quality, and they want an additional "incentive" to buy into its offering.

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Key Rule #2: Debt Discounts, Premiums, and Original Issue Discount (OID)

As discussed in the previous section, the **Market Value** of Debt often differs from its **Face Value** because of changes in interest rates or the company's credit quality (e.g., the company issues Debt at a 5% coupon rate, but similar issuances currently yield 6%).

If the Market Value *upon initial issuance* is different from the Face Value, then investors can buy the Debt directly from the company at a Discount or Premium.

This means what it sounds like: if there's a 5% Discount on a $100 Debt issuance, then investors can buy it for $95 rather than $100.

But if the coupon rate is 5%, those investors will still earn 5% * $100 = $5 in interest per year.

This initial Discount is called **Original Issue Discount (OID)**, and the accounting treatment is similar to the one for Issuance Fees: deduct the OID from Face Value to get the Book Value of Debt and amortize the OID until the Debt's maturity.

If there's a principal repayment or other, early repayment, accelerate the amortization based on the proportion of principal repaid.

Companies can also issue Debt with an **Original Issue Premium (OIP)**, but that is far less common, so we're not going to cover it here (if we did, the signs of everything would be reversed – the OIP Amortization would be "positive Interest" and the Losses would be Gains).

First, let's look at OID *without* Debt Principal Repayments.

For simplicity, we'll ignore the Issuance Fees in this first example:

Debt Schedule and Assumptions:	Units:	Historical Year 0	Projected Year 1	Year 2	Year 3	Year 4	Year 5
Tax Rate:	%	25.0%					
Debt Issuance Fee %:	%	0.0%					
Initial Debt Issuance - Face Value:	$ M	$ 100.0					
Initial Debt Issuance - Issued At:	$ M	90.0					
Debt Maturity:	Year #	10					
Coupon Rate (Fixed Interest Rate):	%	5.0%					
% Annual Debt Principal Repayment:	%	VALID	0.0%	0.0%	0.0%	0.0%	0.0%
Amortization Period Remaining:	# Years	10	9	8	7	6	5
OID Beginning Balance:	$ M		**$ 10.0**	**$ 9.0**	**$ 8.0**	**$ 7.0**	**$ 6.0**
(-) Amortization of OID:	$ M		(1.0)	(1.0)	(1.0)	(1.0)	(1.0)
(-) Loss on Unamortized OID on Repayment:	$ M		-	-	-	-	-
OID Ending Balance:	$ M	**10.0**	**9.0**	**8.0**	**7.0**	**6.0**	**5.0**
Issuance Fee Beginning Balance:	$ M		**-**	**-**	**-**	**-**	**-**
(-) Amortization of Debt Issuance Fees:	$ M		-	-	-	-	-
(-) Loss on Debt Extinguishment:	$ M		-	-	-	-	-
Issuance Fee Ending Balance:	$ M	**-**	**-**	**-**	**-**	**-**	**-**
Beginning Face Value of Debt:	$ M		100.0	100.0	100.0	100.0	100.0
(-) Debt Principal Repayments & Maturities:	$ M		-	-	-	-	-
Ending Face Value of Debt:	$ M	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
Beginning Book Value of Debt:	$ M		90.0	91.0	92.0	93.0	94.0
(+) Amortization of Debt Issuance Fees:	$ M		-	-	-	-	-
(+) Amortization of OID:	$ M		1.0	1.0	1.0	1.0	1.0
(+) Loss on Debt Extinguishment:	$ M		-	-	-	-	-
(+) Loss on Unamortized OID on Repayment:	$ M		-	-	-	-	-
(-) Debt Principal Repayments:	$ M		-	-	-	-	-
Ending Book Value of Debt:	$ M	**90.0**	**91.0**	**92.0**	**93.0**	**94.0**	**95.0**

Discount of $10, and a 10-year maturity, so $10 / 10 = $1 in Amortization per year.

The Debt shown on the Balance Sheet increases by $1 per year, but the company always pays interest based on the $100 Face Value.

On the Income Statement, the Amortization of OID also counts as Interest Expense, and it's a non-cash add-back on the Cash Flow Statement.

If there *are* Debt Principal Repayments before maturity, then these formulas change as follows:

- **Amortization of OID:** = –MIN(OID Beginning Balance This Year, OID Beginning Balance This Year / Remaining Years Until Maturity)

- **Loss on Unamortized OID on Repayment:** = OID Balance After Amortization This Year * Percentage of Beginning Debt Principal This Year Repaid This Year

Just like the Loss on Debt Extinguishment for the Issuance Fees, the Loss on Unamortized OID on Repayment here represents the *acceleration* of Amortization – but for a different line item.

You can also wrap these formulas in IFERROR to handle cases such as repayment of 100% of the principal before maturity.

Here's how it works with 20% principal repayments each year, ignoring Issuance Fees:

		Historical	Projected				
Debt Schedule and Assumptions:	*Units:*	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Tax Rate:	%	25.0%					
Debt Issuance Fee %:	%	0.0%					
Initial Debt Issuance - Face Value:	$ M	$ 100.0					
Initial Debt Issuance - Issued At:	$ M	90.0					
Debt Maturity:	Year #	10					
Coupon Rate (Fixed Interest Rate):	%	5.0%					
% Annual Debt Principal Repayment:	%	VALID	20.0%	20.0%	20.0%	20.0%	20.0%
Amortization Period Remaining:	# Years	10	9	8	7	6	5
OID Beginning Balance:	$ M		**$ 10.0**	**$ 7.2**	**$ 4.8**	**$ 2.8**	**$ 1.2**
(-) Amortization of OID:	$ M		(1.0)	(0.8)	(0.6)	(0.4)	(0.2)
(-) Loss on Unamortized OID on Repayment:	$ M		(1.8)	(1.6)	(1.4)	(1.2)	(1.0)
OID Ending Balance:	$ M	**10.0**	**7.2**	**4.8**	**2.8**	**1.2**	**-**
Issuance Fee Beginning Balance:	$ M		**-**	**-**	**-**	**-**	**-**
(-) Amortization of Debt Issuance Fees:	$ M		-	-	-	-	-
(-) Loss on Debt Extinguishment:	$ M		-	-	-	-	-
Issuance Fee Ending Balance:	$ M	**-**	**-**	**-**	**-**	**-**	**-**
Beginning Face Value of Debt:	$ M		100.0	80.0	60.0	40.0	20.0
(-) Debt Principal Repayments & Maturities:	$ M		(20.0)	(20.0)	(20.0)	(20.0)	(20.0)
Ending Face Value of Debt:	$ M	**100.0**	**80.0**	**60.0**	**40.0**	**20.0**	**-**
Beginning Book Value of Debt:	$ M		90.0	72.8	55.2	37.2	18.8
(+) Amortization of Debt Issuance Fees:	$ M		-	-	-	-	-
(+) Amortization of OID:	$ M		1.0	0.8	0.6	0.4	0.2
(+) Loss on Debt Extinguishment:	$ M		-	-	-	-	-
(+) Loss on Unamortized OID on Repayment:	$ M		1.8	1.6	1.4	1.2	1.0
(-) Debt Principal Repayments:	$ M		(20.0)	(20.0)	(20.0)	(20.0)	(20.0)
Ending Book Value of Debt:	$ M	**90.0**	**72.8**	**55.2**	**37.2**	**18.8**	**-**

Loss worth 20% of the remaining OID is recorded each time a 20% principal repayment happens.

The Book Value keeps decreasing because of the Principal Repayments, but the Amortization and Losses boost it slightly.

Finally, if we now incorporate the Issuance Fees, we can modify the formulas so that they match the ones for OID above:

- **Amortization of Issuance Fees:** = –MIN(Issuance Fee Beginning Balance This Year, Issuance Fee Beginning Balance This Year / Remaining Years Until Maturity)

- **Loss on Debt Extinguishment:** = Issuance Fee Balance After Amortization This Year * Percentage of Beginning Debt Principal This Year Repaid This Year

Here's what it looks like with a 3% Issuance Fee and more "random" Principal Repayments:

		Historical	Projected				
Debt Schedule and Assumptions:	*Units:*	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Tax Rate:	%	25.0%					
Debt Issuance Fee %:	%	3.0%					
Initial Debt Issuance - Face Value:	$ M	$ 100.0					
Initial Debt Issuance - Issued At:	$ M	90.0					
Debt Maturity:	Year #	10					
Coupon Rate (Fixed Interest Rate):	%	5.0%					
% Annual Debt Principal Repayment:	%	VALID	5.0%	0.0%	30.0%	60.0%	5.0%
Amortization Period Remaining:	# Years	10	9	8	7	6	5
OID Beginning Balance:	$ M		**$ 10.0**	**$ 8.6**	**$ 7.6**	**$ 4.6**	**$ 0.3**
(-) Amortization of OID:	$ M		(1.0)	(1.0)	(1.0)	(0.7)	(0.1)
(-) Loss on Unamortized OID on Repayment:	$ M		(0.5)	-	(2.1)	(3.6)	(0.3)
OID Ending Balance:	$ M	**10.0**	**8.6**	**7.6**	**4.6**	**0.3**	**-**
Issuance Fee Beginning Balance:	$ M		**3.0**	**2.6**	**2.3**	**1.4**	**0.1**
(-) Amortization of Debt Issuance Fees:	$ M		(0.3)	(0.3)	(0.3)	(0.2)	(0.0)
(-) Loss on Debt Extinguishment:	$ M		(0.1)	-	(0.6)	(1.1)	(0.1)
Issuance Fee Ending Balance:	$ M	**3.0**	**2.6**	**2.3**	**1.4**	**0.1**	**-**
Beginning Face Value of Debt:	$ M		100.0	95.0	95.0	65.0	5.0
(-) Debt Principal Repayments & Maturities:	$ M		(5.0)	-	(30.0)	(60.0)	(5.0)
Ending Face Value of Debt:	$ M	**100.0**	**95.0**	**95.0**	**65.0**	**5.0**	**-**
Beginning Book Value of Debt:	$ M		87.0	83.9	85.1	59.1	4.6
(+) Amortization of Debt Issuance Fees:	$ M		0.3	0.3	0.3	0.2	0.0
(+) Amortization of OID:	$ M		1.0	1.0	1.0	0.7	0.1
(+) Loss on Debt Extinguishment:	$ M		0.1	-	0.6	1.1	0.1
(+) Loss on Unamortized OID on Repayment:	$ M		0.5	-	2.1	3.6	0.3
(-) Debt Principal Repayments:	$ M		(5.0)	-	(30.0)	(60.0)	(5.0)
Ending Book Value of Debt:	$ M	**87.0**	**83.9**	**85.1**	**59.1**	**4.6**	**(0.0)**

When the Debt is finally repaid in Year 5, EVERYTHING – the OID Balance, Issuance Fee Balance, Face Value, and Book Value – reaches $0.

Here's what happens on the other financial statements in this scenario:

Debt Schedule and Assumptions:	Units:	Historical Year 0	Projected Year 1	Year 2	Year 3	Year 4	Year 5
(+) Interest Expense (Cash):	$ M		5.0	4.8	4.8	3.3	0.3
(+) Amortization of OID (Non-Cash):	$ M		1.0	1.0	1.0	0.7	0.1
(+) Amortization of Debt Issuance Fees (Non-	$ M		0.3	0.3	0.3	0.2	0.0
Total Interest Expense on Income Statement	$ M		**6.3**	**6.0**	**6.0**	**4.1**	**0.3**
Losses on Debt Extinguishment:	$ M	**-**	**0.6**	**-**	**2.7**	**4.7**	**0.3**

Income Statement:	Units:	Historical Year 0	Projected Year 1	Year 2	Year 3	Year 4	Year 5
Total Revenue:	$ M	**$ 200.0**	**$ 200.0**	**$ 200.0**	**$ 200.0**	**$ 200.0**	**$ 200.0**
Cost of Sales:	$ M	75.0	75.0	75.0	75.0	75.0	75.0
Gross Profit:	$ M	**125.0**	**125.0**	**125.0**	**125.0**	**125.0**	**125.0**
Operating Expenses:							
Selling, General & Administrative:	$ M	50.0	50.0	50.0	50.0	50.0	50.0
Depreciation & Amortization:	$ M	25.0	25.0	25.0	25.0	25.0	25.0
Total Operating Expenses:	$ M	**75.0**	**75.0**	**75.0**	**75.0**	**75.0**	**75.0**
Operating Income:	$ M	**50.0**	**50.0**	**50.0**	**50.0**	**50.0**	**50.0**
Operating Margin:	%	*25.0%*	*25.0%*	*25.0%*	*25.0%*	*25.0%*	*25.0%*
Other Income / (Expense):							
(-) Interest Expense:	$ M	-	(6.3)	(6.0)	(6.0)	(4.1)	(0.3)
(-) Losses on Debt Extinguishment:	$ M	-	(0.6)	-	(2.7)	(4.7)	(0.3)
Total Other Income / (Expense):	$ M	**-**	**(6.9)**	**(6.0)**	**(8.7)**	**(8.8)**	**(0.6)**
Pre-Tax Income:	$ M	**50.0**	**43.1**	**44.0**	**41.3**	**41.2**	**49.4**
(-) Provision For Income Taxes:	$ M	(12.5)	(10.8)	(11.0)	(10.3)	(10.3)	(12.3)
Net Income:	$ M	**$ 37.5**	**$ 32.3**	**$ 33.0**	**$ 31.0**	**$ 30.9**	**$ 37.0**

Includes Losses on both Unamortized OID and Issuance Fees.

All these items - Cash Interest, Amortization of OID and Issuance Fees, and Losses - reduce Pre-Tax Income.

On the Cash Flow Statement, the OID and Issuance Fees both appear as outflows in Cash Flow from Financing because the company receives *less in cash proceeds as a result of these items*:

Cash Flow Statement:	Units:	Historical Year 0	Projected Year 1	Year 2	Year 3	Year 4	Year 5
CASH FLOWS FROM OPERATING ACTIVITIES:							
Net Income:	$ M	**$ 37.5**	**$ 32.3**	**$ 33.0**	**$ 31.0**	**$ 30.9**	**$ 37.0**
Adjustments for Non-Cash Charges:							
(+) Depreciation & Amortization:	$ M	25.0	25.0	25.0	25.0	25.0	25.0
(+) Amortization of OID:	$ M	-	1.0	1.0	1.0	0.7	0.1
(+) Amortization of Debt Issuance Fees:	$ M	-	0.3	0.3	0.3	0.2	0.0
(+) Losses on Debt Extinguishment:	$ M	-	0.6	-	2.7	4.7	0.3
(+/-) Deferred Taxes:	$ M	-	-	-	-	-	-
(+/-) Change in Op. Lease Assets:	$ M	-	(2.0)	(2.0)	(2.0)	(2.0)	(2.0)
(+/-) Change in Op. Lease Liabilities:	$ M	-	2.0	2.0	2.0	2.0	2.0
Changes in Operating Assets and Liabilities:							
Inventory:	$ M	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)
Accounts Receivable:	$ M	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)
Accounts Payable:	$ M	5.0	5.0	5.0	5.0	5.0	5.0
Accrued Liabilities:	$ M	5.0	5.0	5.0	5.0	5.0	5.0
Net Cash Provided by Operating Activities:	$ M	**62.5**	**59.2**	**59.2**	**59.9**	**61.4**	**62.4**
CASH FLOWS FROM INVESTING ACTIVITIES:							
(-) Capital Expenditures:	$ M	(30.0)	(32.0)	(34.0)	(36.0)	(38.0)	(40.0)
Net Cash Used in Investing Activities:	$ M	**(30.0)**	**(32.0)**	**(34.0)**	**(36.0)**	**(38.0)**	**(40.0)**
CASH FLOWS FROM FINANCING ACTIVITIES:							
(+) Debt Issuances:	$ M	100.0	-	-	-	-	-
(-) Debt OID & Issuance Fees:	$ M	(13.0)	-	-	-	-	-
(-) Debt Repayments:	$ M	-	(5.0)	-	(30.0)	(60.0)	(5.0)
(-) Dividends Paid:	$ M	(10.0)	(10.0)	(10.0)	(10.0)	(10.0)	(10.0)
(-) Common Stock Repurchases:	$ M	-	-	-	-	-	-
Net Cash Provided by Financing Activities:	$ M	**77.0**	**(15.0)**	**(10.0)**	**(40.0)**	**(70.0)**	**(15.0)**
Change in Cash & Cash Equivalents:	$ M	109.5	12.2	15.2	(16.1)	(46.6)	7.4
Beginning Cash:	$ M	977.5	1,087.0	1,099.2	1,114.5	1,098.4	1,051.8
Ending Cash:	$ M	**$ 1,087.0**	**$ 1,099.2**	**$ 1,114.5**	**$ 1,098.4**	**$ 1,051.8**	**$ 1,059.3**

In terms of **Book vs. Cash Taxes**, it's unclear whether or not these items affect anything, and it may be region-specific.

However, these Amortization and Loss items tend to be so small that the tax impact is also small, which is why we ignored it here.

If you wanted to model this in, you could include the standard Tax Schedule and assume that the Amortization of OID and Issuance Fees and Losses on Debt Extinguishment are *not* deductible for Cash-Tax purposes (which may or may not be true, depending on the region).

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Key Rule #3: Convertible Bond Accounting

In addition to "standard" Debt, such as bonds that companies issue to investors and promise to pay interest on, companies can also issue **Convertible Bonds** that may be "converted" into Common Shares if the company's Share Price exceeds a certain level (the "Conversion Price").

In exchange for this conversion option, investors accept **lower coupon rates**. So, a company might be able to pay 0.5% or 1.0% on a Convertible Bond if it would normally pay 4.0% or 5.0% on a traditional bond.

Convertible Bonds save companies money upfront by reducing their cash interest expense, but they can "cost" more if they convert into Equity later on.

For investors, Convertible Bonds are "hedged equity" – if the company does well, they receive shares in the company. And if it doesn't, at least they receive back their principal upon maturity.

The accounting for Convertible Bonds is tricky, so we're going to focus on the fundamentals here rather than covering every nuance and special case:

Step 1: Initial Convertible Bond Issuance

Under IFRS, the Convertible Bond is split into Liability and Equity components. Under U.S. GAAP, this happens only if there's a "cash-settlement option," but that option usually exists.

So, we will simplify and assume that this same split happens under both accounting systems.

To estimate the Market Value of the Liability component, take the Present Value of the Future Interest Payments and Future Principal Repayment, using a Discount Rate equal to the coupon rate on equivalent, non-convertible debt.

Convertible Bonds rarely, if ever, allow for "partial early repayment" – 100% is due at maturity, the company repays 100% early, or investors convert 100% of the bond into shares early.

But you won't see terms such as 20% principal repayment each year over 4 years, so we ignore that case here.

Once you have the Liability component, the Equity component equals the Face Value of Convertible Bond minus Market Value of Liability Component:

Debt Schedule and Assumptions:	Units:	Historical Year 0	Projected Year 1	Year 2	Year 3	Year 4	Year 5
Tax Rate:	%	25.0%					
Debt Issuance Fee %:	%	3.0%					
Initial Debt Issuance - Face Value:	$ M	$ 100.0					
Coupon Rate (Fixed Interest Rate):	%	0.5%					
Coupon Rate on Equivalent Non-Convertible Debt:	%	4.0%					
(+) PV of Future Interest Payments:	$ M	$ 2.2	$ 0.5	$ 0.5	$ 0.5	$ 0.5	$ 0.5
(+) PV of Principal:	$ M	=NPV($E$15,F18:J18)		-	-	-	100.0
PV of Liability Component:	$ M	**84.4**					
Face Value of Convertible Bond at Issuance:	$ M	100.0					
(-) Market Value of Liability Component:	$ M	(84.4)					
Market Value of Equity Component:	$ M	**15.6**					
Year of Bond's Conversion into Common Shares:	Year #	3	<-- Set to 0 to assume no conversion in projection period.				
Convertible Bond Tenor (Maturity):	Year #	5					

The **Book Value of Debt** equals the Face Value of $100.0 minus the $15.6 Market Value of the Equity Component minus the $3.0 in Issuance Fees, so it's initially $81.4.

Step 2: Handling Maturities and Conversions

With Convertible Bonds, there are two main possibilities:

1) They mature when they're supposed to (Year #5 here), and the company must repay the full principal at that time.

2) Or, they convert into Common Shares on or before that maturity date.

Sometimes, the company might also be able to repay the entire Convertible Bond early, but we're ignoring that case because the accounting is even more complicated/confusing.

The "Debt Conversions" formula in the schedule for the Face Value handles the conversion case:

	A B	C	D	E	F	G	H	I	J
5				0	1	2	3	4	5
6				Historical	Projected				
7		Debt Schedule and Assumptions:	Units:	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
8									
9		Tax Rate:	%	25.0%					
10		Debt Issuance Fee %:	%	3.0%					
11									
12		Initial Debt Issuance - Face Value:	$ M	$ 100.0					
13									
14		Coupon Rate (Fixed Interest Rate):	%	0.5%					
15		Coupon Rate on Equivalent Non-Convertible Debt:	%	4.0%					
16									
25		Year of Bond's Conversion into Common Shares:	Year #	3	<-- Set to 0 to assume no conversion in projection period.				
26		Convertible Bond Tenor (Maturity):	Year #	5					
27									
28		% Annual Debt Principal Repayment:	%		0.0%	0.0%	0.0%	0.0%	100.0%
29									
30		Amortization Period Remaining:	# Years	5	4	3	2	1	0
31									
42		Beginning Face Value of Debt:	$ M		100.0	100.0	100.0	-	-
43		(-) Debt Conversions:	$ M	=IF(F5=$E$25,-MIN(F42,$E$12),0)			(100.0)	-	-
44		(-) Debt Principal Maturities:	$ M		-	-	-	-	-
45		**Ending Face Value of Debt:**	$ M	**100.0**	**100.0**	**100.0**	**-**	**-**	**-**

- **Debt Conversions:** =IF(Current Year = Conversion Year, –MIN(Beginning Face Value This Year, Initial Face Value When Issued), 0)

Using the minimum here ensures that if the bond matures *before* the conversion date, "Debt Conversions" shows a 0 – instead of incorrectly showing a negative $100.

The Debt Principal Maturities formula is simpler:

	A B	C	D	E	F	G	H	I	J
5				0	1	2	3	4	5
6				Historical	Projected				
7		Debt Schedule and Assumptions:	Units:	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
8									
9		Tax Rate:	%	25.0%					
10		Debt Issuance Fee %:	%	3.0%					
11									
12		Initial Debt Issuance - Face Value:	$ M	$ 100.0					
13									
14		Coupon Rate (Fixed Interest Rate):	%	0.5%					
15		Coupon Rate on Equivalent Non-Convertible Debt:	%	4.0%					
16									
25		Year of Bond's Conversion into Common Shares:	Year #	3	<-- Set to 0 to assume no conversion in projection period.				
26		Convertible Bond Tenor (Maturity):	Year #	5					
27									
28		% Annual Debt Principal Repayment:	%		0.0%	0.0%	0.0%	0.0%	100.0%
29									
30		Amortization Period Remaining:	# Years	5	4	3	2	1	0
31									
42		Beginning Face Value of Debt:	$ M		100.0	100.0	100.0	-	-
43		(-) Debt Conversions:	$ M		-	-	(100.0)	-	-
44		(-) Debt Principal Maturities:	$ M		=-F28*SUM(F42:F43)		-	-	-
45		**Ending Face Value of Debt:**	$ M	**100.0**	**100.0**	**100.0**	**-**	**-**	**-**

This formula takes the Repayment Percentage (either 0% or 100%), multiplies it by the remaining Face Value this year (after any conversions), and then flips the sign to make it negative.

This formula means that if there's both a conversion and a maturity in the same year, we "prioritize" the conversion and assume it happens before the maturity.

Issuance Fees work the same way for Convertible Bonds – there's straight-line amortization until maturity, and there's accelerated amortization if there's an early repayment, maturity, or conversion:

		Historical	Projected				
Debt Schedule and Assumptions:	***Units:***	**Year 0**	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
Issuance Fee Beginning Balance:	*$ M*		3.0	2.4	1.8	-	-
(-) Amortization of Debt Issuance Fees:	*$ M*		(0.6)	(0.6)	(0.6)	-	-
(-) Loss on Debt Extinguishment:	*$ M*		-	-	(1.2)	-	-
Issuance Fee Ending Balance:	*$ M*	**3.0**	**2.4**	**1.8**	**-**	**-**	**-**

Accelerated the rest of the Issuance Fee amortization due to the conversion into shares in Year 3.

Step 3: Amortization of the Debt Discount

You could use the straight-line method to amortize the Debt Discount (which initially equals the "Equity Component" calculated in Step 1).

However, companies often use the **effective interest rate method** for Debt issuances with large discounts:

- **Amortization of Debt Discount** = –(Beginning Book Value of Debt This Year + Beginning Issuance Fee Balance This Year) * Coupon Rate on Equivalent, Non-Convertible Debt – Cash Interest Expense This Year)

This formula makes the Amortization higher each year as the Convertible Bond "approaches" a traditional bond:

Debt Schedule and Assumptions:	Units:	Historical Year 0	Projected Year 1	Year 2	Year 3	Year 4	Year 5
Tax Rate:	%	25.0%					
Debt Issuance Fee %:	%	3.0%					
Initial Debt Issuance - Face Value:	$ M	$ 100.0					
Coupon Rate (Fixed Interest Rate):	%	0.5%					
Coupon Rate on Equivalent Non-Convertible Debt:	%	4.0%					
Debt Discount Beginning Balance:	$ M		$ 15.6	$ 12.7	$ 9.7	$ -	$ 0.0
(-) Amortization of Debt Discount:	$ M		(2.9)	(3.0)	(3.1)	0.0	0.0
(-) Remaining Unamortized Discount:	$ M		-	-	(6.6)	-	-
Debt Discount Ending Balance:	$ M	**15.6**	**12.7**	**9.7**	**-**	**0.0**	**0.0**
Issuance Fee Beginning Balance:	$ M		3.0	2.4	1.8	-	-
(-) Amortization of Debt Issuance Fees:	$ M		(0.6)	(0.6)	(0.6)	-	-
(-) Loss on Debt Extinguishment:	$ M		-	-	(1.2)	-	-
Issuance Fee Ending Balance:	$ M	**3.0**	**2.4**	**1.8**	**-**	**-**	**-**
Beginning Book Value of Debt:	$ M		81.4	84.9	88.5	(0.0)	(0.0)
(+) Amortization of Debt Issuance Fees:	$ M		0.6	0.6	0.6	-	-
(+) Amortization of Debt Discount:	$ M		2.9	3.0	3.1	(0.0)	(0.0)
(+) Loss on Debt Extinguishment:	$ M		-	-	1.2	-	-
(+) Remaining Unamortized Discount:	$ M		-	-	6.6	-	-
(-) Debt Conversions:	$ M		-	-	(100.0)	-	-
(-) Debt Principal Maturities:	$ M		-	-	-	-	-
Ending Book Value of Debt:	$ M	**81.4**	**84.9**	**88.5**	**(0.0)**	**(0.0)**	**(0.0)**
(+) Interest Expense (Cash):	$ M		0.5	0.5	0.5	-	-
(+) Amortization of Debt Discount (Non-Cash):	$ M		2.9	3.0	3.1	(0.0)	(0.0)
(+) Amortization of Debt Issuance Fees (Non-Cash):	$ M		0.6	0.6	0.6	-	-
Total Interest Expense on Income Statement:	$ M		**4.0**	**4.1**	**4.2**	**(0.0)**	**(0.0)**
Losses on Debt Extinguishment:	$ M		**-**	**-**	**1.2**	**-**	**-**

Effective interest-rate method makes Amortization increase each year.

The "Remaining Unamortized Discount" part is still handled the same way: accelerate the Amortization of the remaining Discount if there's a conversion into shares.

Step 4: Linking the Statements

When linking this schedule to the rest of the financial statements:

- The **Cash Interest Expense**, **Amortization**, and the **Losses on Debt Extinguishment** all appear on the Income Statement and reduce Pre-Tax Income and Net Income.

- The **non-cash components** (the Amortization and Loss lines) are added back on the Cash Flow Statement.

- If there's a conversion into shares, the "Remaining Unamortized Discount" does **NOT** appear on the Income Statement.

- The **Equity Component** of the Convertible Bond never changes until there's a conversion, in which case it goes to $0. If the Convertible Bond matures without conversion, the Equity Component may also remain the same (though it's also acceptable to reduce it to $0 in that case).

- If there's a conversion, the Book Value of the Convertible Bond *after* Amortization and Losses on Debt Extinguishment in the year and the Equity Component both get "transferred" into Common Shareholders' Equity.

  This step keeps the Balance Sheet in balance because the separate Liability and Equity components go to $0, and Common Shareholders' Equity increases by their sum.

Here's what the Cash Flow Statement looks like with a Year 1 issuance and Year 3 conversion:

		Historical	Projected				
Cash Flow Statement:	*Units:*	**Year 0**	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
CASH FLOWS FROM OPERATING ACTIVITIES:							
Net Income:	*$ M*	**$ 37.5**	**$ 34.5**	**$ 34.4**	**$ 33.4**	**$ 37.5**	**$ 37.5**
Adjustments for Non-Cash Charges:							
(+) Depreciation & Amortization:	*$ M*	25.0	25.0	25.0	25.0	25.0	25.0
(+) Amortization of Debt Discount:	*$ M*	-	2.9	3.0	3.1	(0.0)	(0.0)
(+) Amortization of Debt Issuance Fees:	*$ M*	-	0.6	0.6	0.6	-	-
(+) Losses on Debt Extinguishment:	*$ M*	-	-	-	1.2	-	-
(+/-) Deferred Taxes:	*$ M*	-	-	-	-	-	-
(+/-) Change in Op. Lease Assets:	*$ M*	-	(2.0)	(2.0)	(2.0)	(2.0)	(2.0)
(+/-) Change in Op. Lease Liabilities:	*$ M*	-	2.0	2.0	2.0	2.0	2.0
Changes in Operating Assets and Liabilities:							
Inventory:	*$ M*	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)
Accounts Receivable:	*$ M*	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)
Accounts Payable:	*$ M*	5.0	5.0	5.0	5.0	5.0	5.0
Accrued Liabilities:	*$ M*	5.0	5.0	5.0	5.0	5.0	5.0
Net Cash Provided by Operating Activities:	*$ M*	**62.5**	**63.0**	**63.0**	**63.4**	**62.5**	**62.5**
CASH FLOWS FROM INVESTING ACTIVITIES:							
(-) Capital Expenditures:	*$ M*	(30.0)	(32.0)	(34.0)	(36.0)	(38.0)	(40.0)
Net Cash Used in Investing Activities:	*$ M*	**(30.0)**	**(32.0)**	**(34.0)**	**(36.0)**	**(38.0)**	**(40.0)**
CASH FLOWS FROM FINANCING ACTIVITIES:							
(+) Debt Issuances:	*$ M*	100.0	-	-	-	-	-
(-) Debt Issuance Fees:	*$ M*	(3.0)	-	-	-	-	-
(-) Debt Principal Maturities:	*$ M*	-	-	-	-	-	-
(-) Dividends Paid:	*$ M*	(10.0)	(10.0)	(10.0)	(10.0)	(10.0)	(10.0)
(-) Common Stock Repurchases:	*$ M*	-	-	-	-	-	-
Net Cash Provided by Financing Activities:	*$ M*	**87.0**	**(10.0)**	**(10.0)**	**(10.0)**	**(10.0)**	**(10.0)**
Change in Cash & Cash Equivalents:	*$ M*	119.5	21.0	19.0	17.4	14.5	12.5
Beginning Cash:	*$ M*	977.5	1,097.0	1,118.0	1,137.0	1,154.4	1,168.9
Ending Cash:	*$ M*	**$ 1,097.0**	**$ 1,118.0**	**$ 1,137.0**	**$ 1,154.4**	**$ 1,168.9**	**$ 1,181.4**

Normal add-backs of non-cash Income Statement expenses associated with the Convertible.

The conversion into shares does not make any cash impact! Only issuances, maturities, and fees do.

And here's the Balance Sheet:

Balance Sheet:	Units:	Historical Year 0	Projected Year 1	Year 2	Year 3	Year 4	Year 5
ASSETS:							
Current Assets:							
Cash & Equivalents:	$ M	$ 1,097.0	$ 1,118.0	$ 1,137.0	$ 1,154.4	$ 1,168.9	$ 1,181.4
Inventory:	$ M	200.0	205.0	210.0	215.0	220.0	225.0
Accounts Receivable:	$ M	250.0	255.0	260.0	265.0	270.0	275.0
Total Current Assets:	$ M	**1,547.0**	**1,578.0**	**1,607.0**	**1,634.4**	**1,658.9**	**1,681.4**
Non-Current Assets:							
Net PP&E:	$ M	400.0	407.0	416.0	427.0	440.0	455.0
Operating Lease Assets:	$ M	150.0	152.0	154.0	156.0	158.0	160.0
Total Non-Current Assets:	$ M	**550.0**	**559.0**	**570.0**	**583.0**	**598.0**	**615.0**
Total Assets:	$ M	**$ 2,097.0**	**$ 2,137.0**	**$ 2,177.0**	**$ 2,217.4**	**$ 2,256.9**	**$ 2,296.4**
LIABILITIES AND EQUITY:							
Current Liabilities:							
Accounts Payable:	$ M	$ 150.0	$ 155.0	$ 160.0	$ 165.0	$ 170.0	$ 175.0
Accrued Liabilities:	$ M	200.0	205.0	210.0	215.0	220.0	225.0
Total Current Liabilities:	$ M	**350.0**	**360.0**	**370.0**	**380.0**	**390.0**	**400.0**
Non-Current Liabilities:							
Total Debt:	$ M	81.4	84.9	88.5	-	(0.0)	(0.0)
Operating Lease Liabilities:	$ M	150.0	152.0	154.0	156.0	158.0	160.0
Deferred Tax Liabilities:	$ M	100.0	100.0	100.0	100.0	100.0	100.0
Total Non-Current Liabilities:	$ M	**331.4**	**336.9**	**342.5**	**256.0**	**258.0**	**260.0**
Total Liabilities:	$ M	**681.4**	**696.9**	**712.5**	**636.0**	**648.0**	**660.0**
Common Shareholders' Equity:	$ M	1,400.0	1,424.5	1,448.9	1,581.4	1,608.9	1,636.4
Equity Component of Convertible Bond:	$ M	15.6	15.6	15.6	-	-	-
Total Equity:	$ M	**1,415.6**	**1,440.1**	**1,464.5**	**1,581.4**	**1,608.9**	**1,636.4**
Total Liabilities & Equity:	$ M	**$ 2,097.0**	**$ 2,137.0**	**$ 2,177.0**	**$ 2,217.4**	**$ 2,256.9**	**$ 2,296.4**
Balance Check:		*OK!*	*OK!*	*OK!*	*OK!*	*OK!*	*OK!*

The Liability and Equity components both get "transferred" into Common Shareholders' Equity if there's a conversion - but the Liability component is adjusted for the Amortization and Loss items in Year 3.

Summary of Convertible Bond Accounting

When a Convertible Bond is issued, it's split into **Debt** and **Equity** components. The Equity Component stays the same until conversion (or maturity, though the treatment her varies).

The **Debt Component** keeps increasing due to the Amortization of the Debt Discount (and the Amortization of the Issuance Fees), and it reaches the Face Value if the bond matures without being converted.

If the bond is converted into common shares, the Debt Component and Equity Component are both "transferred" into Common Shareholders' Equity because this capital is now considered 100% Equity, not Debt or "hybrid capital."

We did not look at **changes in the share count** here, but if there's a conversion, the company's common shares outstanding increase, and the Interest Expense goes to $0.

The number of common shares created depends on the terms of the Convertible Bond, but it's equal to approximately Face Value / Conversion Price.

So, if the company's Share Price reaches the Conversion Price, the Face Value of the Convertible Bond is $100 million, and the Conversion Price is $50.00, then $100 million / $50.00 = 2 million new shares will be created.

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Key Rule #4: Types of Debt, PIK Interest, and Debt Schedules in Real Life

So far, we've been lumping all "Debt" or "Bonds" in the same category and assuming a fixed coupon rate with principal repayments over time, or "bullet maturity" (i.e., the entire balance must be repaid on one date in the future).

In reality, however, there are many types of Debt; the two main categories are "Secured" and "Unsecured" Debt.

Secured means it's backed by collateral, and Unsecured means it's backed by nothing.

"Backed by collateral" means that if the company goes bankrupt, and it cannot repay the Debt, lenders can seize some of its assets and sell them to recover some money.

Within these two major categories, there are also different types of Debt; the main characteristics include:

- **Coupon Rate (or "Interest Rate"):** Is it "fixed" (e.g., 10% or 5%) or "floating" (e.g., Fed Funds Rate + 2.0% or Other Benchmark + 5.0%)? Is it higher or lower than rates on other types of Debt?

- **Form of Interest:** Does the company pay it in cash, or does interest accrue to the principal ("Paid-in-Kind" or PIK Interest)?

- **Tenor:** For how long is the Debt outstanding? Five years? Ten years? Something in between?

- **Amortization:** What percentage of the Debt principal must the company repay each year? Or does it not repay anything until maturity?

- **Prepayments:** If a company has extra cash, can it repay some of its outstanding Debt before maturity? Or is it not allowed? Are there penalty fees?

- **Investors:** Who are the typical buyers for this type of Debt? Traditional banks or riskier entities such as hedge funds, merchant banks, and mezzanine funds?

- **Seniority:** In the case of a bankruptcy, who gets paid first? What are the chances of the investors in this tranche of Debt recovering their principal?

- **Call Protection:** Is the company prohibited from repaying the full Debt principal for some time? Is there a penalty fee (a "call premium") for doing so?

- **Covenants:** Does the company have to comply with certain financial ratios (e.g., Debt / EBITDA must stay below 5x), known as "maintenance covenants," or does it have to comply with "incurrence covenants," which prohibit certain *actions*?

Here's a summary:

Debt Type:	Revolver	Term Loan A	Term Loan B	Senior Notes	Subordinated Notes	Mezzanine
Interest Rate:	Lowest	Low	Higher	Higher	Higher	Highest
Floating / Fixed?	Floating			Fixed		
Cash Interest?	Yes					Cash / PIK
Tenor:	3-5 years	4-6 years	4-8 years	7-10 years	8-10 years	8-12 years
Amortization:	None	Straight Line	Minimal	Bullet		
Prepayment?	Yes			No		
Investors:	Conservative Banks			HFs, Merchant Banks, Mezzanine Funds		
Seniority:	Senior Secured			Senior Unsecured	Senior Subordinated	Equity
Secured?	Yes			Sometimes	No	
Call Protection?	No	Sometimes		Yes		
Covenants:	Maintenance			Incurrence		

You know how to model many of these items on the financial statements already, such as the Amortization and Prepayment features (Call Protection and Maintenance Covenants will be covered the LBO and Debt/Equity guides).

But one new feature here is **Paid-in-Kind (PIK) Interest**, also known as **Accrued Interest**.

In real life, student loans work like this: you have no income while in school, so the interest accrues to the loan principal.

So, the loan principal increases each year, which increases the interest you pay each year.

PIK Interest is common on the riskiest, highest-interest-rate forms of Debt, such as Mezzanine.

Mechanics of PIK Interest

If the Interest is **Paid-in-Kind**, it increases both the Face Value and the Book Value of Debt in each period.

You need to be careful with **Debt Repayments & Maturities** when calculating the Face Value of Debt – if there's PIK Interest, and it's the maturity year, then you must repay (Beginning Face Value + PIK Interest This Year):

		0	1	2	3	4	5
		Historical	Projected				
Debt Schedule and Assumptions:	*Units:*	**Year 0**	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
Tax Rate:	%	25.0%					
Debt Issuance Fee %:	%	3.0%					
Initial Debt Issuance - Face Value:	$ M	$ 100.0					
Initial Debt Issuance - Issued At:	$ M	90.0					
Debt Maturity:	Year #	5					
Coupon Rate (Fixed Interest Rate):	%	5.0%					
Paid-in-Kind (PIK) Interest?	Text	Yes					
% Annual Debt Principal Repayment:	%		0.0%	0.0%	0.0%	0.0%	100.0%
Amortization Period Remaining:	# Years	5	4	3	2	1	0
Beginning Face Value of Debt:	$ M		100.0	105.0	110.3	115.8	121.6
(+) Paid-in-Kind (PIK) Interest:	$ M		5.0	5.3	5.5	5.8	6.1
(-) Debt Principal Repayments & Maturities:	$ M		=IF(G5=$E$15,-SUM(G32:G33),0)			-	(127.6)
Ending Face Value of Debt:	$ M	**100.0**	**105.0**	**110.3**	**115.8**	**121.6**	**-**
Beginning Book Value of Debt:	$ M		87.0	94.6	102.5	110.6	119.0
(+) Paid-in-Kind (PIK) Interest:	$ M		5.0	5.3	5.5	5.8	6.1
(+) Amortization of Debt Issuance Fees:	$ M		0.6	0.6	0.6	0.6	0.6
(+) Amortization of OID:	$ M		2.0	2.0	2.0	2.0	2.0
(+) Loss on Debt Extinguishment:	$ M		-	-	-	-	-
(+) Loss on Unamortized OID on Repayment:	$ M		-	-	-	-	-
(-) Debt Principal Repayments:	$ M		-	-	-	-	(127.6)
Ending Book Value of Debt:	$ M	**87.0**	**94.6**	**102.5**	**110.6**	**119.0**	**-**
(+) Interest Expense:	$ M		5.0	5.3	5.5	5.8	6.1
(+) Amortization of OID (Non-Cash):	$ M		2.0	2.0	2.0	2.0	2.0
(+) Amortization of Debt Issuance Fees (Non-Cash):	$ M		0.6	0.6	0.6	0.6	0.6
Total Interest Expense on Income Statement:	$ M		**7.6**	**7.9**	**8.1**	**8.4**	**8.7**
Losses on Debt Extinguishment:	$ M	**-**	**-**	**-**	**-**	**-**	**-**

PIK Interest increases both the Face Value and Book Value each year, and we must include it in the Principal Repayment when the Debt *matures* (i.e., the full amount is owed back to lenders).

PIK Interest still appears within Interest Expense on the Income Statement and reduces Pre-Tax Income and Net Income, and it's added back as a non-cash expense on the Cash Flow Statement.

This CFS add-back then links to Debt on the Balance Sheet, which increases each year until maturity:

LIABILITIES AND EQUITY:							
Current Liabilities:							
Accounts Payable:	$ M	$ 150.0	$ 155.0	$ 160.0	$ 165.0	$ 170.0	$ 175.0
Accrued Liabilities:	$ M	200.0	205.0	210.0	215.0	220.0	225.0
Total Current Liabilities:	$ M	**350.0**	**360.0**	**370.0**	**380.0**	**390.0**	**400.0**
Non-Current Liabilities:							
Total Debt:	$ M	87.0	94.6	102.5	110.6	119.0	-
Operating Lease Liabilities:	$ M	150.0	152.0	154.0	156.0	158.0	160.0
Deferred Tax Liabilities:	$ M	100.0	100.0	100.0	100.0	100.0	100.0
Total Non-Current Liabilities:	$ M	**337.0**	**346.6**	**356.5**	**366.6**	**377.0**	**260.0**
Total Liabilities:	$ M	**687.0**	**706.6**	**726.5**	**746.6**	**767.0**	**660.0**
Common Shareholders' Equity:	$ M	**1,400.0**	**1,421.8**	**1,443.4**	**1,464.8**	**1,486.0**	**1,507.0**
Total Liabilities & Equity:	$ M	**$ 2,087.0**	**$ 2,128.4**	**$ 2,169.9**	**$ 2,211.4**	**$ 2,253.0**	**$ 2,167.0**
Balance Check:		*OK!*	*OK!*	*OK!*	*OK!*	*OK!*	*OK!*

		Historical	Projected				
Cash Flow Statement:	*Units:*	**Year 0**	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
CASH FLOWS FROM OPERATING ACTIVITIES:							
Net Income:	$ M	**$ 37.5**	**$ 31.8**	**$ 31.6**	**$ 31.4**	**$ 31.2**	**$ 31.0**
Adjustments for Non-Cash Charges:							
(+) Depreciation & Amortization:	$ M	25.0	25.0	25.0	25.0	25.0	25.0
(+) PIK Interest:	$ M	-	5.0	5.3	5.5	5.8	6.1
(+) Amortization of OID:	$ M	-	2.0	2.0	2.0	2.0	2.0
(+) Amortization of Debt Issuance Fees:	$ M	-	0.6	0.6	0.6	0.6	0.6
(+) Losses on Debt Extinguishment:	$ M	-	-	-	-	-	-
(+/-) Deferred Taxes:	$ M	-	-	-	-	-	-
(+/-) Change in Op. Lease Assets:	$ M	-	(2.0)	(2.0)	(2.0)	(2.0)	(2.0)
(+/-) Change in Op. Lease Liabilities:	$ M	-	2.0	2.0	2.0	2.0	2.0
Changes in Operating Assets and Liabilities:							
Inventory:	$ M	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)
Accounts Receivable:	$ M	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)
Accounts Payable:	$ M	5.0	5.0	5.0	5.0	5.0	5.0
Accrued Liabilities:	$ M	5.0	5.0	5.0	5.0	5.0	5.0
Net Cash Provided by Operating Activities:	$ M	**62.5**	**64.4**	**64.5**	**64.5**	**64.6**	**64.7**

The usual items link into Debt on the Balance Sheet, including the Amortization ones, Debt Issuances, Repayments, etc.

But the PIK Interest add-back on the CFS, in particular, makes the Total Debt balance keep increasing each year until maturity.

Debt Schedules in Real Life: Toro and Atlassian

In real life, you do not always follow all the accounting rules for the treatment of Debt in this section and the previous ones.

Many of these rules make a *tiny difference* on the financial statements, so you often simplify and skip items like the Debt Issuance Fees and the related Amortization and Loss line items.

You can also simplify the formulas by considering only certain cases.

For example, with Convertible Bonds, you could skip the case where there's a conversion before maturity or early repayment and just focus on what happens at maturity.

And you could make the Amortization of the Debt Discount much simpler by using the straight-line method rather than the effective interest rate method.

Below is an example of a Convertible Bond schedule for Atlassian, where we allow only a full principal repayment upon maturity (though we do use the effective interest rate method for the Amortization of the Debt Discount):

Atlassian - Convertible Bond Schedule:		Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Beginning Face Value of Debt:	$ M		$ 1,000.0	$ 1,000.0	$ 1,000.0	$ 1,000.0	$ 1,000.0
(-) Debt Principal Repayments & Maturities:	$ M		-	-	-	-	(1,000.0)
Ending Face Value of Debt:	$ M	1,000.0	**1,000.0**	**1,000.0**	**1,000.0**	**1,000.0**	**-**
Beginning Book Value of Debt:	$ M		807.5	843.0	880.0	918.4	958.4
(+) Amortization of Debt Discount:	$ M		31.5	32.9	34.4	36.0	37.6
(+) Remaining Unamortized Debt Discount:	$ M		-	-	-	-	0.0
(+) Amortization of Debt Issuance Fees:	$ M		4.0	4.0	4.0	4.0	4.0
(-) Debt Principal Repayments & Maturities:	$ M		-	-	-	-	(1,000.0)
Ending Book Value of Debt:	$ M	**807.5**	**843.0**	**880.0**	**918.4**	**958.4**	**-**
Future Interest Payments:	$ M		6.3	6.3	6.3	6.3	6.3
Principal Repayments:	$ M		-	-	-	-	1,000.0
Market Value of Debt Component:	$ M	**827.5**	**859.0**	=NPV($E$30,H19:$J19)+NPV($E$30,H20:$J20)			
Debt Issuance Fee %:	%	2.0%					
Unamortized Debt Issuance Fees:	$ M	20.0	16.0	12.0	8.0	4.0	-
Unamortized Debt Discount:	$ M	172.5	141.0	108.0	73.6	37.6	-
Debt Maturity:	*Year #*	5					
Coupon Rate (Fixed Interest Rate):	%	0.625%					
Coupon Rate on Equivalent Non-Convertible Debt:	%	4.560%					
Interest Expense (Cash):	$ M		6.3	6.3	6.3	6.3	6.3
Amortization of Debt Discount (Non-Cash):	$ M		31.5	32.9	34.4	36.0	37.6
Amortization of Debt Issuance Fees (Non-Cash):	$ M		4.0	4.0	4.0	4.0	4.0
Total Interest Expense on Income Statement:	$ M		**$ 41.7**	**$ 43.2**	**$ 44.7**	**$ 46.2**	**$ 47.9**

Face Value stays the same at $1,000, until maturity.

But the Book Value keeps increasing due to the Amortization of the huge Debt Discount, and the Amortization of the Issuance Fees.

And the Market Value of the Debt keeps increasing as the Debt approaches maturity and becomes more like a "traditional Bond."

The Interest Expense is huge, but the vast majority is non-cash.

Here's another example for Toro, which has traditional Debt, but which *keeps issuing new Debt* to fund its business each year.

That means that the company must deduct Issuance Fees each year and then amortize those fees over time.

The "deduction" part is easy (Issuance Fee % * Debt Issued), but the Amortization line item needs to be adjusted each year as more Debt is issued.

We use the following approach to do it:

Toro - Debt Schedule:								
Beginning Face Value of Debt:	*$ M*		$ 314.9	$ 314.9	$ 314.9	$ 339.6	$ 413.2	
(+) Debt Issuances:	*$ M*		-	-	24.8	73.6	176.5	
(-) Principal Repayments & Maturities:	*$ M*		-	-	-	-	(91.0)	
Ending Face Value of Debt:	*$ M*	**314.9**	**314.9**	**314.9**	**339.6**	**413.2**	**498.7**	
Beginning Book Value of Debt:	*$ M*		308.6	309.2	309.8	334.7	407.5	
(+) Amortization of Issuance Fees:	*$ M*		0.6	0.6	0.6	=($E$52-SUM($E46:H46))/$E$53		
(-) Deduction for New Issuance Fees:	*$ M*		-	-	(0.5)	(1.5)	(3.5)	
(+) Debt Issuances:	*$ M*		-	-	24.8	73.6	176.5	
(-) Debt Principal Repayments & Maturities:	*$ M*		-	-	-	-	(91.0)	
Ending Book Value of Debt:	*$ M*	**308.6**	**309.2**	**309.8**	**334.7**	**407.5**	**490.3**	
Debt Issuance Fee %:	*%*	2.0%						
Initial Unamortized Debt Issuance Fees:	*$ M*	6.3						
Average Remaining Time to Maturity:	*# Years*	10						
Weighted Average Interest Rate on Debt:	*%*	6.1%						
Interest Expense (Cash):	*$ M*		19.2	19.2	19.2	20.7	25.2	
Amortization of Debt Issuance Fees (Non-Cash):	*$ M*		0.6	0.6	0.6	0.7	0.8	
Total Interest Expense on Income Statement:	*$ M*		**$ 19.8**	**$ 19.8**	**$ 19.8**	**$ 21.4**	**$ 26.0**	

Each year, the Amortization = (Initial Unamortized Issuance Fees + Cumulative Additional Issuance Fees) / Average Remaining Time to Maturity of 10 Years.

This approach is not robust because after 10 years, the Amortization of Issuance Fees for Debt issued 10 years ago should go to $0!

This model only contains 5 years, though, so we do not handle this case where the Amortization *decreases*.

If we had to project 20-30 years into the future, we would have to handle that case, and we'd likely create a separate Amortization schedule to track the total number properly.

However, none of that is necessary for a quick/simple model that covers only a few years.

You can always make Debt schedules more complex, but the added complexity is rarely worthwhile if you're building a quick 3-statement or cash-flow model.

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Key Rule #5: Equity Method of Accounting (Equity Investments)

When a Parent Company (Parent Co.) owns a **minority stake** in another company (Sub Co.), such as 20% or 40% of its common shares outstanding, the Parent Co. must use the **equity method of accounting** to record its stake on the financial statements.

Technically, companies use this method when they have "significant influence" over another company, **but NOT control of that other company**.

In practical terms, that translates to ownership stakes between 20% and 50%, though some companies also use the equity method for ownership stakes below 20%.

If the Parent Co. owns *very little* of Sub Co., such as 1% of the shares, it won't use the equity method because that stake will count as a normal security on the Balance Sheet.

Above 50% ownership, Parent Co. controls Sub Co., so consolidation accounting applies.

The **main idea** of the equity method is that Parent Co. records its Ownership Percentage * Sub Co.'s Net Income on its Income Statement under "Equity Investment Earnings," or a similar name, but *nothing else from Sub Co.* – all the items above it on the IS are only the Parent's.

Then, Parent Co. reverses that line item on its Cash Flow Statement and records Ownership Percentage * Sub Co.'s Dividends as a positive on its CFS.

Both these Cash Flow Statement line items then link into Equity Investments on the Assets side of the Balance Sheet. Here's a simple example of the Income Statement:

Equity Investments / Associate Companies:

Combined Company:	Year 1
Associate Company Market Cap:	$ 100
Ownership in Associate:	30.0%
Value of Equity Investments:	30

Parent Company:	Year 1
Tax Rate:	25.0%

Associate Company:	Year 1
Tax Rate:	25.0%

Combined Company:	Year 1
Revenue:	$ 400
(-) Costs of Goods Sold:	(100)
Gross Profit:	300
(-) Total Operating Expenses:	(200)
(-) Depreciation & Amortization:	(25)
Operating Income:	75
(-) Net Interest Expense:	(15)
Pre-Tax Income:	60
(-) Income Taxes:	(15)
Net Income:	45
(+) Equity Investment Earnings:	5
Net Income to Parent:	$ 50
EBIT:	75
EBITDA:	100

Parent Company:	Year 1
Revenue:	$ 400
(-) Costs of Goods Sold:	(100)
Gross Profit:	300
(-) Total Operating Expenses:	(200)
(-) Depreciation & Amortization:	(25)
Operating Income:	75
(-) Net Interest Expense:	(15)
Pre-Tax Income:	60
(-) Income Taxes:	(15)
Net Income:	$ 45
EBIT:	75
EBITDA:	100

Associate Company:	Year 1
Revenue:	$ 100
(-) Costs of Goods Sold:	(20)
Gross Profit:	80
(-) Total Operating Expenses:	(55)
(-) Depreciation & Amortization:	(5)
Operating Income:	20
(-) Net Interest Expense:	-
Pre-Tax Income:	20
(-) Income Taxes:	(5)
Net Income:	$ 15
EBIT:	20
EBITDA:	25

Multiply by the 30% Ownership and list on the Parent's IS.

To demonstrate the treatment on the full financial statements, we'll start with a simple example of a Parent Co. with $400 million in revenue, growing to $600 million in Year 5.

It's about 10x the size of Sub Co., which has $40 million in revenue growing to $60 million in the same period.

We'll assume that Parent Co. acquires a **30%** stake in Sub Co. when Sub Co.'s Market Cap is $100 million, at the end of Year 2, and that it uses 50% Cash and 50% Debt to do the deal.

Here's what the Income Statement looks like:

Model Assumptions:					
Parent Company Tax Rate:	25.0%				
Interest Rate on Debt:	5.0%				
Minority-Stake Investments:		**End of Year 2:**	**End of Year 3:**	**End of Year 4:**	**End of Year 5:**
Market Cap of Associate Company:		$ 100	$ 100	$ 100	$ 100
Ownership in Associate Becomes:	0.0%	30.0%	30.0%	30.0%	30.0%
Percentage Change in Equity Investments:		30.0%	0.0%	0.0%	0.0%
Change in Equity Investment Dollar Amount:		30	-	-	-
Cost Basis Just Before Change:					
Gain or Loss on Sale:					
% Cash Used in Acquisition:		50.0%	50.0%	50.0%	50.0%
% Debt Used in Acquisition:		50.0%	50.0%	50.0%	50.0%
Ownership DURING the Period:	0.0%	0.0%	30.0%	30.0%	30.0%

Income Statement	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue:	$ 400	$ 450	$ 500	$ 550	$ 600
(-) Costs of Goods Sold:	(100)	(120)	(140)	(160)	(180)
Gross Profit:	300	330	360	390	420
(-) Total Operating Expenses:	(200)	(220)	(235)	(245)	(255)
(-) Depreciation & Amortization:	(25)	(30)	(35)	(40)	(45)
Operating Income:	75	80	90	105	120
(-) Net Interest Expense:	-	-	(1)	(1)	(1)
(+) Gains / (-) Losses on Eq. Inv. Sales:					
Pre-Tax Income:	75	80	89	104	119
(-) Income Taxes:	(19)	(20)	(22)	(26)	(30)
Net Income:	56	60	67	78	89
(+) Equity Investment Earnings:	-	-	2	2	3
Net Income to Parent:	$ 56	$ 60	$ 69	$ 81	$ 92

New Interest Expense on Debt used to fund the acquisition of this 30% stake.

30% of Sub Co.'s Net Income starting in Year 3, just after the acquisition closes.

If the Market Cap of the Associate Company (Sub Co.) changes, that does not appear on the Income Statement unless Parent Co. sells some of its stake based on that new Market Cap.

In other words, *Unrealized Gains and Losses* on Equity Investments do not appear on the financial statements – only Realized Gains and Losses (see the next section).

Parent Co.'s Cash Flow Statement looks like this:

Cash Flow Statement:	Year 1	Year 2	Year 3	Year 4	Year 5
CASH FLOWS FROM OPERATING ACTIVITIES:					
Net Income:	$ 56	$ 60	$ 69	$ 81	$ 92
Adjustments for Non-Cash Charges:					
(+) Depreciation & Amortization:	25	30	35	40	45
(+/-) Deferred Taxes:	4	4	4	5	5
(-) Gains / (+) Losses on Eq. Inv. Sales:					
(-) Equity Investment Earnings:	-	-	(2)	(2)	(3)
(+) Dividends from Equity Investments:	-	-	1	1	1
(+/-) Change in Op. Lease Assets:	(20)	(20)	(20)	(20)	(20)
(+/-) Change in Op. Lease Liabilities:	20	20	20	20	20
Changes in Operating Assets and Liabilities:					
Accounts Receivable:	(40)	(40)	(40)	(40)	(40)
Inventory:	(30)	(30)	(30)	(30)	(30)
Accounts Payable:	30	30	30	30	30
Accrued Liabilities:	20	20	20	20	20
Net Cash Provided by Operating Activities:	**65**	**74**	**87**	**104**	**120**
CASH FLOWS FROM INVESTING ACTIVITIES:					
(-) Capital Expenditures:	(30)	(35)	(40)	(45)	(50)
(-) Purchase / (+) Sale of Equity Investments:	-	(30)	-	-	-
Net Cash Used in Investing Activities:	**(30)**	**(65)**	**(40)**	**(45)**	**(50)**
CASH FLOWS FROM FINANCING ACTIVITIES:					
(+) Debt Issuances:	-	15	-	-	-
(-) Debt Repayments:	-	-	-	-	-
(-) Common Dividends Paid:	(20)	(22)	(24)	(27)	(30)
(+) Stock Issuances:	-	-	-	-	-
(-) Stock Repurchases:	-	-	-	-	-
Net Cash Provided by Financing Activities:	**(20)**	**(7)**	**(24)**	**(27)**	**(30)**
Change in Cash & Cash Equivalents:	**15**	**2**	**23**	**32**	**40**
Beginning Cash:	100	115	117	140	172
Ending Cash:	$ 115	$ 117	$ 140	$ 172	$ 212

Parent Co. reverses the Equity Investment Earnings and records Ownership Percentage * Sub Co. Dividends as a positive cash inflow on the CFS.

And purchases and sales of stakes in Sub Co. show up within Cash Flow from Investing, with Debt and Stock Issuances to fund them in Cash Flow from Financing.

On the Balance Sheet, the initial Purchase of Equity Investments at the end of Year 2 creates the "Equity Investments" line item on the Assets side.

This line item is also called "Associate Companies" or "Associates" in some regions.

After it's created, it increases by the Equity Investment Earnings and decreases by Dividends from Equity Investments:

Balance Sheet:	Year 1	Year 2	Year 3	Year 4	Year 5
ASSETS:					
Current Assets:					
Cash & Equivalents:	$ 115	$ 117	$ 140	$ 172	$ 212
Accounts Receivable:	300	340	380	420	460
Inventory:	150	180	210	240	270
Total Current Assets:	**565**	**637**	**730**	**832**	**942**
Non-Current Assets:					
Net PP&E:	500	505	510	515	520
Operating Lease Assets:	200	220	240	260	280
Equity Investments:	-	30	31	33	35
Total Non-Current Assets:	**700**	**755**	**781**	**808**	**835**
Total Assets:	**$ 1,265**	**$ 1,392**	**$ 1,511**	**$ 1,640**	**$ 1,777**
LIABILITIES AND EQUITY:					
Current Liabilities:					
Accounts Payable:	$ 300	$ 330	$ 360	$ 390	$ 420
Accrued Liabilities:	200	220	240	260	280
Total Current Liabilities:	**500**	**550**	**600**	**650**	**700**
Non-Current Liabilities:					
Total Debt:	-	15	15	15	15
Operating Lease Liabilities:	200	220	240	260	280
Deferred Tax Liabilities:	150	154	158	163	168
Total Non-Current Liabilities:	**350**	**389**	**413**	**438**	**463**
Total Liabilities:	**850**	**939**	**1,013**	**1,088**	**1,163**
Common Shareholders' Equity:	**415**	**453**	**498**	**552**	**614**
Total Liabilities & Equity:	**$ 1,265**	**$ 1,392**	**$ 1,511**	**$ 1,640**	**$ 1,777**
Balance Check:	*OK!*	*OK!*	*OK!*	*OK!*	*OK!*

Initial balance, and then *subtract* the Equity Investment Earnings on the CFS and the Dividends from Equity Investments on the CFS.

By subtracting, we're really *adding* the Earnings and subtracting the Dividends due to their signs on the CFS.

The rationale for this treatment is that if Parent Co. does **NOT** control Sub Co., then it does not "get" Sub Co.'s Net Income in cash each year.

It owns less than 50% of the company, so it can't walk over to Sub Co. and say, "Hey! Please give us all your Net Income right now. Forget about your other shareholders."

Therefore, Earnings from Equity Investments are reversed on the CFS, which means they're subtracted if they're positive on the IS.

Parent Co. receives in cash only Sub Co.'s Dividends * its Ownership Percentage.

If Parent Co. **changes** its ownership in Sub Co., the accounting gets more complicated.

Before explaining the rules, we'll impose a significant **constraint:** Parent Co. can own between 0% and 49% of Sub Co., but no more than that.

If Parent Co. moves into a control position (>= 50% ownership), consolidation accounting in the next section applies.

To make the model support changes in the ownership stake, we need Sub Co.'s Market Cap and the new Ownership Percentage in each year (assuming end-of-year changes only):

Minority-Stake Investments:		End of Year 2:	End of Year 3:	End of Year 4:	End of Year 5:
Market Cap of Associate Company:		$ 100	$ 50	$ 150	$ 200
Ownership in Associate Becomes:	0.0%	30.0%	0.0%	40.0%	25.0%

The **easy part** is determining the Percentage Change in Equity Investments and the Change in the Equity Investment Dollar Amount.

The Percentage Change is a simple subtraction, while the change in the Equity Investment Dollar Amount is Percentage Change in Period * Market Cap of Associate Company in Period.

If the Parent Co. is **increasing** its stake, we can stop there and record this as a cash outflow on the CFS that links into the Equity Investments line item on the BS.

But if Parent Co.'s stake in Sub Co. *decreases*, that means it has sold some of its stake, which means there will almost always be a Realized Gain or Loss.

To calculate this Realized Gain or Loss, we need the Cost Basis right before the change takes place, as well as the market value at which the stake was sold.

- **Cost Basis** = Equity Investments in Previous Period – Equity Investment Earnings This Period – Dividends from Equity Investments This Period

Because of the signs of these items on the CFS, we are *adding* Equity Investment Earnings and *subtracting* Dividends from Equity Investments.

To calculate the Gain or Loss, we use this formula:

- **Realized Gain or Loss** = IF(Percentage Change is Negative, (Sub Co. Market Cap * Previous Ownership Percentage – Cost Basis) * –Percentage Change / Previous Ownership Percentage, 0)

Here it is in Excel:

	A	B	C	D	E	F	G	H	I	J
9										
10			**Minority-Stake Investments:**		**End of Year 2:**	**End of Year 3:**	**End of Year 4:**	**End of Year 5:**		
11			Market Cap of Associate Company:		$ 100	$ 50	$ 150	$ 200		
12			Ownership in Associate Becomes:	0.0%	30.0%	0.0%	40.0%	25.0%		
13										
14			Percentage Change in Equity Investments:		30.0%	(30.0%)	40.0%	(15.0%)		
15			Change in Equity Investment Dollar Amount:		30	(15)	60	(30)		
16										
17			Cost Basis Just Before Change:		-	31	-	62		
18			Gain or Loss on Sale:		-	(16)	=IF(G14<0,(G11*F12-G17)*-G14/F12,0)			
19										
20			% Cash Used in Acquisition:		50.0%	50.0%	50.0%	50.0%		
21			% Debt Used in Acquisition:		50.0%	50.0%	50.0%	50.0%		
22										
23			Ownership DURING the Period:	0.0%	0.0%	30.0%	0.0%	40.0%		

The **logic** in this formula is as follows:

First, we cannot possibly have a Gain or Loss if the Ownership Percentage has increased. So, if the Percentage Change is >= 0, we set the Gain or Loss to 0.

The next part is easiest to understand with a specific example.

Let's say the Sub Co. Market Cap is $150, the Previous Ownership is 30%, the Cost Basis is $30, and the New Ownership is 15%.

Therefore, in this formula:

(Sub Co. Market Cap * Previous Ownership Percentage – Cost Basis) * –Percentage Change / Previous Ownership Percentage

Sub Co. Market Cap * Previous Ownership Percentage = $150 * 30% = $45.

The Cost Basis is $30, so the Total Gain or Loss is $45 – $30 = $15.

However, Parent Co. is not selling its *entire* stake in Sub Co.! It's moving from 30% ownership to 15% ownership.

Therefore, we need to adjust this Total Gain or Loss with the last part: Percentage Change / Previous Ownership Percentage.

In this case, it's equal to 15% / 30% = 50%, so the Gain is $15 * 50% = $7.5 instead.

The negative sign is in front because the Percentage Change will be negative when the ownership decreases, so we flip it to a positive with that sign.

To link the statements, we put these Realized Gains and Losses on the Income Statement:

Minority-Stake Investments:		End of Year 2:	End of Year 3:	End of Year 4:	End of Year 5:
Market Cap of Associate Company:		$ 100	$ 50	$ 150	$ 200
Ownership in Associate Becomes:	0.0%	30.0%	0.0%	40.0%	25.0%
Percentage Change in Equity Investments:		30.0%	(30.0%)	40.0%	(15.0%)
Change in Equity Investment Dollar Amount:		30	(15)	60	(30)
Cost Basis Just Before Change:		-	31	-	62
Gain or Loss on Sale:		-	(16)	-	7
% Cash Used in Acquisition:		50.0%	50.0%	50.0%	50.0%
% Debt Used in Acquisition:		50.0%	50.0%	50.0%	50.0%
Ownership DURING the Period:	0.0%	0.0%	30.0%	0.0%	40.0%

Income Statement	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue:	**$ 400**	**$ 450**	**$ 500**	**$ 550**	**$ 600**
(-) Costs of Goods Sold:	(100)	(120)	(140)	(160)	(180)
Gross Profit:	**300**	**330**	**360**	**390**	**420**
(-) Total Operating Expenses:	(200)	(220)	(235)	(245)	(255)
(-) Depreciation & Amortization:	(25)	(30)	(35)	(40)	(45)
Operating Income:	**75**	**80**	**90**	**105**	**120**
(-) Net Interest Expense:	-	-	(1)	(1)	(2)
(+) Gains / (-) Losses on Eq. Inv. Sales:	-	-	(16)	-	7
Pre-Tax Income:	**75**	**80**	**73**	**104**	**124**
(-) Income Taxes:	(19)	(20)	(18)	(26)	(31)
Net Income:	**56**	**60**	**55**	**78**	**93**
(+) Equity Investment Earnings:	-	-	2	-	4
Net Income to Parent:	**$ 56**	**$ 60**	**$ 57**	**$ 78**	**$ 97**

The Equity Investment Earnings line keeps changing because the Ownership goes from 0% to 30% to 0% to 40%, and Sub Co.'s Net Income also changes during the period.

On the Cash Flow Statement, we reverse the Realized Gains or Losses in Cash Flow from Operations, record the entire Change in Equity Investment Dollar Amount in Cash Flow from Investing, and record Debt or Stock issuances for additional purchases in Cash Flow from Financing.

If there's a decrease in ownership, the Realized Gain or Loss + Change in Equity Investment Dollar Amount correspond to the change in the Balance Sheet line item:

Cash Flow Statement:	Year 1	Year 2	Year 3	Year 4	Year 5
CASH FLOWS FROM OPERATING ACTIVITIES:					
Net Income:	$ 56	$ 60	$ 57	$ 78	$ 97
Adjustments for Non-Cash Charges:					
(+) Depreciation & Amortization:	25	30	35	40	45
(+/-) Deferred Taxes:	4	4	4	5	5
(-) Gains / (+) Losses on Eq. Inv. Sales:	-	-	16	-	(7)
(-) Equity Investment Earnings:	-	-	(2)	-	(4)
(+) Dividends from Equity Investments:	-	-	1	-	1
(+/-) Change in Op. Lease Assets:	(20)	(20)	(20)	(20)	(20)
(+/-) Change in Op. Lease Liabilities:	20	20	20	20	20
Changes in Operating Assets and Liabilities:					
Accounts Receivable:	(40)	(40)	(40)	(40)	(40)
Inventory:	(30)	(30)	(30)	(30)	(30)
Accounts Payable:	30	30	30	30	30
Accrued Liabilities:	20	20	20	20	20
Net Cash Provided by Operating Activities:	**65**	**74**	**91**	**103**	**118**
CASH FLOWS FROM INVESTING ACTIVITIES:					
(-) Capital Expenditures:	(30)	(35)	(40)	(45)	(50)
(-) Purchase / (+) Sale of Equity Investments:	-	(30)	15	(60)	30
Net Cash Used in Investing Activities:	**(30)**	**(65)**	**(25)**	**(105)**	**(20)**
CASH FLOWS FROM FINANCING ACTIVITIES:					
(+) Debt Issuances:	-	15	-	30	-
(-) Debt Repayments:	-	-	-	-	-
(-) Common Dividends Paid:	(20)	(22)	(24)	(27)	(30)
(+) Stock Issuances:	-	-	-	-	-
(-) Stock Repurchases:	-	-	-	-	-
Net Cash Provided by Financing Activities:	**(20)**	**(7)**	**(24)**	**3**	**(30)**
Change in Cash & Cash Equivalents:	**15**	**2**	**42**	**1**	**68**
Beginning Cash:	100	115	117	159	160
Ending Cash:	$ 115	$ 117	$ 159	$ 160	$ 228

As always, we reverse all Gains and Losses on the CFS; normal Equity Investment line items also still show up here.

Equal to Percentage Change in Equity Investment Ownership * Market Cap of Sub Co. this year.

Any Debt or Stock issued to fund increases in the stake.

On the Balance Sheet, Equity Investments change to reflect these ownership changes:

Balance Sheet:	Year 1	Year 2	Year 3	Year 4	Year 5
ASSETS:					
Current Assets:					
Cash & Equivalents:	$ 115	$ 117	$ 159	$ 160	$ 228
Accounts Receivable:	300	340	380	420	460
Inventory:	150	180	210	240	270
Total Current Assets:	**565**	**637**	**749**	**820**	**958**
Non-Current Assets:					
Net PP&E:	500	505	510	515	520
Operating Lease Assets:	200	220	240	260	280
Equity Investments:	-	30	-	60	39
Total Non-Current Assets:	**700**	**755**	**750**	**835**	**839**
Total Assets:	**$ 1,265**	**$ 1,392**	**$ 1,499**	**$ 1,655**	**$ 1,797**

All 4 line items from the CFS flow in, so Earnings, Dividends, Gains and Losses, and Purchases / Sales of Equity Investments all affect this.

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Key Rule #6: Consolidation Accounting (Noncontrolling Interests)

When a Parent Company (Parent Co.) owns 50% or more of another company, thereby **controlling** it, it must use **consolidation accounting** to record this stake.

In this scenario, the financial statements of the Parent Co. and Sub Co. are **consolidated 100%**, and there are small adjustments for Sub Co.'s Net Income and Dividends.

On the Income Statement, there's a deduction for (1 – Ownership Percentage) * Sub Co.'s Net Income because a portion of Sub Co.'s Net Income is **NOT** attributable to Parent Co.:

Consolidation Accounting for Noncontrolling Interests - Introduction
($ in Million Except Per Share Data)

Noncontrolling Interest Example:

Combined Company:	Year 1
Subsidiary Co. Market Cap:	$ 100
Ownership in Sub Co.:	70.0%
Value of Stake:	70

Parent Company:	Year 1
Tax Rate:	25.0%

Subsidiary Company:	Year 1
Tax Rate:	25.0%

Combined Company:	Year 1
Revenue:	**$ 500**
(-) Costs of Goods Sold:	(120)
Gross Profit:	**380**
(-) Total Operating Expenses:	(255)
(-) Depreciation & Amortization:	(30)
Operating Income:	**95**
(-) Net Interest Expense:	(15)
Pre-Tax Income:	**80**
(-) Income Taxes:	(20)
Net Income:	**60**
(-) Net Income Attributable to NCI:	(5)
Net Income to Parent:	**$ 56**
EBIT:	95
EBITDA:	125

Parent Company:	Year 1
Revenue:	**$ 400**
(-) Costs of Goods Sold:	(100)
Gross Profit:	**300**
(-) Total Operating Expenses:	(200)
(-) Depreciation & Amortization:	(25)
Operating Income:	**75**
(-) Net Interest Expense:	(15)
Pre-Tax Income:	**60**
(-) Income Taxes:	(15)
Net Income:	**$ 45**
EBIT:	75
EBITDA:	100

Subsidiary Company:	Year 1
Revenue:	**$ 100**
(-) Costs of Goods Sold:	(20)
Gross Profit:	**80**
(-) Total Operating Expenses:	(55)
(-) Depreciation & Amortization:	(5)
Operating Income:	**20**
(-) Net Interest Expense:	-
Pre-Tax Income:	**20**
(-) Income Taxes:	(5)
Net Income:	**$ 15**
EBIT:	20
EBITDA:	25

We literally add together the Income Statements of Parent Co. and Sub Co., and we deduct (1 - Ownership Percentage) * Sub Co. Net Income at the bottom.

We then reverse this line item on the Cash Flow Statement because when Parent Co. controls Sub Co., it *does* get Sub Co.'s Net Income in cash.

And on the CFS, we combine the Dividends of Parent Co. and Sub Co., but we also add back Sub Co.'s Dividends * Ownership Percentage because *those Dividends go to the Parent*.

To understand the full process of consolidation from beginning to end, let's start by assuming that Parent Co. already owns 30% of Sub Co.

Therefore, Parent Co. has an **Equity Investment** on its Balance Sheet that represents this 30% stake in Sub Co.

If Parent Co. wants to increase its stake to 70%, we need to make the calculations required for a full Balance Sheet combination first.

In this case – acquiring a **controlling stake** in another company – Goodwill will be created, along with Other Intangible Assets, Asset Write-Ups, Deferred Tax Liabilities, etc.

Another issue is that Parent Co. most likely acquired its existing 30% stake in Sub Co. *at a different price than Sub Co.'s current Market Cap*.

For example, the Parent here purchased this 30% stake in Sub Co. when Sub Co.'s Market Cap was $50, but not it plans to acquire an additional 40% when Sub Co.'s Market Cap is $100.

When this happens, we need to calculate the **Premium or Discount to the Book Value of the Existing Minority Stake** on the Balance Sheet:

Model Assumptions:	
Initial Stake in Sub Co. in Year 0:	30.0%
Cost Basis for 100% of Sub Co. in Year 0:	$ 50
Minimum Stake to Consolidate:	50.0%
Interest Rate on New Debt:	5.0%
% Cash Used in Acquisition:	50.0%
% Debt Used in Acquisition:	50.0%
Ownership in Sub Co.:	**End of Year 1:**
Market Cap of Sub Co.:	$ 100
Ownership in Sub Co. Becomes:	70.0%
Premium / (Discount) to Book Value of Existing Stake:	14
Purchase Price for Additional Stake:	40

Equal to $100 * 30% - $16 Equity Investment on the Balance Sheet currently.

If you're wondering why the Sub Co. Equity Investment is $16 rather than $15, Sub Co.'s Net Income and Dividends have changed the Equity Investment's value slightly.

We need this Premium or Discount because the Combined Balance Sheet will not balance without this adjustment.

The "Purchase Price for Additional Stake" is straightforward: (70% – 30%) * $100 = $40.

Next, we calculate Goodwill based on 100% of Sub Co.'s Equity – even though the Parent will own only 70% of the company:

Ownership in Sub Co.:	End of Year 1:
Market Cap of Sub Co.:	$ 100
Ownership in Sub Co. Becomes:	70.0%
Premium / (Discount) to Book Value of Existing Stake:	14
Purchase Price for Additional Stake:	40

Goodwill Calculation:		
Equity Purchase Price for 100% of Target:		**100**
(-) Seller Common Shareholders' Equity:		(42)
(+) Write-Off of Existing Goodwill:		-
Total Goodwill Created:	**$**	**58**

Goodwill is still based on 100% of Sub Co.'s Equity, even though we'll own only 70% of it.

Then, on the Balance Sheet, we add all of Sub Co.'s Assets and Liabilities and leave out its Common Shareholders' Equity, which is written down in the deal.

Sub Co. no longer exists as an independent entity, so its "Equity" is no longer reported as a separate line item.

We also add the new Goodwill, eliminate the old Equity Investments, deduct the Cash used to fund the deal, and add any Debt and Common Stock issued to fund it.

And we add a **Noncontrolling Interest** equal to (1 – New Ownership Percentage) * Sub Co.'s Market Cap on the L&E side within Equity.

Within Equity, we also record the Premium or Discount to the Book Value of the old Equity Investment, which was calculated above.

Remember that on the Assets side, "Debit" means "Add" and "Credit" means "Subtract"; it's the opposite on the L&E side:

Balance Sheet:	Year 1	Pro-Forma Adjustments: Debit	Credit	Post-Deal
ASSETS:				
Current Assets:				
Cash & Equivalents:	$ 116	$ 12	$ (20)	$ 107
Accounts Receivable:	300	30	-	330
Inventory:	150	15	-	165
Total Current Assets:	**566**			**602**
Non-Current Assets:				
Net PP&E:	500	50	-	550
Operating Lease Assets:	200	20	-	220
Goodwill:	-	58	-	58
Equity Investments:	16	-	(16)	-
Total Non-Current Assets:	**716**			**828**
Total Assets:	**$ 1,282**			**$ 1,431**
LIABILITIES AND EQUITY:				
Current Liabilities:				
Accounts Payable:	$ 300	$ -	$ 30	$ 330
Accrued Liabilities:	200	-	20	220
Total Current Liabilities:	**500**			**550**
Non-Current Liabilities:				
Total Debt:	-	-	-	-
New Transaction Debt:	-	-	20	20
Operating Lease Liabilities:	200	-	20	220
Deferred Tax Liabilities:	150	-	15	165
Total Non-Current Liabilities:	**350**			**405**
Total Liabilities:	**850**			**955**
Equity:				
Common Shareholders' Equity:	**432**	**-**	**14**	**446**
Noncontrolling Interests:	-	-	30	30
Total Equity:	**432**			**476**
Total Liabilities & Equity:	**$ 1,282**			**$ 1,431**
Balance Check:	*OK!*			*OK!*

Add Sub Co.'s Assets, deduct Cash used, add new Goodwill, and eliminate old Equity Investments since we now own 70%, not 30%.

Add Sub Co.'s Liabilities and any new Debt used to fund the deal.

Do NOT add Sub Co.'s CSE because it's written down; create a new Noncontrolling Interest to represent the portion that Parent Co. does NOT own. And adjust for the Premium or Discount in CSE.

In Step 2 of this process, we'll assume that Parent Co.'s ownership in Sub Co. remains at 70% in Years 2 – 4.

So, we combine the financial statements 100% and make a few adjustments to Net Income and Dividends:

Ownership DURING the Period:	30.0%				70.0%	70.0%	70.0%
		Pro-Forma Adjustments:					
Income Statement:	**Year 1**	**Debit**	**Credit**	**Post-Deal**	**Year 2**	**Year 3**	**Year 4**
Revenue:	$ 400				$ 495	$ 550	$ 605
(-) Costs of Goods Sold:	(100)				(132)	(154)	(176)
Gross Profit:	300				363	396	429
(-) Total Operating Expenses:	(200)				(242)	(259)	(270)
(-) Depreciation & Amortization:	(25)				(33)	(39)	(44)
Operating Income:	75				88	99	116
(-) Existing Net Interest Expense:	-				-	-	-
(-) Interest Expense on New Debt:	-				(1)	(1)	(1)
(-) Goodwill Impairment:	-				-	-	-
Pre-Tax Income:	75				87	98	115
(-) Income Taxes:	(19)				(22)	(25)	(29)
Net Income:	56				65	74	86
(+) Equity Investment Earnings:	2				-	-	-
(-) Net Income Attributable to NCI:	-				(2)	(2)	(2)
Net Income to Parent:	$ 58				$ 63	$ 71	$ 84

Adding 100% of Parent Co.'s and 100% of Sub Co.'s figures together and reflecting Interest Expense on New Debt.

Deduct (1 - Ownership Percentage) * Sub Co.'s Net Income.

The Cash Flow Statement is a bit trickier because we need to modify a few items:

- **Dividends:** Technically, we should combine 100% of Parent Co.'s and 100% of Sub Co.'s Dividends in Cash Flow from Financing. *However, for modeling purposes, it's easier to make these two separate line items because they'll link to separate items on the Balance Sheet*.

- **Equity Investment Earnings and Net Income Attributable to Noncontrolling Interests:** These line items from the Income Statement must be reversed on the Cash Flow Statement.

- **Dividends Received from Partially Owned Companies:** These are positive cash inflows typically listed in Cash Flow from Operations, and they're based on Ownership Percentage in Sub Co. * Sub Co.'s Dividends. **This formula is the same regardless of whether the Parent owns a majority or minority stake in Sub Co.**

- **Cash:** Remember that the "Beginning Cash" in the first post-transaction year must link to the Cash number *immediately after the deal closes in the "Post-Deal" column!*

Cash Flow Statement:	Year 1	Pro-Forma Adjustments: Debit	Credit	Post-Deal	Year 2	Year 3	Year 4
CASH FLOWS FROM OPERATING ACTIVITIES:							
Net Income to Parent:	$ 58				$ 63	$ 71	$ 84
Adjustments for Non-Cash Charges:							
(+) Depreciation & Amortization:	25				33	39	44
(+/-) Deferred Taxes:	4				4	4	6
(+) Goodwill Impairment:	-				-	-	-
(-) Equity Investment Earnings:	(2)				-	-	-
(+) Net Income Attributable to NCI:	-				2	2	2
(+) Dividends Rec'd from Partially Owned Co's:	1				2	2	2
(+/-) Change in Op. Lease Assets:	(20)				(22)	(22)	(22)
(+/-) Change in Op. Lease Liabilities:	20				22	22	22
Changes in Operating Assets and Liabilities:							
Accounts Receivable:	(40)				(44)	(44)	(44)
Inventory:	(30)				(33)	(33)	(33)
Accounts Payable:	30				33	33	33
Accrued Liabilities:	20				22	22	22
Net Cash Provided by Operating Activities:	**66**				**82**	**96**	**115**
CASH FLOWS FROM INVESTING ACTIVITIES:							
(-) Capital Expenditures:	(30)				(39)	(44)	(50)
Net Cash Used in Investing Activities:	**(30)**				**(39)**	**(44)**	**(50)**
CASH FLOWS FROM FINANCING ACTIVITIES:							
(+) Debt Issuances:	-				-	-	-
(-) Debt Repayments:	-				-	-	-
(+) New Debt Issuances:	-				-	-	-
(-) Parent - Common Dividends Paid:	(20)				(22)	(24)	(27)
(-) Sub Co. - Common Dividends Paid:	-				(2)	(2)	(3)
(+) Stock Issuances:	-				-	-	-
(-) Stock Repurchases:	-				-	-	-
Net Cash Provided by Financing Activities:	**(20)**				**(24)**	**(26)**	**(30)**
Change in Cash & Cash Equivalents:	**16**				**19**	**26**	**36**
Beginning Cash:	100				107	127	153
Ending Cash:	$ 116				$ 127	$ 153	$ 189

Reverse Net Income to NCI; record Ownership % * Sub Co. Dividends.

Technically, should be one line, but this makes the model easier.

Post-transaction Cash!

If you're wondering about the **rationale** for these adjustments, when Parent Co. **controls** Sub Co., the Cash balances are also combined.

Therefore, Parent Co. "gets" all the Net Income generated by Sub Co., so *in cash terms*, subtracting (1 – Ownership Percentage) * Sub Co.'s Net Income is not accurate. That's why we reverse this part.

Then, under "Common Dividends Paid," we deduct 100% of both companies' Dividends… but *70% of Sub Co.'s Dividends go to the Parent!*

So, in cash terms, Parent Co. is not actually "losing" 100% of Sub Co.'s Dividends.

Therefore, we record Ownership Percentage * Sub Co.'s Dividends for "Dividends Rec'd from Partially Owned Co's" in Cash Flow from Operations.

The Combined Balance Sheet is straightforward; most of the Assets and Liabilities are simple combinations of Parent Co. and Sub Co. line items, and items like Cash flow in from the Cash Flow Statement:

		Pro-Forma Adjustments:					
Balance Sheet:	**Year 1**	**Debit**	**Credit**	**Post-Deal**	**Year 2**	**Year 3**	**Year 4**
ASSETS:							
Current Assets:							
Cash & Equivalents:	$ 116	$ 12	$ (20)	$ 107	$ 127	$ 153	$ 189
Accounts Receivable:	300	30	-	33[illegible]	374	418	462
Inventory:	150	15	-	16[illegible]	198	231	264
Total Current Assets:	**566**			[illegible]**2**	**699**	**802**	**915**
Non-Current Assets:							
Net PP&E:	500	50	-	550	556	561	567
Operating Lease Assets:	200	20	-	220	242	264	286
Goodwill:	-	58	-	58	58	58	58
Equity Investments:	16	-	(16)	-	-	-	-
Total Non-Current Assets:	**716**			**828**	**856**	**883**	**911**
Total Assets:	**$ 1,282**			**$ 1,431**	**$ 1,555**	**$ 1,685**	**$ 1,826**

From the bottom of the CFS.

Still here and linked to Equity Investment Earnings and Dividends (but Dividends only link in if the Ownership is < 50%).

The Liabilities & Equity side is more complicated because of the links for Common Shareholders' Equity and Noncontrolling Interests:

- **Common Shareholders' Equity:** Old CSE + Net Income to Parent + Parent Co. Dividends + Stock Issuances + Stock Repurchases.

- **Noncontrolling Interests:** Old NCI + Net Income to NCI + Dividends Rec'd from Partially Owned Co's + Sub Co. Dividends. We also need to check the Ownership Percentage and link in "Dividends Rec'd from Partially Owned Co's" *only* if the Parent currently owns 50% or more of Sub Co.

The "+" signs here refer to the Excel formulas, not the actual signs. Parent Co. Dividends and Sub Co. Dividends are negative, so we're subtracting them in these formulas.

LIABILITIES AND EQUITY:							
Current Liabilities:							
Accounts Payable:	$ 300	$ -	$ 30	$ 330	$ 363	$ 396	$ 429
Accrued Liabilities:	200	-	20	220	242	264	286
Total Current Liabilities:	**500**			**550**	**605**	**660**	**715**
Non-Current Liabilities:							
Total Debt:	-	-	-	-	-	-	-
New Transaction Debt:	-	-	20	20	20	20	20
Operating Lease Liabilities:	200	-	20	220	242	264	286
Deferred Tax Liabilities:	150	-	15	165	169	174	179
Total Non-Current Liabilities:	**350**			**405**	**431**	**458**	**485**
Total Liabilities:	**850**			**955**	**1,036**	**1,118**	**1,200**
Equity:							
Common Shareholders' Equity:	**432**	**-**	**14**	**446**	**487**	**535**	**591**
Noncontrolling Interests:	-	-	30	30	31	32	34
Total Equity:	**432**			**476**	**518**	**567**	**625**
Total Liabilities & Equity:	**$ 1,282**			**$ 1,431**	**$ 1,555**	**$ 1,685**	**$ 1,826**
Balance Check:	*OK!*			*OK!*	*OK!*	*OK!*	*OK!*

Simple Parent Co. + Sub Co. line items, or New Debt that stays constant.

Only Parent Co.'s Dividends are subtracted!

Only Sub Co.'s Dividends and Dividends Received from Partially Owned Co's... if Parent owns >= 50% of Sub Co.

The **rationale** is that we deduct only the portion of Sub Co.'s Dividends that *go to the minority shareholders in Sub Co. in the NCI line item.*

The Noncontrolling Interest acts as a "mini-Shareholders' Equity" for *the minority shareholders* in Sub Co.

So, *their* Net Income is added, and *their* Dividends are subtracted.

In **Step 3**, we'll look at the **deconsolidation process:** what happens when Parent Co. sells part of its stake in Sub Co. and moves back to ownership below 50%.

In this step, the Noncontrolling Interest is removed, and a new Equity Investment line item is created.

The biggest challenge is **determining the Gain or Loss** on this transaction.

It's more complicated than calculating Gains or Losses on Equity Investments because of the Balance Sheet changes, the new Goodwill, and the combined financial statements.

We start by assuming a Market Cap (Equity Value) for the Sub Co. when this ownership change takes place at the end of Year 4.

Based on that and the new Ownership Percentage, we can calculate the Market Value of the Stake Sold and the value of the new Equity Investment. Then:

- **Gain or Loss on Stake Sold** = Market Value of Stake Sold + Market Value of New Equity Investment + Book Value of NCI Being Removed – Net Assets of Sub Co., Including Goodwill Created in Previous Deal

Here's what it looks like in Excel:

	G	H	I	J	K
11					
12					
13					
14					
15					
16				**End of Year 4:**	
17		Market Cap of Sub Co.:		$ 150	
18		Ownership in Sub Co. Becomes:		40.0%	
19					
20		Percentage Sold:		30.0%	
21		Market Value of Stake Sold:		45	
22		New Equity Investment:		60	
23					
24		Gain or Loss on Stake Sold:		=+J21+J22+J85-(Sub!G37+J62-Sub!G51)	
25		After-Tax Gain or Loss:		19	
26		Total Cash Proceeds:		39	

The intuition here is that the first two parts give us the Market Value of the 70% the Parent owned *before*, the J85 for the Book Value of the NCI accounts for the remaining 30%, and then the Net Assets of Sub Co., including new Goodwill, represent the "Cost Basis."

We use the Book Value of Noncontrolling Interests (cell J85) rather than the Market Value because when Parent Co. sells some of its stake, it doesn't get the Market Value for all 100% of Sub Co.!

Parent Co. never "owned" that Noncontrolling Interest, so it doesn't make sense to use its Market Value in this calculation.

The company must also pay taxes on this Gain (or get a tax benefit if there's a Loss), so we multiply by (1 – Tax Rate) to calculate the After-Tax Gain or Loss.

Total Cash Proceeds are based on (Market Value of Stake Sold – Gain or Loss) + After-Tax Gain or Loss.

That's because Parent Co. *pays taxes only on the Gain*. If it just recovers its Cost Basis, that is not taxed.

It's similar to individuals buying and selling stocks: they pay taxes only if they sell higher and realize a Gain – if they buy a stock for $10.00 and sell it for $10.00, there are no taxes.

The next part of this process is the **deconsolidation of the Balance Sheet**.

To do this, we remove all of Sub Co.'s Assets and Liabilities, including its Cash. These removals will be Credits on the Assets side and Debits on the L&E side.

Also, we remove the Goodwill created in the majority-stake acquisition and the Noncontrolling Interest created in the same transaction.

We show the Cash Received as an increase to Cash, we record the new Equity Investment line item, and we put the After-Tax Gain or Loss in Common Shareholders' Equity.

This last bit is a "shortcut" – normally, Realized Gains and Losses show up on the Income Statement and flow into CSE via their impact on Net Income.

But we want the Balance Sheet to balance *immediately after the deal closes*, so we set it up this way:

		Pro-Forma Adjustments:			
Balance Sheet:	**Year 4**	**Debit**	**Credit**	**Post-Deal**	**Year 5**
ASSETS:					
Current Assets:					
Cash & Equivalents:	$ 189	$ 39	$ (19)	$ 209	$ 249
Accounts Receivable:	46	-	(42)	420	460
Inventory:	264	-	(24)	240	270
Total Current Assets:	**815**			**869**	**979**
Non-Current Assets:					
Net PP&E:	567	-	(52)	515	520
Operating Lease Assets:	286	-	(26)	260	280
Goodwill:	58	-	(58)	-	-
Equity Investments:	-	60	-	60	62
Total Non-Current Assets:	**911**			**835**	**862**
Total Assets:	**$ 1,826**			**$ 1,704**	**$ 1,841**

Remove Sub Co.'s Assets, including the Goodwill created previously.

Add Total Cash Proceeds Received.

New Equity Investment for 40% * Market Cap of Sub Co.

Balance Sheet:	Year 4	Pro-Forma Adjustments: Debit	Credit	Post-Deal	Year 5
LIABILITIES AND EQUITY:					
Current Liabilities:					
Accounts Payable:	$ 429	$ (39)	$ -	$ 390	$ 420
Accrued Liabilities:	286	(26)	-	260	280
Total Current Liabilities:	**715**			**650**	**700**
Non-Current Liabilities:					
Total Debt:	-	-	-	-	-
New Transaction Debt:	20	-	-	20	20
Operating Lease Liabilities:	286	(26)	-	260	280
Deferred Tax Liabilities:	179	(16)	-	163	168
Total Non-Current Liabilities:	**485**			**443**	**468**
Total Liabilities:	**1,200**			**1,093**	**1,168**
Equity:					
Common Shareholders' Equity:	**591**	**-**	**19**	**611**	**673**
Noncontrolling Interests:	34	(34)	-	-	-
Total Equity:	**625**			**611**	**673**
Total Liabilities & Equity:	**$ 1,826**			**$ 1,704**	**$ 1,841**
Balance Check:	*OK!*			*OK!*	*OK!*

Remove all of Sub Co.'s Liabilities.

Remove the NCI and record the After-Tax Gain or Loss in CSE.

Next, we have to complete the financial statements in Year 5, after the deconsolidation takes place and Parent Co. owns only 40% of Sub Co.:

Ownership DURING the Period: 70.0% 40.0%

Income Statement:	Year 4	Pro-Forma Adjustments: Debit	Credit	Post-Deal	Year 5
Revenue:	**$ 605**				**$ 600**
(-) Costs of Goods Sold:	(176)				(180)
Gross Profit:	**429**				**420**
(-) Total Operating Expenses:	(270)				(255)
(-) Depreciation & Amortization:	(44)				(45)
Operating Income:	**116**				**120**
(-) Existing Net Interest Expense	-				-
(-) Interest Expense on New Deb	(1)				(1)
(-) Goodwill Impairment:	-				-
Pre-Tax Income:	**115**				**119**
(-) Income Taxes:	(29)				(30)
Net Income:	**86**				**89**
(+) Equity Investment Earnings:	-				4
(-) Net Income Attributable to N(	(2)				-
Net Income to Parent:	**$ 84**				**$ 93**

Parent Co.'s standalone numbers now, with adjustments for new Interest Expense and Ownership Percentage * Sub Co. Net Income.

The same thing happens on the Cash Flow Statement, which is mostly Parent Co.'s numbers with a few adjustments for the Equity Investment Earnings and the Dividend lines:

Cash Flow Statement:	Year 4	Pro-Forma Adjustments: Debit	Credit	Post-Deal	Year 5
CASH FLOWS FROM OPERATING ACTIVITIES:					
Net Income to Parent:	$ 84				$ 93
Adjustments for Non-Cash Charges:					
(+) Depreciation & Amortization:	44				45
(+/-) Deferred Taxes:	6				5
(+) Goodwill Impairment:	-				-
(-) Equity Investment Earnings:	-				(4)
(+) Net Income Attributable to NCI:	2				-
(+) Dividends Rec'd from Partially Owned Co's:	2				1
(+/-) Change in Op. Lease Assets:	(22)				6
(+/-) Change in Op. Lease Liabilities:	22				(6)
Changes in Operating Assets and Liabilities:					
Accounts Receivable:	(44)				(40)
Inventory:	(33)				(30)
Accounts Payable:	33				30
Accrued Liabilities:	22				20
Net Cash Provided by Operating Activities:	**115**				**120**
CASH FLOWS FROM INVESTING ACTIVITIES:					
(-) Capital Expenditures:	(50)				(50)
Net Cash Used in Investing Activities:	**(50)**				**(50)**
CASH FLOWS FROM FINANCING ACTIVITIES:					
(+) Debt Issuances:	-				-
(-) Debt Repayments:	-				-
(+) New Debt Issuances:	-				-
(-) Parent - Common Dividends Paid:	(27)				(30)
(-) Sub Co. - Common Dividends Paid:	(3)				-
(+) Stock Issuances:	-				-
(-) Stock Repurchases:	-				-
Net Cash Provided by Financing Activities:	**(30)**				**(30)**
Change in Cash & Cash Equivalents:	**36**				**40**
Beginning Cash:	153				209
Ending Cash:	$ 189				$ 249

Back to Equity Method Accounting for the 40% stake.

And we now record only Parent Co.'s Dividends here.

Finally, on the Balance Sheet, many items come directly from the Parent's Balance Sheet (such as the Working Capital line items and Operating Lease Assets and Liabilities).

The rest use formulas similar to the ones in Years 2 – 4.

Cash comes from the bottom of the CFS, and the links for Net PP&E, Equity Investments, Noncontrolling Interests, Debt, etc., are all the same:

Balance Sheet:	Year 4	Pro-Forma Adjustments: Debit	Credit	Post-Deal	Year 5
ASSETS:					
Current Assets:					
Cash & Equivalents:	$ 189	$ 39	$ (19)	$ 209	$ 249
Accounts Receivable:	462	-	(42)	420	460
Inventory:	264	-	(24)	240	270
Total Current Assets:	**915**			**869**	**979**
Non-Current Assets:					
Net PP&E:	567	-	(52)	515	520
Operating Lease Assets:	286	-	(26)	260	280
Goodwill:	58	-	(58)	-	-
Equity Investments:	-	60	-	60	62
Total Non-Current Assets:	**911**			**835**	**862**
Total Assets:	**$ 1,826**			**$ 1,704**	**$ 1,841**

Back to Equity Investment Earnings and Dividends linking in here, with a check to make sure the Ownership Percentage is < 50% before linking in the Dividends.

Balance Sheet:	Year 4	Pro-Forma Adjustments: Debit	Credit	Post-Deal	Year 5
LIABILITIES AND EQUITY:					
Current Liabilities:					
Accounts Payable:	$ 429	$ (39)	$ -	$ 390	$ 420
Accrued Liabilities:	286	(26)	-	260	280
Total Current Liabilities:	**715**			**650**	**700**
Non-Current Liabilities:					
Total Debt:	-	-	-	-	-
New Transaction Debt:	20	-	-	20	20
Operating Lease Liabilities:	286	(26)	-	260	280
Deferred Tax Liabilities:	179	(16)	-	163	168
Total Non-Current Liabilities:	**485**			**443**	**468**
Total Liabilities:	**1,200**			**1,093**	**1,168**
Equity:					
Common Shareholders' Equity:	**591**	**-**	**19**	**611**	**673**
Noncontrolling Interests:	34	(34)	-	-	-
Total Equity:	**625**			**611**	**673**
Total Liabilities & Equity:	**$ 1,826**			**$ 1,704**	**$ 1,841**
Balance Check:	*OK!*			*OK!*	*OK!*

The most difficult part of this entire process is the **deconsolidation** in this last step, but in real life, you rarely show it in financial models (unless you're modeling a spin-off).

Most models assume that a company's ownership stakes in other companies stay about the same over time, so the Net Income and Dividend links are the most important points.

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Key Rule #7: Stock-Based Compensation and Excess Tax Benefits

Normally in models, we assume that Stock-Based Compensation is **not** Cash-Tax Deductible: it appears as an Income Statement expense but does **not** reduce the company's Cash Taxes.

We model this as a change in Deferred Taxes on the Cash Flow Statement.

But this treatment is only for the **initial** grant of stock options, shares, or RSUs to employees.

When the employees finally receive their shares or exercise their options, the accounting gets more complicated – and it differs under U.S. GAAP and IFRS.

To make things even worse, the rules changed in 2017 under U.S. GAAP (ASC 718), which made the treatment under both accounting systems even more different.

In real-life models, we almost always ignore this part of the Stock-Based Compensation cycle because the timing is unpredictable, and it's difficult to forecast when employees will exercise their options.

However, it's still useful to understand these accounting rules because they also explain some of the strange behavior that you've probably seen on companies' Income Statements:

- How can Net Income exceed Pre-Tax Income?
- How can Book Income Taxes be *positive* on the Income Statement rather than negative?

To explain the treatment under U.S. GAAP, we'll take a simple example where $20 of SBC is issued in Year 1 and then increases in value from Year 2 through Year 5.

The employees then exercise their options in Year 5 and receive their shares, at which point the company can take the deduction and reduce its Cash Taxes.

But since the value of the SBC has *increased* by Year 5, the company also gets an "Excess Tax Benefit" and can save even more in taxes.

Here's the basic schedule for the $20 of SBC, which rises in value to $80 by Year 5:

Stock-Based Compensation, Tax Benefits, and Excess Tax Benefits (U.S. GAAP Treatment)
($ in Millions Except Per Share and Per Unit Data)

Model Assumptions and Inputs:	Units:	Historical Year 0	Projected Year 1	Year 2	Year 3	Year 4	Year 5
Tax Rate:	%	25.0%					
Interest Rate on Debt:	%	5.0%					
Value of SBC Granted to Executives:	$ M		$ 20.0				
Year # of SBC Exercise and Receipt:	$ M	5					
Value of SBC Upon Exercise:	$ M			20.0	40.0	60.0	80.0
Amount to Deduct for Cash-Tax Purposes:	$ M			-	-	-	20.0
Amount for Excess Tax Benefits / (Deficiencies):	$ M			-	-	-	60.0

Initial grant is NOT Cash-Tax Deductible, so it creates a DTA...

The company then gets the deduction in Year 5, when its DTA goes to $0.

Here's the Tax Schedule for this scenario:

Tax Schedule:	Units:	Historical Year 0	Projected Year 1	Year 2	Year 3	Year 4	Year 5
Book Pre-Tax Income:		$ 45.0	$ 25.0	$ 45.0	$ 45.0	$ 45.0	$ 45.0
(+) Stock-Based Compensation:		-	20.0	-	-	-	-
(-) Deduction for Exercise of Stock Options:		-	-	-	-	-	(20.0)
Cash Taxable Income:		**45.0**	**45.0**	**45.0**	**45.0**	**45.0**	**25.0**
Cash Taxes Payable:		(11.3)	(11.3)	(11.3)	(11.3)	(11.3)	(6.3)
Deferred Income Taxes:		**$ -**	**$ (5.0)**	**$ -**	**$ -**	**$ -**	**$ 5.0**

This concept is straightforward: the company can't take the tax deduction in Year 1, but it can use it in Year 5 when the employees exercise their options, so it is deferred until then.

However, since the SBC's value has increased to $80 by Year 5, we also need to record the "Excess Tax Benefits" from it.

Under the accounting rules that were put in place in 2017, this shows up as "Excess Tax Benefits" or "Excess Tax Deficiencies" on the Income Statement as a direct reduction or addition to Income Taxes:

Amount to Deduct for Cash-Tax Purposes:	$ M				-	-	-	20.0
Amount for Excess Tax Benefits / (Deficiencies):	$ M				-	-	-	60.0
		Historical	Projected					
Income Statement:	*Units:*	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	
Total Revenue:	$ M	$ 200.0	$ 200.0	$ 200.0	$ 200.0	$ 200.0	$ 200.0	
Cost of Sales:	$ M	75.0	75.0	75.0	75.0	75.0	75.0	
Gross Profit:	$ M	**125.0**	**125.0**	**125.0**	**125.0**	**125.0**	**125.0**	
Operating Expenses:								
Selling, General & Administrative:	$ M	50.0	50.0	50.0	50.0	50.0	50.0	
Depreciation & Amortization:	$ M	25.0	25.0	25.0	25.0	25.0	25.0	
Stock-Based Compensation:	$ M	-	20.0	-	-	-	-	
Total Operating Expenses:	$ M	**75.0**	**95.0**	**75.0**	**75.0**	**75.0**	**75.0**	
Operating Income:	$ M	**50.0**	**30.0**	**50.0**	**50.0**	**50.0**	**50.0**	
Operating Margin:	%	*25.0%*	*15.0%*	*25.0%*	*25.0%*	*25.0%*	*25.0%*	
(-) Interest Expense:	$ M	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)	
Pre-Tax Income:	$ M	**45.0**	**25.0**	**45.0**	**45.0**	**45.0**	**45.0**	
(-) Provision For Income Taxes:	$ M	(11.3)	(6.3)	(11.3)	(11.3)	(11.3)	(11.3)	
(+) Excess Tax Benefits / (-) Deficiencies from SBC:	$ M	-	-	-	-	-	=+J17*Tax_Rate	
Net Income:	$ M	$ 33.8	$ 18.8	$ 33.8	$ 33.8	$ 33.8	$ 48.8	

Yes, this treatment creates some strange outcomes. Here, for example, Net Income exceeds Pre-Tax Income because "Income Taxes" becomes a positive $3.8.

In the old method (pre-2017), companies recorded this Excess Tax Benefit only on the Cash Flow Statement, and it linked to Common Shareholders' Equity on the Balance Sheet.

Linking the statements with this new treatment is straightforward: SBC is a non-cash add-back on the CFS, as always, and the Deferred Tax adjustment will be negative in Year 1 and positive in Year 5.

The Net DTA on the Balance Sheet increases by $5 in Year 1 and decreases by $5 in Year 5:

Cash Flow Statement:	Units:	Historical Year 0	Projected Year 1	Year 2	Year 3	Year 4	Year 5
CASH FLOWS FROM OPERATING ACTIVITIES:							
Net Income:	$ M	**$ 33.8**	**$ 18.8**	**$ 33.8**	**$ 33.8**	**$ 33.8**	**$ 48.8**
Adjustments for Non-Cash Charges:							
(+) Depreciation & Amortization:	$ M	25.0	25.0	25.0	25.0	25.0	25.0
(+) Stock-Based Compensation:	$ M	-	20.0	-	-	-	-
(+/-) Deferred Taxes:	$ M	-	(5.0)	-	-	-	5.0
(+/-) Change in Op. Lease Assets:	$ M	-	(2.0)	(2.0)	(2.0)	(2.0)	(2.0)
(+/-) Change in Op. Lease Liabilities:	$ M	-	2.0	2.0	2.0	2.0	2.0
Changes in Operating Assets and Liabilities:							
Inventory:	$ M	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)
Accounts Receivable:	$ M	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)
Accounts Payable:	$ M	5.0	5.0	5.0	5.0	5.0	5.0
Accrued Liabilities:	$ M	5.0	5.0	5.0	5.0	5.0	5.0
Net Cash Provided by Operating Activities:	$ M	**58.8**	**58.8**	**58.8**	**58.8**	**58.8**	**78.8**
CASH FLOWS FROM INVESTING ACTIVITIES:							
(-) Capital Expenditures:	$ M	(30.0)	(32.0)	(34.0)	(36.0)	(38.0)	(40.0)
Net Cash Used in Investing Activities:	$ M	**(30.0)**	**(32.0)**	**(34.0)**	**(36.0)**	**(38.0)**	**(40.0)**
CASH FLOWS FROM FINANCING ACTIVITIES:							
(+) Debt Issuances:	$ M	-	-	-	-	-	-
(-) Debt Repayments:	$ M	-	-	-	-	-	-
(-) Dividends Paid:	$ M	(10.0)	(10.0)	(10.0)	(10.0)	(10.0)	(10.0)
(-) Common Stock Repurchases:	$ M	-	-	-	-	-	-
Net Cash Provided by Financing Activities:	$ M	**(10.0)**	**(10.0)**	**(10.0)**	**(10.0)**	**(10.0)**	**(10.0)**
Change in Cash & Cash Equivalents:	$ M	18.8	16.8	14.8	12.8	10.8	28.8
Beginning Cash:	$ M	977.5	996.3	1,013.0	1,027.8	1,040.5	1,051.3
Ending Cash:	$ M	**$ 996.3**	**$ 1,013.0**	**$ 1,027.8**	**$ 1,040.5**	**$ 1,051.3**	**$ 1,080.0**

Balance Sheet:	Units:	Historical Year 0	Projected Year 1	Year 2	Year 3	Year 4	Year 5
ASSETS:							
Current Assets:							
Cash & Equivalents:	$ M	$ 996.3	$ 1,013.0	$ 1,027.8	$ 1,040.5	$ 1,051.3	$ 1,080.0
Inventory:	$ M	200.0	205.0	210.0	215.0	220.0	225.0
Accounts Receivable:	$ M	250.0	255.0	260.0	265.0	270.0	275.0
Total Current Assets:	$ M	**1,446.3**	**1,473.0**	**1,497.8**	**1,520.5**	**1,541.3**	**1,580.0**
Non-Current Assets:							
Net PP&E:	$ M	300.0	307.0	316.0	327.0	340.0	355.0
Net Deferred Tax Asset:	$ M	-	5.0	5.0	5.0	5.0	-
Operating Lease Assets:	$ M	150.0	152.0	154.0	156.0	158.0	160.0
Total Non-Current Assets:	$ M	**450.0**	**464.0**	**475.0**	**488.0**	**503.0**	**515.0**
Total Assets:	$ M	**$ 1,896.3**	**$ 1,937.0**	**$ 1,972.8**	**$ 2,008.5**	**$ 2,044.3**	**$ 2,095.0**
LIABILITIES AND EQUITY:							
Current Liabilities:							
Accounts Payable:	$ M	$ 150.0	$ 155.0	$ 160.0	$ 165.0	$ 170.0	$ 175.0
Accrued Liabilities:	$ M	200.0	205.0	210.0	215.0	220.0	225.0
Total Current Liabilities:	$ M	**350.0**	**360.0**	**370.0**	**380.0**	**390.0**	**400.0**
Non-Current Liabilities:							
Total Debt:	$ M	100.0	100.0	100.0	100.0	100.0	100.0
Operating Lease Liabilities:	$ M	150.0	152.0	154.0	156.0	158.0	160.0
Total Non-Current Liabilities:	$ M	**250.0**	**252.0**	**254.0**	**256.0**	**258.0**	**260.0**
Total Liabilities:	$ M	**600.0**	**612.0**	**624.0**	**636.0**	**648.0**	**660.0**
Common Shareholders' Equity:	$ M	**1,296.3**	**1,325.0**	**1,348.8**	**1,372.5**	**1,396.3**	**1,435.0**
Total Liabilities & Equity:	$ M	**$ 1,896.3**	**$ 1,937.0**	**$ 1,972.8**	**$ 2,008.5**	**$ 2,044.3**	**$ 2,095.0**

The **result** of this treatment is that if the company's stock price has increased significantly, meaning that SBC granted in earlier years has also appreciated, then it will receive significant tax savings when employees exercise their options.

On the other hand, if the SBC has *decreased* in value, then the company pays more in taxes when employees get their shares.

IFRS Treatment

The main difference under IFRS is that the Deferred Tax Asset created by the initial Stock-Based Compensation is **revalued** whenever the value of the SBC changes.

That reflects the changing value of the future Cash-Tax deductions when the SBC value changes.

On the L&E side of the Balance Sheet, APIC within Common Shareholders' Equity changes to balance this revaluation.

Also, under IFRS, as soon as the SBC meets or exceeds its initial value, the company records "Tax Benefits" for it on the Income Statement.

Finally, if the SBC's value *decreases*, then it's recorded as a "Tax Deficiency" on the Income Statement, equal to – Change in Value * Tax Rate. Here's the schedule:

Stock-Based Compensation, Tax Benefits, and Excess Tax Benefits (IFRS Treatment)
($ in Millions Except Per Share and Per Unit Data)

		Historical	Projected				
Model Assumptions and Inputs:	***Units:***	**Year 0**	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
Tax Rate:	%	25.0%					
Interest Rate on Debt:	%	5.0%					
Value of SBC Granted to Executives:	$ M		$ 20.0				
Initial Expected Future Tax Benefit:	$ M		5.0				
Year # of SBC Exercise and Receipt:	$ M	5					
Value of SBC Upon Exercise:	$ M			25.0	30.0	35.0	40.0
Change in Value of SBC:	$ M			5.0	5.0	5.0	5.0
Tax Benefits to Record on IS:	$ M			5.0	-	-	-
Tax Deficiencies to Record on IS:	$ M			-	-	-	-
Increase in Deferred Tax Asset:	$ M			1.3	1.3	1.3	1.3
Excess Tax Benefit Recognized in Equity:	$ M			1.3	1.3	1.3	1.3
Amount to Deduct for Cash-Tax Purposes:	$ M			-	-	-	40.0

Under IFRS, Tax Benefits on the IS show up only when SBC first meets or exceeds its grant value. Tax Deficiencies appear if its value falls.

DTA increases only if SBC's value increases.

The Tax Schedule under IFRS looks like this:

Tax Schedule:	Units:	Historical Year 0	Projected Year 1	Year 2	Year 3	Year 4	Year 5
Book Pre-Tax Income:	$ M	$ 45.0	$ 25.0	$ 45.0	$ 45.0	$ 45.0	$ 45.0
(+) Initial Stock-Based Compensation:	$ M	-	20.0	-	-	-	-
(+/-) Change in Value of SBC:	$ M	-	-	5.0	5.0	5.0	5.0
(-) Deduction for Exercise of Stock Options:	$ M	-	-	-	-	-	(40.0)
Cash Taxable Income:	$ M	45.0	45.0	50.0	50.0	50.0	10.0
Cash Taxes Payable:	$ M	(11.3)	(11.3)	(12.5)	(12.5)	(12.5)	(2.5)
Deferred Income Taxes:	$ M	$ -	$ (5.0)	$ (1.3)	$ (1.3)	$ (1.3)	$ 8.8

Each time the value of SBC increases, the tax benefits are deferred into the future, so the DTA increases. When it finally becomes deductible, the entire DTA goes to $0 as the company applies the entire benefit built up in the DTA.

The Income Statement does not change much with this treatment because only the initial Tax Benefits ($20 * 25% = $5) show up in Year 2, reducing the company's Income Taxes:

Income Statement:	Units:	Historical Year 0	Projected Year 1	Year 2	Year 3	Year 4	Year 5
Total Revenue:	$ M	$ 200.0	$ 200.0	$ 200.0	$ 200.0	$ 200.0	$ 200.0
Cost of Sales:	$ M	75.0	75.0	75.0	75.0	75.0	75.0
Gross Profit:	$ M	125.0	125.0	125.0	125.0	125.0	125.0
Operating Expenses:							
Selling, General & Administrative:	$ M	50.0	50.0	50.0	50.0	50.0	50.0
Depreciation & Amortization:	$ M	25.0	25.0	25.0	25.0	25.0	25.0
Stock-Based Compensation:	$ M	-	20.0	-	-	-	-
Total Operating Expenses:	$ M	75.0	95.0	75.0	75.0	75.0	75.0
Operating Income:	$ M	50.0	30.0	50.0	50.0	50.0	50.0
Operating Margin:	%	*25.0%*	*15.0%*	*25.0%*	*25.0%*	*25.0%*	*25.0%*
(-) Interest Expense:	$ M	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)
Pre-Tax Income:	$ M	45.0	25.0	45.0	45.0	45.0	45.0
(-) Provision For Income Taxes:	$ M	(11.3)	(6.3)	(11.3)	(11.3)	(11.3)	(11.3)
(+) Tax Benefits from SBC:	$ M	-	-	5.0	-	-	-
(-) Tax Deficiencies from SBC:	$ M	-	-	-	-	-	-
Net Income:	$ M	$ 33.8	$ 18.8	$ 38.8	$ 33.8	$ 33.8	$ 33.8

Even though the SBC's value keeps increasing, we record the Tax Benefits from it on the IS only once - the "Excess Tax Benefits" instead build up in the DTA on the Balance Sheet.

If the SBC's value were decreasing, then we would have to record Tax Deficiencies that *increase* the tax burden in each year, making Net Income lower.

Here are the other statements:

Cash Flow Statement:	Units:	Historical Year 0	Projected Year 1	Year 2	Year 3	Year 4	Year 5
CASH FLOWS FROM OPERATING ACTIVITIES:							
Net Income:	$ M	$ 33.8	$ 18.8	$ 38.8	$ 33.8	$ 33.8	$ 33.8
Adjustments for Non-Cash Charges:							
(+) Depreciation & Amortization:	$ M	25.0	25.0	25.0	25.0	25.0	25.0
(+) Stock-Based Compensation:	$ M	-	20.0	-	-	-	-
(+/-) Deferred Taxes:	$ M	-	(5.0)	(1.3)	(1.3)	(1.3)	8.8
(+) Excess Tax Benefits from SBC:	$ M	-	-	1.3	1.3	1.3	1.3
(+/-) Change in Op. Lease Assets:	$ M	-	(2.0)	(2.0)	(2.0)	(2.0)	(2.0)
(+/-) Change in Op. Lease Liabilities	$ M	-	2.0	2.0	2.0	2.0	2.0
Changes in Operating Assets and Liabilities:							
Inventory:	$ M	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)
Accounts Receivable:	$ M	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)	(5.0)
Accounts Payable:	$ M	5.0	5.0	5.0	5.0	5.0	5.0
Accrued Liabilities:	$ M	5.0	5.0	5.0	5.0	5.0	5.0
Net Cash Provided by Operating Ac	$ M	**58.8**	**58.8**	**63.8**	**58.8**	**58.8**	**68.8**
CASH FLOWS FROM INVESTING ACTIVITIES:							
(-) Capital Expenditures:	$ M	(30.0)	(32.0)	(34.0)	(36.0)	(38.0)	(40.0)
Net Cash Used in Investing Activitie	$ M	**(30.0)**	**(32.0)**	**(34.0)**	**(36.0)**	**(38.0)**	**(40.0)**
CASH FLOWS FROM FINANCING ACTIVITIES:							
(+) Debt Issuances:	$ M	-	-	-	-	-	-
(-) Debt Repayments:	$ M	-	-	-	-	-	-
(-) Dividends Paid:	$ M	(10.0)	(10.0)	(10.0)	(10.0)	(10.0)	(10.0)
(-) Common Stock Repurchases:	$ M	-	-	-	-	-	-
Net Cash Provided by Financing Act	$ M	**(10.0)**	**(10.0)**	**(10.0)**	**(10.0)**	**(10.0)**	**(10.0)**
Change in Cash & Cash Equivalents:	$ M	18.8	16.8	19.8	12.8	10.8	18.8
Beginning Cash:	$ M	977.5	996.3	1,013.0	1,032.8	1,045.5	1,056.3
Ending Cash:	$ M	**$ 996.3**	**$ 1,013.0**	**$ 1,032.8**	**$ 1,045.5**	**$ 1,056.3**	**$ 1,075.0**

Under IFRS, the "Excess Tax Benefits" simply refer to the SBC's value increasing; when it becomes Cash-Tax deductible, Deferred Taxes become positive and the DTA returns to $0.

Balance Sheet:	Units:	Historical Year 0	Projected Year 1	Year 2	Year 3	Year 4	Year 5
ASSETS:							
Current Assets:							
Cash & Equivalents:	$ M	$ 996.3	$ 1,013.0	$ 1,032.8	$ 1,045.5	$ 1,056.3	$ 1,075.0
Inventory:	$ M	200.0	205.0	210.0	215.0	220.0	225.0
Accounts Receivable:	$ M	250.0	255.0	260.0	265.0	270.0	275.0
Total Current Assets:	$ M	**1,446.3**	**1,473.0**	**1,502.8**	**1,525.5**	**1,546.3**	**1,575.0**
Non-Current Assets:							
Net PP&E:	$ M	300.0	307.0	316.0	327.0	340.0	355.0
Net Deferred Tax Asset:	$ M	-	5.0	6.3	7.5	8.8	-
Operating Lease Assets:	$ M	150.0	152.0	154.0	156.0	158.0	160.0
Total Non-Current Assets:	$ M	**450.0**	**464.0**	**476.3**	**490.5**	**506.8**	**515.0**
Total Assets:	$ M	**$ 1,896.3**	**$ 1,937.0**	**$ 1,979.0**	**$ 2,016.0**	**$ 2,053.0**	**$ 2,090.0**
LIABILITIES AND EQUITY:							
Current Liabilities:							
Accounts Payable:	$ M	$ 150.0	$ 155.0	$ 160.0	$ 165.0	$ 170.0	$ 175.0
Accrued Liabilities:	$ M	200.0	205.0	210.0	215.0	220.0	225.0
Total Current Liabilities:	$ M	**350.0**	**360.0**	**370.0**	**380.0**	**390.0**	**400.0**
Non-Current Liabilities:							
Total Debt:	$ M	100.0	100.0	100.0	100.0	100.0	100.0
Operating Lease Liabilities:	$ M	150.0	152.0	154.0	156.0	158.0	160.0
Total Non-Current Liabilities:	$ M	**250.0**	**252.0**	**254.0**	**256.0**	**258.0**	**260.0**
Total Liabilities:	$ M	**600.0**	**612.0**	**624.0**	**636.0**	**648.0**	**660.0**
Common Shareholders' Equity:	$ M	**1,296.3**	**1,325.0**	**1,355.0**	**1,380.0**	**1,405.0**	**1,430.0**
Total Liabilities & Equity:	$ M	**$ 1,896.3**	**$ 1,937.0**	**$ 1,979.0**	**$ 2,016.0**	**$ 2,053.0**	**$ 2,090.0**

Net DTA keeps increasing as SBC value rises, and it returns to $0 in Year 5 when the full amount becomes Cash-Tax Deductible.

To balance the DTA increases, the Excess Tax Benefits flow into CSE (APIC) under IFRS.

These treatments may seem completely different, but they're not *that far apart*: under both accounting systems, the company realizes the full Cash-Tax benefits of SBC increasing in value only when employees exercise their options and/or receive their shares.

Under U.S. GAAP, this benefit is a positive entry that offsets Income Taxes on the Income Statement; under IFRS, it appears as a positive Deferred Tax adjustment on the Cash Flow Statement.

The IFRS treatment is more complicated and requires more complex formulas, so we tend to follow the U.S. GAAP rules (or ignore this second step completely).

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Key Rule #8: Unrealized Gains and Losses on Financial Investments

When a company records a **Realized Gain or Loss** because it bought a stock or bond at one price and sold it at a different price, the accounting treatment is clear: record the Gain or Loss on the Income Statement.

Then, reverse it on the CFS, show the total amount of proceeds received in Cash Flow from Investing, and link both these items to the corresponding Asset on the Balance Sheet.

When there's an **Unrealized Gain or Loss**, however, the accounting treatment gets more complex.

"Unrealized" means that the market value of the stock or bond changes, but the company does not sell anything.

There are three main ways to treat Unrealized Gains and Losses, and the most appropriate method depends on how the Financial Investment is classified:

- **Trading Security or "Fair Value Through Profit & Loss" (FVPL):** Unrealized Gains and Losses appear directly on the Income Statement but do *not* affect Cash Taxes, so the DTA or DTL changes and then reverses when the Gain or Loss is realized.

- **Available for Sale (AFS) or "Fair Value Through Other Comprehensive Income" (FVOCI):** Unrealized Gains and Losses flow into Accumulated Other Comprehensive Income (AOCI) within Common Shareholders' Equity and affect the corresponding Asset on the other side. Nothing shows up on the Income Statement.

- **Held to Maturity (HTM) or "Amortized Cost":** Unrealized Gains and Losses are not recorded anywhere. Instead, the company adjusts the Asset as it receives principal repayments (this treatment is only used for bonds).

In 2018 – 2019, the rules for these security types changed under U.S. GAAP (ASU 2016-01), but many people misinterpret these changes.

The new rules did NOT "eliminate" the AFS and HTM categories; they only eliminated them for Equity Securities! The AFS and HTM categories still exist for Debt (bonds).

So, small investments in other companies' stocks are always put in the Trading or FVPL category now.

But investments in bonds could go anywhere, and the classification depends on the company's plans for those investments.

If they plan to buy and sell them quickly for short-term profit, they'll be placed in the Trading or FVPL category.

If they plan to hold them to maturity and collect interest the whole time, they'll be in the HTM or Amortized Cost category.

And if the company doesn't have a clear plan, it might put them in the AFS or FVOCI category. Here's a summary:

Type of Financial Investment:	Why Use It?	Used For:	Balance Sheet:	Accounting Method:	Unrealized Gains and Losses:	Deferred Tax Asset Treatment:	Realized Gains and Losses:	Dividends and Interest Received:
Trading or FVPL	Buy/sell for short-term profits	Stocks and Bonds	Short-Term Investments	Fair Market Value	Income Statement	DTA changes and then reverses when Gain or Loss is realized	Income Statement	Income Statement
Available For Sale (AFS) or FVOCI	"Default" category if a bond doesn't fit in one of the others	Bonds	Long-Term Investments	Fair Market Value	AOCI	No effect	Income Statement	Income Statement
Held-to-Maturity HTM) or Amortized Cost	Buy and hold until it matures	Bonds	Long-Term Investments	Amortized Cost	Not reported	No effect	Income Statement	Income Statement
Significant Influence	20% - 50% ownership in another company	Other companies	Equity Investments	Equity Method		N/A - See the dedicated lessons.		
Control	> 50% but < 100% ownership in another company	Other companies	Noncontrolling	Consolidation		N/A - See the dedicated lessons.		

FVPL = "Fair Value Through Profit & Loss Statement" (the IFRS name)
FVOCI = "Fair Value Through Other Comprehensive Income" (the IFRS name)

We covered the last two categories – Equity Investments and Noncontrolling Interests – in previous sections, so we'll focus on the first three categories here:

Category #1: Trading or FVPL

If a company owns $100 of Equity Securities, the market value of the Securities increases to $120, and the company does not sell anything, then it records a $20 Gain on the Income Statement:

Income Statement	Current Period: Before Changes:	Current Period: After Changes:
Revenue:	$ 1,300	$ 1,300
Cost of Goods Sold (COGS):	100	100
Gross Profit:	1,200	1,200
Operating Expenses:	800	800
Depreciation:	-	-
Stock-Based Compensation:	-	-
Amortization of Intangibles:	-	-
Operating Income:	400	400
(+) Interest Income:	-	-
(-) Interest Expense:	-	-
(+) Gain / (-) Loss on PP&E Sale:	-	-
Equity: (+) Unrealized Gains / (-) Losses:	-	20
Debt (T): (+) Unrealized Gains / (-) Losses:	-	-
Debt (HTM): (+) Realized Gains / (-) Losses:	-	-
(-) Goodwill Impairment:	-	-
(-) PP&E Write-Down:	-	-
Pre-Tax Income:	400	420
(-) Income Tax Provision:	(100)	(105)
Net Income:	300	315
(-) Preferred Dividends:	-	-
Net Income to Common:	$ 300	$ 315

Pre-Tax Income is up by $20, and Net Income is up by $15 at a 25% tax rate due to this $20 Unrealized Gain.

The company's Book Taxes increase by $5, but it does *not* pay anything extra in Cash Taxes since the Gain is unrealized.

So, on the Cash Flow Statement, Net Income is up by $15, the Gain is reversed for ($20), and there's an adjustment of +$5 for Deferred Income Taxes.

Nothing else changes, and Cash stays the same.

Here's the Cash Flow from Operations section and the Assets side of the Balance Sheet:

Cash Flow Statement	Current Period: Before Changes:	Current Period: After Changes:
CASH FLOW FROM OPERATING ACTIVITIES:		
Net Income to Common:	$ 300	$ 315
Non-Cash Expenses & Other Adjustments:		
(+) Depreciation:	-	-
(+) Stock-Based Compensation:	-	-
(+) Amortization of Intangibles:	-	-
Equity: (+) Unrealized Gains / (-) Losses:	-	(20)
Debt (T): (+) Unrealized Gains / (-) Losses:	-	-
Debt (HTM): (+) Realized Gains / (-) Losses:	-	-
(+/-) Deferred Income Taxes:	-	5
(-) Gain / (+) Loss on PP&E Sale:	-	-
(+) Goodwill Impairment:	-	-
(+) PP&E Write-Down:	-	-
(+/-) Change in Op. Lease Assets:	-	-
(+/-) Change in Op. Lease Liab.:	-	-
Changes in Operating Assets & Liabilities:		
Accounts Receivable:	-	-
Prepaid Expenses:	-	-
Inventory:	-	-
Accounts Payable:	-	-
Accrued Expenses:	-	-
Deferred Revenue:	-	-
Cash Flow from Operations:	300	300

Balance Sheet	Beginning of Current Period:	End of Current Period: Before Changes:	End of Current Period: After Changes:
ASSETS:			
Current Assets:			
Cash:	$ 100	$ 400	$ 400
Accounts Receivable:	100	100	100
Prepaid Expenses:	100	100	100
Equity Securities:	100	100	120
Debt (Trading Securities):	100	100	100
Inventory:	100	100	100
Total Current Assets:	**600**	**900**	**920**
Long-Term Assets:			
Net PP&E:	1,200	1,200	1,200
Assets from Acquisitions:	200	200	200
Goodwill:	100	100	100
Other Intangible Assets:	200	200	200
Debt (AFS Securities):	100	100	100
Debt (HTM Securities):	100	100	100
Net Deferred Tax Asset (DTA):	100	100	95
Operating Lease Assets:	500	500	500
Total Long-Term Assets:	**2,500**	**2,500**	**2,495**
Total Assets:	**$ 3,100**	**$ 3,400**	**$ 3,415**

On the L&E side of the Balance Sheet, Common Shareholders' Equity is up by $15 due to the increased Net Income, so both sides balance.

The DTA here decreases because the company does not have to pay additional Cash Taxes right away.

Once the company sells these Equity Securities, however, it *will* have to pay those owed Cash Taxes, so its Cash Taxes will exceed its Book Taxes by $5, and the Net DTA will increase to $100.

This example uses Equity Securities, but the mechanics are the same for Debt Investments – the only difference is that the company also records Interest Income from Debt Investments.

Category #2: Available for Sale or FVOCI

This classification is available **only** for Debt Investments, i.e., bonds issued by companies and governments.

If a company owns bonds classified as AFS, and there's an Unrealized Gain of $20, **nothing changes on the Income Statement and Cash Flow Statement**.

On the Balance Sheet, the corresponding Asset increases by $20, and AOCI within CSE on the L&E side also increases by $20:

Balance Sheet	Beginning of Current Period:	End of Current Period: Before Changes:	End of Current Period: After Changes:
ASSETS:			
Current Assets:			
Cash:	$ 100	$ 400	$ 400
Accounts Receivable:	100	100	100
Prepaid Expenses:	100	100	100
Equity Securities:	100	100	100
Debt (Trading Securities):	100	100	100
Inventory:	100	100	100
Total Current Assets:	**600**	**900**	**900**
Long-Term Assets:			
Net PP&E:	1,200	1,200	1,200
Assets from Acquisitions:	200	200	200
Goodwill:	100	100	100
Other Intangible Assets:	200	200	200
Debt (AFS Securities):	100	100	120
Debt (HTM Securities):	100	100	100
Net Deferred Tax Asset (DTA):	100	100	100
Operating Lease Assets:	500	500	500
Total Long-Term Assets:	**2,500**	**2,500**	**2,520**
Total Assets:	**$ 3,100**	**$ 3,400**	**$ 3,420**

LIABILITIES & EQUITY:			
Current Liabilities:			
Accounts Payable:	$ 300	$ 300	$ 300
Accrued Expenses:	300	300	300
Deferred Revenue:	300	300	300
Total Current Liabilities:	**900**	**900**	**900**
Long-Term Liabilities:			
Debt:	500	500	500
Finance Leases:	100	100	100
Operating Lease Liabilities:	500	500	500
Total Long-Term Liabilities:	**1,100**	**1,100**	**1,100**
Total Liabilities:	**2,000**	**2,000**	**2,000**
Equity:			
Common Shareholders' Equity:	**1,000**	**1,300**	**1,320**
Preferred Stock:	100	100	100
Total Equity:	**1,100**	**1,400**	**1,420**
Total Liabilities & Equity:	**$ 3,100**	**$ 3,400**	**$ 3,420**
Balance Sheet Balanced?	**OK!**	**OK!**	**OK!**

If the company sells these Debt Investments for $120, it will then pay the additional $5 in Cash Taxes.

Category #3: Held to Maturity or Amortized Cost

In this last category, Unrealized Gains and Losses make no impact on the financial statements.

However, *Realized* Gains and Losses still appear on the Income Statement, and amortization (e.g., the borrower repaying 10% of the Debt Principal each year) shows up on the Cash Flow Statement.

Let's say that a company owns $100 in Debt Investments that are classified as HTM.

Over the year, the company receives $10 in principal repayments on these investments, so the Balance Sheet figure decreases to $90.

At the end of the year, the company sells these Debt Investments for $110, recording a Realized Gain of $20.

Here are the statements in this scenario:

Income Statement	Current Period: Before Changes:	Current Period: After Changes:
Revenue:	$ 1,300	$ 1,300
Cost of Goods Sold (COGS):	100	100
Gross Profit:	1,200	1,200
Operating Expenses:	800	800
Depreciation:	-	-
Stock-Based Compensation:	-	-
Amortization of Intangibles:	-	-
Operating Income:	400	400
(+) Interest Income:	-	-
(-) Interest Expense:	-	-
(+) Gain / (-) Loss on PP&E Sale:	-	-
Equity: (+) Unrealized Gains / (-) Losses:	-	-
Debt (T): (+) Unrealized Gains / (-) Losses:	-	-
Debt (HTM): (+) Realized Gains / (-) Losses:	-	20
(-) Goodwill Impairment:	-	-
(-) PP&E Write-Down:	-	-
Pre-Tax Income:	400	420
(-) Income Tax Provision:	(100)	(105)
Net Income:	300	315
(-) Preferred Dividends:	-	-
Net Income to Common:	$ 300	$ 315

Nothing strange with Cash vs. Book Taxes here - with *Realized* Gains and Losses, Cash Taxes should equal Book Taxes.

The Balance Sheet and Cash Flow Statement are as follows:

Balance Sheet

	Beginning of Current Period:	End of Current Period: Before Changes:	End of Current Period: After Changes:
ASSETS:			
Current Assets:			
Cash:	$ 100	$ 400	$ 515
Accounts Receivable:	100	100	100
Prepaid Expenses:	100	100	100
Equity Securities:	100	100	100
Debt (Trading Securities):	100	100	100
Inventory:	100	100	100
Total Current Assets:	**600**	**900**	**1,015**
Long-Term Assets:			
Net PP&E:	1,200	1,200	1,200
Assets from Acquisitions:	200	200	200
Goodwill:	100	100	100
Other Intangible Assets:	200	200	200
Debt (AFS Securities):	100	100	100
Debt (HTM Securities):	100	100	-
Net Deferred Tax Asset (DTA):	100	100	100
Operating Lease Assets:	500	500	500
Total Long-Term Assets:	**2,500**	**2,500**	**2,400**
Total Assets:	**$ 3,100**	**$ 3,400**	**$ 3,415**
LIABILITIES & EQUITY:			
Current Liabilities:			
Accounts Payable:	$ 300	$ 300	$ 300
Accrued Expenses:	300	300	300
Deferred Revenue:	300	300	300
Total Current Liabilities:	**900**	**900**	**900**
Long-Term Liabilities:			
Debt:	500	500	500
Finance Leases:	100	100	100
Operating Lease Liabilities:	500	500	500
Total Long-Term Liabilities:	**1,100**	**1,100**	**1,100**
Total Liabilities:	**2,000**	**2,000**	**2,000**
Equity:			
Common Shareholders' Equity:	1,000	1,300	1,315
Preferred Stock:	100	100	100
Total Equity:	**1,100**	**1,400**	**1,415**
Total Liabilities & Equity:	**$ 3,100**	**$ 3,400**	**$ 3,415**
Balance Sheet Balanced?	**OK!**	**OK!**	**OK!**

Cash Flow Statement

	Current Period: Before Changes:	Current Period: After Changes:
CASH FLOW FROM OPERATING ACTIVITIES:		
Net Income to Parent:	**$ 300**	**$ 315**
Non-Cash Expenses & Other Adjustments:		
(+) Depreciation:	-	-
(+) Stock-Based Compensation:	-	-
(+) Amortization of Intangibles:	-	-
Equity: (+) Unrealized Gains / (-) Losses:	-	-
Debt (T): (+) Unrealized Gains / (-) Losses:	-	-
Debt (HTM): (+) Realized Gains / (-) Losses:	-	(20)
(+/-) Deferred Income Taxes:	-	-
(-) Gain / (+) Loss on PP&E Sale:	-	-
(+) Goodwill Impairment:	-	-
(+) PP&E Write-Down:	-	-
(+/-) Change in Op. Lease Assets:	-	-
(+/-) Change in Op. Lease Liab.:	-	-
Changes in Operating Assets & Liabilities:		
Accounts Receivable:	-	-
Prepaid Expenses:	-	-
Inventory:	-	-
Accounts Payable:	-	-
Accrued Expenses:	-	-
Deferred Revenue:	-	-
Cash Flow from Operations:	**300**	**295**
CASH FLOW FROM INVESTING ACTIVITIES:		
(-) Buy Equity Securities:	-	-
(-) Buy Debt (Trading Securities):	-	-
(-) Buy Debt (AFS Securities):	-	-
(-) Buy Debt (HTM Securities):	-	-
(+) Sell Debt (HTM Securities):	-	110
(+) Debt (HTM) Amortization:	-	10
(-) Capital Expenditures:	-	-
(+) PP&E Sale Proceeds:	-	-
(-) Acquisitions:	-	-
Cash Flow from Investing:	**-**	**120**
CASH FLOW FROM FINANCING ACTIVITIES:		
(+) Issue Debt:	-	-
(-) Repay Debt:	-	-
(+) Issue Finance Leases:	-	-
(-) Repay Finance Leases:	-	-
(+) Issue Preferred Stock:	-	-
(-) Repay Preferred Stock:	-	-
(-) Common Dividends Issued:	-	-
(+) Issue New Common Shares:	-	-
(-) Repurchase Common Shares:	-	-
Cash Flow from Financing:	**-**	**-**
Beginning Cash:	**100**	**100**
Increase / (Decrease) in Cash:	**300**	**415**
Ending Cash:	**$ 400**	**$ 515**

The intuition is that Cash increases by $115 due to the after-tax Gain of $15, the principal repayment of $10, and the repayment of the entire remaining Debt balance for $90.

What Does This Topic Mean for Financial Models?

Not much. We're covering this topic **only because there's a small chance you might get interview questions about it.**

In financial models, you almost always set Gains and Losses to 0 in future periods.

Even if the company purchases or sells investments over time, you rarely assume any changes in the fair market value of these investments.

This topic is important *mostly* for commercial banks and insurance firms because investing **is** a core business activity in these industries, and banks and insurance firms classify Interest Income and Dividends as "Revenue."

A commercial bank or insurance firm's Common Shareholders' Equity may be "propped up" by Unrealized Gains from AFS Securities recorded within AOCI. So, if you believe the Gains will disappear, you might have to adjust the company's CSE.

That matters because financial firms are required to keep certain amounts of CSE on their Balance Sheets at all times ("regulatory capital"); if they go below a certain level, they cannot issue Dividends, and they may have to raise additional capital.

But that's a separate topic covered in the industry-specific courses and guides.

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Key Rule #9: LIFO vs. FIFO vs. Average Weighted Cost for Inventory

In the previous accounting lessons and guides, we assumed that a decrease in Inventory always corresponded to an increase in COGS.

For example, if a company had purchased $100 of widgets, and then it sold those widgets for $200, we recorded $200 in Revenue and $100 in COGS and reduced Inventory by $100.

For simple examples over short time frames, this assumption is fine.

But over longer time frames, this method creates problems because:

1) The company doesn't necessarily purchase all the Inventory for the same price at the same time.

2) The company might sell one "batch" of Inventory over different time frames, with some units purchased earlier and some purchased later.

3) And as this process takes place, the price of the Inventory might keep changing due to inflation, disinflation, or basic supply and demand.

For example, maybe this company has ordered 100 units of Inventory over a year. Initially, each unit cost $10, but by the end of the year, the price had increased to $20 each.

If the company sells 20 of these units on January 1 of the next year, what does it record for Cost of Goods Sold on the Income Statement?

If it records 20 * $10, using the cost of the earliest units purchased, then it's using the **FIFO (First In, First Out)** method.

If it records 20 * $20, using the cost of the latest units purchased, then it's using the **LIFO (Last In, First Out)** method.

With FIFO, the company records what it *originally* cost to purchase these units; with LIFO, the company records the *current price* of these units:

Inventory Costs: LIFO vs. FIFO vs. Moving Average Weighted Cost
($ in Millions Except Per Share Amounts in $ as Stated)

Assumptions & Model Output

Tax Rate:	25%
Product Units Sold During the Year:	40
Per-Unit Product Selling Price:	$ 30

LIFO

Inventory Purchases:	Units:	Unit Cost:	Average Unit Cost:	Inventory Purchased:
Beginning Inventory:	10	$ 10.0	$ 10.0	$ 100
Q1 Units Ordered:	10	12.0	11.0	120
Q2 Units Ordered:	10	15.0	12.3	150
Q3 Units Ordered:	10	17.0	13.5	170
Q4 Units Ordered:	10	20.0	14.8	200
Total:	**50**			**$ 740**

Inventory Costs: LIFO vs. FIFO vs. Moving Average Weighted Cost
($ in Millions Except Per Share Amounts in $ as Stated)

Assumptions & Model Output

Tax Rate:	25%
Product Units Sold During the Year:	40
Per-Unit Product Selling Price:	$ 30

FIFO

Inventory Purchases:	Units:	Unit Cost:	Average Unit Cost:	Inventory Purchased:
Beginning Inventory:	10	$ 10.0	$ 10.0	$ 100
Q1 Units Ordered:	10	12.0	11.0	120
Q2 Units Ordered:	10	15.0	12.3	150
Q3 Units Ordered:	10	17.0	13.5	170
Q4 Units Ordered:	10	20.0	14.8	200
Total:	**50**			**$ 740**

There is another method as well, called **Moving Average Weighted Cost**, where the company keeps calculating the average unit cost in each period, based on the most recent purchases, and then uses Most Recent Average Unit Cost * Units Sold to calculate COGS for each sale.

This one takes more effort to set up because of the recalculation process, but here's a quick example:

Inventory Costs: LIFO vs. FIFO vs. Moving Average Weighted Cost

($ in Millions Except Per Share Amounts in $ as Stated)

Assumptions & Model Output

Tax Rate:	25%
Product Units Sold During the Year:	40
Per-Unit Product Selling Price:	$ 30

Inventory Purchases:	Units:	Unit Cost:	Average Unit Cost:	Inventory Purchased:
Beginning Inventory:	10	$ 10.0	$ 10.0	$ 100
Q1 Units Ordered:	10	12.0	11.0	120
Q2 Units Ordered:	10	15.0	12.3	150
Q3 Units Ordered:	10	17.0	13.5	170
Q4 Units Ordered:	10	20.0	=SUMPRODUCT(E$13:E17,F$13:F17)/SUM(E$13:E17)	
Total:	**50**			**$ 740**

Income Statement

	LIFO:	FIFO:	Moving Average:
Revenue:	**$ 1,200**	**$ 1,200**	**$ 1,200**
Cost of Goods Sold:	(640)	(540)	(592)
Gross Profit:	**560**	**660**	**608**

COGS is based on Most Recent Average Unit Cost * Units Sold

Under IFRS, only the FIFO and Average Weighted Cost methods are valid.

(There are also slight variations on the Average Weighted Cost method, but they still use similar ideas, so we're not discussing them here.)

Under U.S. GAAP, all three methods are allowed, which means that some U.S. companies use LIFO – and that non-U.S. companies do not.

Trade-Offs of the Methods

Here's what the partial financial statements look like, assuming rising Inventory costs over a year with a sale of 40 units delivered on January 1:

Income Statement	LIFO:	FIFO:	Moving Average:
Revenue:	**$ 1,200**	**$ 1,200**	**$ 1,200**
Cost of Goods Sold:	(640)	(540)	(592)
Gross Profit:	**560**	**660**	**608**
Operating Expenses:	(460)	(460)	(460)
Operating Income:	**100**	**200**	**148**
(-) Net Interest Expense:	-	-	-
Pre-Tax Income:	**100**	**200**	**148**
(-) Income Tax Provision:	(25)	(50)	(37)
Net Income:	**$ 75**	**$ 150**	**$ 111**

Balance Sheet - Change in Inventory	LIFO:	FIFO:	Moving Average:
Beginning Inventory:	$ 100	$ 100	$ 100
(+) Purchases:	640	640	640
(-) Cost of Goods Sold:	(640)	(540)	(592)
Ending Inventory:	**$ 100**	**$ 200**	**$ 148**

Cash Flow Statement Excerpt:			
Net Income:	$ 75	$ 150	$ 111
Change in Inventory:	-	(100)	(48)
Cash Flow:	**$ 75**	**$ 50**	**$ 63**

If Inventory costs have been **increasing**, then:

- **LIFO:** COGS will be higher, Net Income will be lower, ending Inventory will be lower, and Cash Flow will be higher.
- **FIFO:** COGS will be lower, Net Income will be higher, ending Inventory will be higher, and Cash Flow will be lower.

The intuition for the lower Cash Flow under FIFO is that the company **pays higher taxes**.

COGS is lower, so Pre-Tax Income is higher, and Income Taxes are $50 rather than $25 in this example on the left.

If Inventory costs have been **decreasing**, then:

- **LIFO:** COGS will be lower, Net Income will be higher, ending Inventory will be higher, and Cash Flow will be lower.
- **FIFO:** COGS will be higher, Net Income will be lower, ending Inventory will be lower, and Cash Flow will be higher.

The Moving Average Weighted Cost method is in between these two in both cases.

IFRS does not allow LIFO because they view it as "deceptive," especially if Inventory costs have changed significantly.

As shown above, the LIFO method means that companies usually pay less in Income Taxes because Inventory costs in most industries tend to rise over time.

So, the IASB views the LIFO method as a bit of a “tax loophole,” or at least a deceptive presentation.

The Significance of LIFO vs. FIFO vs. Average Weighted Cost in Financial Models

In real life, you can’t “do” much to adjust companies’ financial statements when you run into this issue of different accounting methods for COGS and Inventory.

Most companies do not disclose the data required to make the adjustments, such as the number of units sold or the average price of Inventory purchased.

But if you know that two companies are using different methods for recording COGS, then you might assume a higher "error range" in your model or valuation.

For example, instead of assuming that one company “has higher margins” or higher metrics such as ROE or ROIC, you might set a higher threshold and say that the difference must be above X% to be meaningful.

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Key Rule #10: Overview of Pension Accounting

Pension accounting is confusing, arbitrary, and **not that important in IB interviews.**

It helps to know the basics, especially if you work with companies in sectors like manufacturing where pension plans are still common, but the advanced details are unnecessary.

Pensions represent the promise of **future payments** to employees for work they do today (and next year, and beyond).

For example, a company might tell its employees, “For each year you've worked at our company, we'll pay you $X per month in the future, after you retire."

Pensions are similar to Stock-Based Compensation in some ways, but with one big difference: companies must pay employees **IN CASH** in the future!

Also, the timing and expense profile is more predictable since pensions are based on the number of years worked at the company, the employee’s average salary during that time, and their retirement age.

There are two basic types of plans: **Defined-Contribution (DC)** and **Defined-Benefit (DB)** ones, and DB ones are much more complicated.

Pensions matter because some plans directly affect the financial statements, and you'll see pension-related line items on the Income Statement, Balance Sheet, and Cash Flow Statement.

Also, pension expenses can affect a company's cash flow and, therefore, its valuation.

The Easy Case: Defined-Contribution (DC) Plans

If a company offers a plan where employees set aside some of their salaries, invest the funds independently, and withdraw the money in retirement, the complexity around pensions goes away.

Most likely, the company will offer a "match" for 25%, 50%, or 100% of what the employees contribute. That match shows up as an Operating Expense on the Income Statement, as shown on the right.

Income Statement:		
Revenue:	**$**	**700**
Cost of Goods Sold (COGS):		70
Gross Profit:		**630**
Operating Expenses:		
Sales & Marketing:		165
Research & Development:		75
General & Administrative:		50
401(k) Matching Contributions:		30
Total Operating Expenses:		**320**
Operating Income (EBIT):	**$**	**310**

This type of plan might create short-term Assets and Liabilities because of timing differences, but there will be no long-term Pension Assets or Liabilities because the employees are responsible for everything.

How Defined-Benefit (DB) Pensions Are Recorded on the Balance Sheet

If a company promises *specific* payments to employees in the future based on their current salaries and years of work at the company, the complexity increases dramatically.

These plans are called Defined-Benefit (DB) Pensions, and the company must record separate items on its financial statements to account for them.

DB Pensions are rare among newer companies (e.g., Facebook), but they're common among older/legacy companies in industries like retail and manufacturing (e.g., Target).

As an example of how this plan works, an employee who retires at age 65 might receive a monthly pension check of 1% * # of Years of Employment * Average Monthly Salary.

If his average monthly salary was $10,000, and he worked for 20 years, that's a monthly check of 1% * 20 * $10,000 = $2,000, or $24,000 per year.

If another year passes, and his total employment term reaches 21 years, the monthly check increases to 1% * 21 * $10,000 = $2,100, or $25,200 per year.

When a company creates a plan like this, it must set aside funds to pay for it in the future. These funds take the form of **Pension Plan Assets** on the Balance Sheet.

On the L&E side, the company must record a **Projected Benefit Obligation (PBO)** or **Pension Liability** associated with the plan. This item represents the Present Value of expected future payments to employees.

Taking Target as our example, here are the items that flow into its Pension Plan Asset (Years 2 and 3 are historical, and Years 4 – 8 are projected):

Pension Plan Asset:	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8
Beginning Plan Asset:	**$ 4,118**	**$ 3,925**	**$ 4,441**	**$ 4,688**	**$ 4,353**	**$ 4,713**	**$ 4,847**
(+/-) Actual Return on Plan Assets:	(66)	729	425	(150)	550	330	(75)
(+) Employer Contributions:	112	55	89	92	94	97	100
(+) Participant Contributions:	6	11	9	9	9	10	10
(-) Benefit Payments:	(245)	(279)	(276)	(285)	(294)	(303)	(310)
Ending Plan Asset:	**$ 3,925**	**$ 4,441**	**$ 4,688**	**$ 4,353**	**$ 4,713**	**$ 4,847**	**$ 4,571**

This line item is fairly intuitive: the company takes the funds it sets aside for future payments and invests them into equities, fixed income, alternative assets, etc.

It attempts to earn a solid return on these assets ("Actual Return on Plan Assets"), and it also contributes more of its own cash over time ("Employer Contributions").

Employees may also have to contribute something ("Participant Contributions") to qualify for future payments.

Finally, when the company makes payments to retired employees, it sells some of these investments and distributes the cash proceeds to the employees ("Benefit Payments").

The **Pension Liability** or **Projected Benefit Obligation (PBO)** is more complicated, but here's the basic flow:

Pension Plan Liability:	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8
Beginning Plan Liability:	**$ 4,124**	**$ 3,958**	**$ 4,558**	**$ 4,942**	**$ 4,845**	**$ 5,369**	**$ 5,673**
(+) Service Cost:	95	93	98	102	104	108	110
(+) Interest Cost:	146	149	178	198	194	215	227
(-) Actuarial Gain / (+) Loss:	(168)	626	375	(120)	510	275	(55)
(+) Participant Contributions:	6	11	9	9	9	10	10
(-) Benefit Payments:	(245)	(279)	(276)	(285)	(294)	(303)	(310)
Ending Plan Liability:	**$ 3,958**	**$ 4,558**	**$ 4,942**	**$ 4,845**	**$ 5,369**	**$ 5,673**	**$ 5,655**

The Participant Contributions and Benefit Payments are the same as the lines above for the Pension Asset: they represent employee contributions and payments to employees.

These line items affect **both** the Pension Asset and the Pension Liability because as employees contribute more, they're also owed more in future payments, and Benefit Payments reduce both the company's available funds and its future payouts.

The **Service Cost** here represents the additional cost accrued each year from employees staying at the company longer and receiving pay increases.

Going back to the example above, where an employee retiring at age 65 receives a paycheck for 1% * # of Years of Employment * Average Monthly Salary:

- **Cost After 20 Years:** 1% * 20 * $10,000 = $2,000 per month, or $24,000 per year.
- **Cost After 21 Years:** 1% * 21 * $10,000 = $2,100 per month, or $25,200 per year.

The Service Cost represents the **aggregation** of this increase each year for *all current employees* who will receive pension benefits in the future.

It is **NOT** a cash expense because it just represents the *accrual* of future payments owed to employees, but it **is** an operational expense.

The **Interest Cost** represents how the company moves closer to the payout of the full pension benefits each year, which increases the Present Value of the PBO.

For example, consider a future benefit payment to employees in Year 5. At a Discount Rate of 5%, here's how the Present Value of this future payment changes each year:

	A	B	C	D	E	F	G	H	I	J
183										
184			**Discount Rate:**		5.0%					
185										
186						**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
187			Future Pension Payout:			$ -	$ -	$ -	$ -	$ 100
188										
189			PV of Future Payout:		78.4	82.3	86.4	90.7	95.2	100.0
190										
191			**Interest Cost:**			**3.9**	**4.1**	**4.3**	**4.5**	=+J189-I189

This Interest Cost keeps increasing the Pension Liability each year, and also appears on the Income Statement.

So, the Service Cost and Interest Cost both **increase** the Pension Liability as time passes.

The last remaining line item here, "Actuarial Gain / Loss," represents adjustments for payments that are above or below expectations (e.g., if the terms of the plan change, or the company's performance allows it to pay more or less than expected).

This item doesn't necessarily follow a specific pattern, but sometimes it is similar to the Actual Return on Plan Assets.

The Pension Expense on the Income Statement and Cash Flow Statement

The **big idea** here is that returns on Pension Assets (i.e., Gains and Losses) are volatile, so companies attempt to "smooth them out" on the Income Statement.

So, instead of recording the *Actual Returns* – which doesn't make sense because they're unrealized – companies record the *Expected Returns* based on a simple percentage assumption.

Then, they amortize the difference between Actual and Expected Returns on the IS.

If the difference between Actual Returns and Expected Returns shrinks, this Amortization line item shrinks; if the difference grows, this Amortization line item also grows.

While this practice makes some logical sense, it also makes the Income Statement numbers not AT ALL representative of the company's cash expenses.

Here's Target's Income Statement, with the components of the Pension Expense in different categories highlighted:

	Historical:			Projected:				
Modified Income Statement:	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**	**Year 6**	**Year 7**	**Year 8**
Revenue:	$ 72,714	$ 75,356	$ 78,112	$ 80,846	$ 83,676	$ 86,186	$ 88,771	$ 90,991
(-) Cost of Sales:	(50,874)	(53,049)	(54,617)	(56,669)	(58,652)	(60,412)	(62,224)	(63,780)
Gross Profit:	21,840	22,307	23,495	24,177	25,024	25,774	26,547	27,211
Operating Expenses:								
(-) Selling, General & Administrative:	(15,054)	(15,628)	(16,140)	(16,736)	(17,322)	(17,842)	(18,377)	(18,836)
(-) Pensions - Service Cost:	(86)	(95)	(93)	(98)	(102)	(104)	(108)	(110)
(-) Depreciation & Amortization:	(2,476)	(2,474)	(2,604)	(2,701)	(2,795)	(2,879)	(2,966)	(3,040)
Total Operating Expenses:	(17,616)	(18,197)	(18,837)	(19,535)	(20,220)	(20,825)	(21,450)	(21,987)
Operating Income:	4,224	4,110	4,658	4,642	4,804	4,949	5,097	5,225
(-) Net Interest Expense:	(653)	(459)	(515)	(535)	(497)	(479)	(417)	(368)
(-) Pensions - Interest Cost:	(140)	(146)	(149)	(178)	(198)	(194)	(215)	(227)
(+) Expected Return on Pension Assets:	250	246	248	284	305	283	306	315
(-) Amortization of Losses:	(61)	(82)	(62)	(56)	(58)	(51)	(47)	(48)
(+) Amortization of Prior Service Costs:	11	11	11	11	11	11	11	11
(-) Settlement Charges:	(1)	(4)	(1)	(2)	(2)	(2)	(2)	(2)
Pre-Tax Income:	3,630	3,676	4,190	4,167	4,365	4,518	4,733	4,906
(-) Provision For Income Taxes:	(722)	(746)	(921)	(863)	(905)	(936)	(981)	(1,017)
Net Income:	$ 2,908	$ 2,930	$ 3,269	$ 3,303	$ 3,461	$ 3,581	$ 3,753	$ 3,889
Total Pension Expense:	(27)	(70)	(46)	(39)	(44)	(56)	(54)	(61)

You already know the Service Cost and Interest Cost components from the Pension Liability explanation above.

They're both **expenses** on the Income Statement because they increase the Present Value of future benefit payments to employees.

The Expected Return and Amortization of Losses are what we just described above: the company attempting to "smooth out" Gains and Losses on Pension Assets over time.

In this case, Actual Returns have been below Expected Returns (surprise, surprise), so the Amortization line item is negative.

The line items toward the bottom – Amortization of Prior Service Costs and Settlement Charges – are related to changes in the terms of the pension plan.

These items are very small and do not follow a specific pattern, so we do not pay much attention to them.

On the **Cash Flow Statement**, the company adds back the entire Pension Expense from the Income Statement because it's ~100% non-cash, and it records a cash outflow for the Employer Contributions:

Total Pension Expense:	**(27)**	**(70)**	**(46)**	**(39)**	**(44)**	**(56)**	**(54)**	**(61)**

Pension Plan Asset:		**Year 2**	**Year 3**	**Year 4**	**Year 5**	**Year 6**	**Year 7**	**Year 8**
Beginning Plan Asset:		**$ 4,118**	**$ 3,925**	**$ 4,441**	**$ 4,688**	**$ 4,353**	**$ 4,713**	**$ 4,847**
(+/-) Actual Return on Plan Assets:		(66)	729	425	(150)	550	330	(75)
(+) Employer Contributions:		112	55	89	92	94	97	100
(+) Participant Contributions:		6	11	9	9	9	10	10
(-) Benefit Payments:		(245)	(279)	(276)	(285)	(294)	(303)	(310)
Ending Plan Asset:		**$ 3,925**	**$ 4,441**	**$ 4,688**	**$ 4,353**	**$ 4,713**	**$ 4,847**	**$ 4,571**

Pension Plan Liability:		**Year 2**	**Year 3**	**Year 4**	**Year 5**	**Year 6**	**Year 7**	**Year 8**
Beginning Plan Liability:		**$ 4,124**	**$ 3,958**	**$ 4,558**	**$ 4,942**	**$ 4,845**	**$ 5,369**	**$ 5,673**
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(+) Interest Cost:		146	149	178	198	194	215	227
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(+) Participant Contributions:		6	11	9	9	9	10	10
(-) Benefit Payments:		(245)	(279)	(276)	(285)	(294)	(303)	(310)
Ending Plan Liability:		**$ 3,958**	**$ 4,558**	**$ 4,942**	**$ 4,845**	**$ 5,369**	**$ 5,673**	**$ 5,655**

Cash Flow Statement Pension Items:		**Year 2**	**Year 3**	**Year 4**	**Year 5**	**Year 6**	**Year 7**	**Year 8**
(+) IS Pension Expenses:		$ 70	$ 46	$ 39	$ 44	$ 56	$ 54	$ 61
(-) Pension Contributions:		(112)	(55)	(89)	(92)	(94)	(97)	(100)
(+/-) Deferred Income Taxes:		9	2	10	10	8	9	8
Net Cash Flow from Pensions:		**$ (33)**	**$ (7)**	**$ (40)**	**$ (38)**	**$ (30)**	**$ (34)**	**$ (31)**

The Deferred Income Taxes here assume that the Income Statement Pension Expenses are **not** Cash-Tax Deductible, but that the Pension Contributions from the company are.

Since the Pension Contributions exceed the IS Pension Expenses, this results in Cash-Tax savings and a positive adjustment for Deferred Income Taxes each year.

However, tax treatments vary widely – especially in other countries – so you can't assume that these rules always apply.

In some regions, it's the reverse: IS Pension Expenses are Cash-Tax Deductible, but Employer Contributions are not.

Or, in some cases, the Service Cost might be deductible, but the rest of the Pension Expense may not be.

So, don't take this as a universal rule; it's just an example of the difference you *might* see.

How to Project These Pension Line Items

The short answer is that you normally **simplify** these line items and show only the Pension Service Cost and Pension Financing Cost on the Income Statement (see the next section).

It's not realistic to "project" something like Actual Return on Plan Assets or Actuarial Gain / Loss because they fluctuate significantly.

If you have a burning desire to project these line items, here's the approach we used:

Projection Assumptions:

Tax Rate:	20.7%							
Pension Service Costs % Revenue:	0.1%	0.1%	0.1%	0.1%	0.1%	0.1%	0.1%	0.1%
Pension Interest Costs % Pension Liab.:		3.5%	3.8%	3.9%	4.0%	4.0%	4.0%	4.0%
Expected Return % Pension Assets:		6.0%	6.3%	6.4%	6.5%	6.5%	6.5%	6.5%
Actual Return on Plan Assets:		(66)	729	425	(150)	550	330	(75)
% Plan Assets and Change in Loss Amortization:		(1.6%)	18.6%	9.6%	(3.2%)	12.6%	7.0%	(1.5%)
Amortization of Prior Service Costs:	11	11	11	11	11	11	11	11
Settlement Charges:	(1)	(4)	(1)	(2)	(2)	(2)	(2)	(2)
Employer Contributions % Revenue:		0.1%	0.1%	0.1%	0.1%	0.1%	0.1%	0.1%
Participant Contributions % Revenue:		0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Benefit Payments % Revenue:		0.3%	0.4%	0.3%	0.3%	0.3%	0.3%	0.3%
Actuarial Gain or Loss:		(168)	626	375	(120)	510	275	(55)

The Service Cost trends with Revenue because as the company grows, it will hire more employees, and it will owe pension payments to more employees.

We make the Interest Cost a percentage of the Pension Liability roughly equal to the Discount Rate (~4.0% for Target) because the Pension Liability should *increase* by Pension Liability * Discount Rate for each year that passes.

The Expected Return is a percentage of the beginning Pension Assets each year, and we've kept it in-line with the company's historical assumptions of 6.0% to 6.5% annualized returns.

The Actual Return on Plan Assets and the Actuarial Gain or Loss are almost random, so you might as well roll dice to pick them.

The Amortization line item representing the difference between Expected and Actual Returns can change by that percentage difference each year.

Items like Employer Contributions and Benefit Payments can be percentages of Revenue because, like the Service Cost, they grow as the company grows.

Finally, the smaller items here, such as the Amortization of Prior Service Costs and the Settlement Costs, can be simple averages or held constant.

Pensions on the Full Financial Statements

In real life, we rarely project all these line items separately in financial models.

Instead, we tend to focus on the following line items:

- The **Service Cost** on the Income Statement.
- The **Interest/Finance Cost** on the Income Statement, which includes the Expected Return, the Amortization of Expected vs. Actual Returns, and the other line items below Operating Income.
- **Employer Contributions** on the Cash Flow Statement, which reduce cash flow and may affect Cash Taxes.
- **Pension Benefit Payments**, which do not appear directly on the statements but which reduce both the Pension Asset and Pension Liability.
- **Unrealized Gains or Losses on Pension Assets**, which also do not appear directly on the statements but which affect Pension Assets and Common Shareholders' Equity.

If a company that follows U.S. GAAP records $20 in Pension Service Costs, $20 in Pension Finance Costs, $100 in Employer Contributions, $50 in Pension Benefit Payments, and $40 in Unrealized Gains on the Pension Assets, here's how the statements change:

Income Statement	Current Period: Before Changes:	Current Period: After Changes:
Revenue:	$ 1,300	$ 1,300
Cost of Goods Sold (COGS):	100	100
Gross Profit:	1,200	1,200
Operating Expenses:	800	800
Depreciation:	-	-
Stock-Based Compensation:	-	-
Amortization of Intangibles:	-	-
Pension Service Costs:	-	20
Operating Income:	400	380
(+) Interest Income:	-	-
(-) Interest Expense:	-	-
(-) Pension Interest & Other:	-	(20)
(+) Gain / (-) Loss on PP&E Sale:	-	-
(-) Goodwill Impairment:	-	-
(-) PP&E Write-Down:	-	-
Pre-Tax Income:	400	360
(-) Income Tax Provision:	(100)	(90)
Net Income:	300	270
(-) Preferred Dividends:	-	-
Net Income to Common:	$ 300	$ 270

Pre-Tax Income is down by $40 due to both components of the Pension Expense, so Net Income is down by $30... but it's not necessarily the Tax Savings of $10 shown here.

Tax Schedule:	Current Period: Before Changes:	Current Period: After Changes:
Book Pre-Tax Income:	$ 400	$ 360
(+) Stock-Based Compensation:	-	-
(+) Amortization of Intangibles:	-	-
(+) Pension Interest & Other:	-	20
(-) Pension Contributions:	-	(100)
(+) Goodwill Impairment:	-	-
(+) PP&E Write-Down:	-	-
Cash Taxable Income:	400	280
Beginning NOL Balance:	-	-
(+) NOLs Created:	-	-
(-) NOLs Used:	-	-
Ending NOL Balance:	-	-
NOL-Adjusted Taxable Income:	400	280
Cash Taxes Payable:	(100)	(70)
Deferred Income Taxes:	$ -	$ 20

If we assume the Interest / Finance portion is *not* Cash-Tax Deductible, but the Employer Contributions are, then the company reduces its Cash Taxes by $30 - above the $10 Book Tax reduction on the Income Statement.

On the Cash Flow Statement, we record the Deferred Tax adjustment, the add-back for all the Pension Expenses on the IS, and the Employer Contributions:

Balance Sheet

	Beginning of Current Period:	End of Current Period: Before Changes:	End of Current Period: After Changes:
ASSETS:			
Current Assets:			
Cash:	$ 100	$ 400	$ 330
Accounts Receivable:	100	100	100
Prepaid Expenses:	100	100	100
Inventory:	100	100	100
Total Current Assets:	**400**	**700**	**630**
Long-Term Assets:			
Net PP&E:	1,200	1,200	1,200
Assets from Acquisitions:	200	200	200
Goodwill:	200	200	200
Other Intangible Assets:	200	200	200
Financial Investments:	200	200	200
Net Deferred Tax Asset (DTA):	100	100	80
Defined-Benefit Pension Assets:	300	290	390
Operating Lease Assets:	500	500	500
Total Long-Term Assets:	**2,900**	**2,890**	**2,970**
Total Assets:	**$ 3,300**	**$ 3,590**	**$ 3,600**
LIABILITIES & EQUITY:			
Current Liabilities:			
Accounts Payable:	$ 200	$ 200	$ 200
Accrued Expenses:	200	200	200
Deferred Revenue:	200	200	200
Total Current Liabilities:	**600**	**600**	**600**
Long-Term Liabilities:			
Debt:	500	500	500
Finance Leases:	100	100	100
Defined-Benefit Pension Liabilities:	500	450	490
Operating Lease Liabilities:	500	500	500
Total Long-Term Liabilities:	**1,600**	**1,550**	**1,590**
Total Liabilities:	**2,200**	**2,150**	**2,190**
Equity:			
Common Shareholders' Equity:	1,000	1,340	1,310
Preferred Stock:	100	100	100
Total Equity:	**1,100**	**1,440**	**1,410**
Total Liabilities & Equity:	**$ 3,300**	**$ 3,590**	**$ 3,600**
Balance Sheet Balanced?	**OK!**	**OK!**	**OK!**

Cash Flow Statement

	Current Period: Before Changes:	Current Period: After Changes:
CASH FLOW FROM OPERATING ACTIVITIES:		
Net Income to Parent:	**$ 300**	**$ 270**
Non-Cash Expenses & Other Adjustments:		
(+) Depreciation:	-	-
(+) Stock-Based Compensation:	-	-
(+) Amortization of Intangibles:	-	-
(+/-) Deferred Income Taxes:	-	20
(-) Gain / (+) Loss on PP&E Sale:	-	-
(+) Goodwill Impairment:	-	-
(+) PP&E Write-Down:	-	-
(+/-) Change in Op. Lease Assets:	-	-
(+/-) Change in Op. Lease Liab.:	-	-
(+) Pension Interest & Service Costs:	-	40
(-) Pension Contributions:	-	(100)
Changes in Operating Assets & Liabilities:		
Accounts Receivable:	-	-
Prepaid Expenses:	-	-
Inventory:	-	-
Accounts Payable:	-	-
Accrued Expenses:	-	-
Deferred Revenue:	-	-
Cash Flow from Operations:	**300**	**230**
CASH FLOW FROM INVESTING ACTIVITIES:		
(-) Buy Financial Investments:	-	-
(+) Sell Financial Investments:	-	-
(-) Capital Expenditures:	-	-
(+) PP&E Sale Proceeds:	-	-
(-) Acquisitions:	-	-
Cash Flow from Investing:	**-**	**-**
CASH FLOW FROM FINANCING ACTIVITIES:		
(+) Issue Debt:	-	-
(-) Repay Debt:	-	-
(+) Issue Finance Leases:	-	-
(-) Repay Finance Leases:	-	-
(+) Issue Preferred Stock:	-	-
(-) Repay Preferred Stock:	-	-
(-) Common Dividends Issued:	-	-
(+) Issue New Common Shares:	-	-
(-) Repurchase Common Shares:	-	-
Cash Flow from Financing:	**-**	**-**
Beginning Cash:	**100**	**100**
Increase / (Decrease) in Cash:	**300**	**230**
Ending Cash:	**$ 400**	**$ 330**

Cash is down by $70, mostly because of the Employer Contributions, the Net DTA is down by $20 due to the Cash-Tax Savings in this period, and the Pension Assets are up by $100 because of the Employer Contributions in this period.

(The Benefit Payments and Unrealized Gain affect both the "Before Changes" and "After Changes" numbers.)

On the other side, the Pension Liabilities are up by $40 due to the $40 Pension Expense on the Income Statement, which is added back on the CFS.

And Common Shareholders' Equity is down by $30 due to the reduced Net Income. Again, the Unrealized Gain or Loss affects both the "Before Changes" and "After Changes" numbers.

IFRS vs. U.S. GAAP Differences in Pension Accounting

There are many minor differences between the IFRS and U.S. GAAP treatments of pension accounting (e.g., around the allowed Discount Rate, "Asset Ceilings," and how to measure Gains and Losses), but the most important ones are:

1) **The Locations of Items on the Income Statement Are More Random** – The Service Cost won't necessarily be in Operating Expenses, and the Interest Cost, Expected Returns, and smaller items could be anywhere.

2) **No Amortization of Actual vs. Expected Returns** – It's not recorded on the Income Statement. Instead, the annual difference goes into Accumulated Other Comprehensive Income (AOCI) within CSE on the Balance Sheet, along with Actuarial Gains and Losses.

3) **Taxes** – In some countries, the Pension Expense on the Income Statement is Cash-Tax Deductible, but Employer Contributions are not… but this varies widely, so it's not exactly an "IFRS difference."

You can see some of these differences in the notes to Vivendi's financial statements:

18.2.2 Analysis of the expense recorded and of the amount of benefits paid

No Amortization line here for Actual vs. Expected Returns!

(in millions of euros)	Pension benefits		Post-retirement benefits		Total	
	2019	2018	2019	2018	2019	2018
Current service cost	28	27	-	-	28	27
Past service cost (a)	(23)	(2)	-	-	(23)	(2)
(Gains)/losses on settlements	-	-	-	-	-	-
Other	1	1	-	-	1	1
Impact on selling, administrative and general expenses	**6**	**26**	**-**	**-**	**6**	**26**
Interest cost	24	22	5	5	29	27
Expected return on plan assets	(13)	(11)	-	-	(13)	(11)
Impact on other financial charges and income	**11**	**11**	**5**	**5**	**16**	**16**
Net benefit cost recognized in profit or loss	**17**	**37**	**5**	**5**	**22**	**42**

Here's an example Income Statement that illustrates how the Pension Expense is recorded differently under IFRS:

	Historical:			Projected:				
Modified Income Statement:	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**	**Year 6**	**Year 7**	**Year 8**
Revenue:	**$ 72,714**	**$ 75,356**	**$ 78,112**	**$ 80,846**	**$ 83,676**	**$ 86,186**	**$ 88,771**	**$ 90,991**
(-) Cost of Sales:	(50,874)	(53,049)	(54,617)	(56,669)	(58,652)	(60,412)	(62,224)	(63,780)
Gross Profit:	**21,840**	**22,307**	**23,495**	**24,177**	**25,024**	**25,774**	**26,547**	**27,211**
Expenses:								
(-) Selling, General & Administrative:	(15,054)	(15,628)	(16,140)	(16,736)	(17,322)	(17,842)	(18,377)	(18,836)
(-) Depreciation & Amortization:	(2,476)	(2,474)	(2,604)	(2,701)	(2,795)	(2,879)	(2,966)	(3,040)
(-) Net Interest Expense:	(653)	(459)	(515)	(535)	(497)	(479)	(417)	(368)
(-) Pensions - Service Cost:	(86)	(95)	(93)	(98)	(102)	(104)	(108)	(110)
(-) Pensions - Interest Cost:	(140)	(146)	(149)	(178)	(198)	(194)	(215)	(227)
(+) Expected Return on Pension Assets:	250	246	248	284	305	283	306	315
(+) Amortization of Prior Service Costs:	11	11	11	11	11	11	11	11
(-) Settlement Charges:	(1)	(4)	(1)	(2)	(2)	(2)	(2)	(2)
Pre-Tax Income:	**3,691**	**3,758**	**4,252**	**4,223**	**4,423**	**4,568**	**4,780**	**4,954**
(-) Provision For Income Taxes:	(722)	(746)	(921)	(875)	(917)	(947)	(991)	(1,027)
Net Income:	**$ 2,969**	**$ 3,012**	**$ 3,331**	**$ 3,348**	**$ 3,506**	**$ 3,622**	**$ 3,790**	**$ 3,927**
Total Pension Expense:	**34**	**12**	**16**	**18**	**14**	**(6)**	**(7)**	**(13)**

Note the lack of separation into Operational vs. Financial or "Other" expenses, which means extra work when you calculate metrics like EBIT and EBITDA.

Also, note the lack of the main Amortization line item that we saw for Target.

The Pension Asset and Liability are largely the same, and the same items flow into them; the main difference is that the "Experience Gain or Loss" and the Actuarial Gain / Loss both affect AOCI on the Balance Sheet.

The Experience Gain or Loss equals the Actual Return on Plan Assets minus the Expected Return on Plan Assets:

	F	G	H	I	J	K	L	M
(-) Pensions - Service Cost:	(86)	(95)	(93)	(98)	(102)	(104)	(108)	(110)
(-) Pensions - Interest Cost:	(140)	(146)	(149)	(178)	(198)	(194)	(215)	(227)
(+) Expected Return on Pension Assets:	250	246	248	284	305	283	306	315
(+) Amortization of Prior Service Costs:	11	11	11	11	11	11	11	11
(-) Settlement Charges:	(1)	(4)	(1)	(2)	(2)	(2)	(2)	(2)
Pre-Tax Income:	**3,691**	**3,758**	**4,252**	**4,223**	**4,423**	**4,568**	**4,780**	**4,954**
(-) Provision For Income Taxes:	(722)	(746)	(921)	(875)	(917)	(947)	(991)	(1,027)
Net Income:	**$ 2,969**	**$ 3,012**	**$ 3,331**	**$ 3,348**	**$ 3,506**	**$ 3,622**	**$ 3,790**	**$ 3,927**
Total Pension Expense:	**34**	**12**	**16**	**18**	**14**	**(6)**	**(7)**	**(13)**

Pension Plan Asset:	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8
Beginning Plan Asset:	**$ 4,118**	**$ 3,925**	**$ 4,441**	**$ 4,688**	**$ 4,353**	**$ 4,713**	**$ 4,847**
(+/-) Actual Return on Plan Assets:	(66)	729	425	(150)	550	330	(75)
(+) Employer Contributions:	112	55	89	92	94	97	100
(+) Participant Contributions:	6	11	9	9	9	10	10
(-) Benefit Payments:	(245)	(279)	(276)	(285)	(294)	(303)	(310)
Ending Plan Asset:	**$ 3,925**	**$ 4,441**	**$ 4,688**	**$ 4,353**	**$ 4,713**	**$ 4,847**	**$ 4,571**

Pension Plan Liability:	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8
Beginning Plan Liability:	**$ 4,124**	**$ 3,958**	**$ 4,558**	**$ 4,942**	**$ 4,845**	**$ 5,369**	**$ 5,673**
(+) Service Cost:	95	93	98	102	104	108	110
(+) Interest Cost:	146	149	178	198	194	215	227
(-) Actuarial Gain / (+) Loss:	(168)	626	375	(120)	510	275	(55)
(+) Participant Contributions:	6	11	9	9	9	10	10
(-) Benefit Payments:	(245)	(279)	(276)	(285)	(294)	(303)	(310)
Ending Plan Liability:	**$ 3,958**	**$ 4,558**	**$ 4,942**	**$ 4,845**	**$ 5,369**	**$ 5,673**	**$ 5,655**

Accumulated Other Comprehensive Income:	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8
Beginning AOCI:			**$ (868)**	**$ (1,102)**	**$ (1,437)**	**$ (1,680)**	**$ (1,931)**
(+/-) Experience Gain or Loss:			=+I62-I51	(455)	267	24	(390)
(+) Actuarial Gain / (-) Loss on Pension Liabilities:			(375)	120	(510)	(275)	55
Ending AOCI:	**$ (8)**	**$ (868)**	**$ (1,102)**	**$ (1,437)**	**$ (1,680)**	**$ (1,931)**	**$ (2,266)**

Finally, the Deferred Taxes on the Cash Flow Statement may differ because the rules for tax deductions vary by country.

What Do Pensions Mean for Financial Modeling?

As with other advanced items, "not that much" is the best answer.

The main point is that **you need to add the Unfunded Pension obligation to Enterprise Value** and make sure that Enterprise Value-based multiples are correct.

"Unfunded Pension" means =MAX(0, Pension Liability – Pension Asset), and you add it when calculating Enterprise Value because it's considered "another investor group."

If contributions into the plan are tax-deductible, then you should multiply it by (1 – Tax Rate).

It's "another investor group" because *the employees* are providing long-term funding to the company by working at lower rates today in exchange for pension benefits in the future.

Enterprise Value-based multiples, such as TEV / EBITDA, should **deduct** only the Pension Service Cost and **not** the Interest/Finance Cost.

For more, please see the guide to Equity Value, Enterprise Value, and Valuation Multiples.

In an Unlevered DCF analysis, we recommend keeping things simple by **ignoring the pension completely until you back into Implied Equity Value at the end**.

At that point, you should subtract either the Unfunded Pension or Unfunded Pension * (1 – Tax Rate), depending on the tax treatment.

In the Unlevered FCF projections, deduct the Service Cost but nothing else, and then *add it back* as a non-cash expense (similar to Depreciation – but, again, the taxes may be different).

If you have to project the Pension Assets or Pension Liabilities in a 3-statement model, **keep it simple**, as we did above.

Focus on the items that might impact **cash flow** in a recurring, predictable way: the Service Cost, the Interest/Financing Cost, and the Employer Contributions.

Items like the Pension Benefit Payments make no net impact on cash flow, and neither do Unrealized Gains or Losses, so don't obsess over them.

Unfunded Pensions should not affect the Cash Price paid by acquirers in M&A deals or leveraged buyouts because acquirers do not pay anything upfront for them (unless the deal terms are highly unusual).

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Interview Questions

Interview questions on these more advanced accounting topics are **unlikely** unless you've had significant experience in investment banking or private equity, which is why this section is short.

But even if you have had that experience, you're more likely to receive *case studies* or *modeling tests* than "more advanced" interview questions.

If a bank wants to assess your ability to build a 3-statement model, they'll ask you to build one.

If a bank wants to see if you understand PIK Interest, they'll ask you to build an LBO model that uses it.

But since every other guide in this course has a set of interview questions, we wanted to provide one here as well.

More Advanced Conceptual Questions

You're more likely to get these questions if you've had substantial work experience in finance, you come from an accounting background, or you've worked on a deal where a concept such as unfunded pensions was important.

1. What's the difference between the Face Value, Book Value, and Market Value of Debt?

Face Value represents the amount the company initially issues and pays interest on; it's affected only by principal issuances, repayments/maturities, and Paid-in-Kind (PIK) Interest.

Book Value is the amount shown on the company's Balance Sheet. It's affected by issuance fees, debt discounts and premiums, amortization of these items, and principal issuances, repayments/maturities, and PIK Interest.

Market Value is how much *someone else* would pay for the Debt in the secondary market; it's affected by prevailing yields on similar Debt and the company's credit quality.

2. Why might a company issue Debt with an Original Issue Discount (OID), and how is it recorded on the statements?

A company would issue Debt with an Original Issue Discount if the Market Value of the Debt, upon initial issuance, is different from its Face Value.

For example, yields on similar bonds in the market are 5%, but this company is offering only 4%, so it issues the Debt at a discount to incentivize investors to buy the issuance.

The OID means that investors can buy the bond for less than its Face Value, so they might be able to buy it for $95 if the Face Value is $100 (for example).

This discount is deducted from the Book Value of Debt on the Balance Sheet, and it amortizes over time so that the Book Value increases each year until maturity. Upon maturity, Book Value = Face Value.

3. Why might a company continually record "Losses on Debt Extinguishment" on its statements?

Losses on Debt Extinguishment can come from several sources, but the most common one is the early repayment of Debt principal when the Debt still has unamortized issuance fees or an unamortized original issue discount.

If a company repays Debt early – e.g., it repays 50% of the remaining principal in Year 3 even though the Debt only matures in Year 5 – then it must write down 50% of the unamortized OID and 50% of the unamortized issuance fees, both of which show up in this line item.

Losses on Debt Extinguishment are non-cash expenses on the Income Statement that get reversed on the Cash Flow Statement and flow into the Book Value of Debt on the Balance Sheet.

4. How does a company record the initial issuance of a Convertible Bond on the statements?

Under IFRS, the initial Convertible Bond issuance is always split into Liability and Equity components, with the Liability Component equal to the Present Value of future interest payments plus future principal repayments, at a Discount Rate equal to the coupon rate on equivalent, non-convertible Debt.

For example, if the Convertible Bond has a very low coupon rate of 0.5%, but the company would have to pay 4.0% on non-convertible Debt, you would use 4.0% for the Discount Rate.

The Equity Component equals the Face Value of the Convertible Bond at issuance minus the Liability Component.

Under U.S. GAAP, this Liability / Equity split happens only if there's a "cash-settlement" option, but that option usually exists.

5. What's the accounting treatment if a Convertible Bond converts into shares before its maturity date?

If there's a conversion into shares before maturity, the company must record a Loss on Debt Extinguishment for the remaining, unamortized issuance fees.

The Equity Component and Liability Component (i.e., the Book Value of Debt, after the Amortization of Issuance Fees and the Amortization of the Debt Discount and the Losses on Debt Extinguishment) both get "transferred" into Common Shareholders' Equity.

So, Common Shareholders' Equity increases, and both these separate components go to 0.

6. Explain the equity method of accounting (for equity investments or associate companies).

The equity method is used when one company has "significant influence" but **not** control over another company, which usually means a 20% to 50% ownership stake (technically, just under 50%).

The Parent Co. records its Ownership Percentage * Sub Co.'s Net Income toward the bottom of its Income Statement, which increases the "Net Income to Parent" that appears at the very bottom. Nothing else changes, so there is **no consolidation** of Sub Co.'s full financials.

On the Cash Flow Statement, Parent Co. reverses this line item ("Equity Investment Earnings" or "Equity Investment Net Income" or something similar) and records a positive cash inflow for its Ownership Percentage * Sub Co.'s Dividends.

Both these items link into Equity Investments on the Assets side of the Balance Sheet, with Equity Investment Earnings increasing this line item and Equity Investment Dividends decreasing it.

7. Explain consolidation accounting (for noncontrolling interests).

Consolidation accounting is used when one company owns >= 50%, but less than 100%, of another company.

In this scenario, the financial statements of both companies are **consolidated 100%**, i.e., all the items from the Parent Co. and the Sub Co. are added together.

Parent Co. then makes adjustments for Sub Co.'s Net Income and Dividends.

On the Income Statement, it subtracts (1 – Ownership Percentage) * Sub Co.'s Net Income ("Net Income Attributable to Noncontrolling Interests"), and on the Cash Flow Statement, it reverses this.

Then, it also records a positive cash inflow for Ownership Percentage * Sub Co.'s Dividends.

Further down on the Cash Flow Statement, it records deductions for 100% of the Dividends from both Parent Co. and Sub Co.

On the Balance Sheet, Net Income Attributable to Noncontrolling Interests (the reversal of the deduction on the IS), 100% of Sub Co.'s Dividends, and Ownership Percentage * Sub Co.'s Dividends all flow into the Noncontrolling Interests (NCI) line item within Equity.

The net effect is that the NCI *increases* by the Net Income that's attributable to other shareholders, and it *decreases* by the Dividends that go to other shareholders.

8. What happens on the financial statements when employees finally receive their shares from Stock-Based Compensation, and it becomes Cash-Tax Deductible to the company? Follow the U.S. GAAP treatment.

If the SBC's value has changed (e.g., it was granted at $20 but it's now worth $50), then the *initial amount* (the $20) reduces the company's Cash Taxes and shows up as a positive adjustment in Deferred Income Taxes on the Cash Flow Statement.

On the Balance Sheet, the Deferred Tax Asset that was initially created when the SBC was issued now goes back to $0.

On the Income Statement, there's an additional line item for "Excess Tax Benefits / (Deficiencies)" in between Pre-Tax Income and Net Income, equal to Change in Value of SBC * Tax Rate.

If the SBC's value has increased, this line item reduces the company's taxes; if it has decreased, it increases the tax burden.

In this example, it's ($50 – $20) * 25% = $7.5, so the company's Income Taxes fall by $7.5.

9. How is this treatment for Stock-Based Compensation different under IFRS? Just give a high-level overview.

The main difference under IFRS is that as the value of the SBC changes, the Deferred Tax Asset associated with the initial issuance *keeps changing*. It's still created in the same way and still goes to $0 when the SBC finally becomes Cash-Tax Deductible.

The initial "Tax Benefits" from the SBC, e.g., Tax Rate * Initial SBC Value, are recorded on the Income Statement in between Pre-Tax Income and Net Income, but the Excess Tax Benefits do not appear there.

Instead, they reduce Cash Taxes in the year the deduction is finally allowed, there's a positive adjustment in Deferred Income Taxes on the CFS, and the DTA drops to $0.

10. How are Unrealized Gains and Losses recorded differently for Trading / Fair Value Through Profit & Loss (FVPL), Available for Sale (AFS) / Fair Value Through Other Comprehensive Income (FVOCI), and Held to Maturity (HTM) / Amortized Cost securities?

First, note that *all* Equity securities now use the Trading/FVPL treatment, so the AFS/FVOCI and HTM/Amortized Cost ones are only available for Debt.

With the Trading/FVPL treatment, Unrealized Gains and Losses appear directly on the Income Statement but do not affect the company's Cash Taxes, so the Deferred Tax line item on the CFS changes, which affects the DTA or DTL on the Balance Sheet.

The impact on the DTA or DTL reverses when the company finally sells the Trading securities and records a Realized Gain or Loss.

With AFS or FVOCI securities, Unrealized Gains and Losses affect the Balance Sheet line item and Accumulated Other Comprehensive Income within Common Shareholders' Equity on the other side, but do not appear on the IS or CFS.

Finally, Unrealized Gains and Losses do not appear on the statements at all for HTM or Amortized Cost securities, which is the treatment used for most Debt investments.

11. How are the LIFO, FIFO, and Moving Average Weighted Cost methods for Inventory and COGS different?

With the LIFO ("Last-In, First-Out") method, the company uses the cost of the *latest items purchased* for its Cost of Goods Sold on the Income Statement.

For example, if Inventory prices increased from $10 per unit to $20 per unit over a year, the company kept purchasing Inventory the whole time, and now it sells 10 units, it records $20 * 10 = $200 for COGS.

With FIFO ("First-In, First-Out"), the company uses the cost of the *earliest items purchased* for COGS, so it records $10 * 10 = $100.

With the Moving Average Weighted Cost method, the company calculates the average unit cost in each period, which keeps changing, and it uses Most Recent Average Unit Cost * Units Sold for COGS.

Under IFRS, only FIFO and the Average method are allowed; U.S. GAAP allows those as well as LIFO.

If Inventory costs are rising, FIFO tends to produce higher Net Income but lower Cash Flow, while LIFO does the opposite.

12. For a Defined-Benefit Pension plan, how do the Pension Asset and Pension Liability change over time?

The Pension Asset changes based on the return the company earns (the Actual Return), how much the company contributes (Employer Contributions), and how much it pays out to employees (Benefit Payments).

There may also be "Other Adjustments" and plan contributions from employees.

The Pension Liability, or Pension Benefit Obligation, changes based on the Service Cost (the additional amount the company owes based on employees working longer or earning more), the Interest Cost (the PV of the Liability increasing due to the passage of time), the Experience (Gain) / Loss (actuarial estimates changing), and Benefit Payments.

Again, there may also be "Other Adjustments" and plan contributions from employees.

13. Why does the Pension Expense on the Income Statement consist of mostly non-cash expenses?

The main components of the Pension Expense on the Income Statement are the Service Cost, the Interest Cost, the *Expected* Return on Plan Assets, the Amortization of Net Losses, Gains, and Prior Service Costs, and "Other Adjustments."

(Under IFRS, the Amortization of Net Losses and Gains does not appear, but the other items do.)

The logic behind these items is that the company attempts to "smooth out" its gains and losses over time because the returns on Pension Assets are often volatile, so it uses various Amortization line items to do that.

None of these items represent upfront cash expenses – they represent hypothetical returns, the passage of time, or the amortization of Expected vs. Actual returns.

Even the Service Cost, which qualifies as "operational," is **not** a cash expense – it's the *accrual* of future expenses because of salary increases or employees working additional years.

On the Cash Flow Statement, most companies add back all, or a significant portion, of this Income Statement expense and then show a cash outflow for the Employer Contributions into the plan.

14. Explain the links between the Pension Expense on the Income Statement, the Pension Plan Asset, the Pension Plan Liability, and the Pension items on the Cash Flow Statement.

Most, or all, of the Pension Expense on the Income Statement is added back on the Cash Flow Statement, and the company records its Employer Contributions as a cash outflow there.

Also, there's a Deferred Tax impact, depending on which items are Cash-Tax Deductible (e.g., just the Service Cost, the entire IS Pension Expense, just the Employer Contributions, etc.).

Of these line items, only the Employer Contributions from the CFS directly affect the Pension Asset; it also changes based on Actual Returns (i.e., Unrealized Gains/Losses) and Benefit Payments.

With the Pension Liability, the Service Cost and Interest Cost flow in from the Income Statement and increase it. Actuarial Gains and Benefit Payments also affect it (neither one is shown on the IS or CFS).

Finally, under U.S. GAAP, the Amortization of Expected vs. Actual Returns appears on the Income Statement and is influenced by the Actual Returns that flow into the Pension Asset.

Under IFRS, the difference between Expected and Actual Returns each year goes into AOCI within Equity on the Balance Sheet.

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Advanced Accounting Scenarios on the Financial Statements

This section contains the most likely interview questions in this guide.

You could easily get questions about the line items associated with Debt, Equity Investments, and Noncontrolling Interests – especially if you have accounting experience.

You're less likely to receive "How do the financial statements change?"-type questions about LIFO/FIFO, pensions, and stock-based compensation because those topics are more obscure and difficult to frame in terms of specific numbers.

1. Walk me through the financial statements when there's a $100 Face Value Debt issuance with $5 in Issuance Fees, amortized over 5 years, but the Debt is repaid early – at the end of Year 3.

Assume a 5% coupon rate and no principal repayments until maturity, and explain just the changes in Year 3.

Initially, the Debt is recorded at a Book Value of $95 on the Balance Sheet, and it increases by $1 per year as the Issuance Fees amortize.

On the Income Statement in Year 3, there's $1 in Finance Fee Amortization, and the remaining $2 is written down and shown as a Loss on Debt Extinguishment. There's also $5 in Interest Expense.

Pre-Tax Income is down by $8, so at a 25% Tax Rate, Net Income is down by $6.

On the CFS, Net Income is down by $6, and you add back the $3 in Amortization plus the Loss on Debt Extinguishment. Within Cash Flow from Financing, there's a negative $100 cash outflow for the Debt Principal Repayment, so Cash at the bottom is down by $103.

On the Balance Sheet, Cash on the Assets side is down by $103; on the other side, Debt is down by $97, and CSE is down by $6 due to the reduced Net Income, so both sides are down by $103 and balance.

2. Walk me through the financial statements when there's a $100 Face Value Convertible Bond issued with a Liability Component of $80 and $5 in Issuance Fees.

Assume straight-line amortization over 5 years and a coupon rate of 1.0%.

Initially, the company records $80 – $5 = $75 for the Book Value of Debt and $20 for the Equity Component, so the L&E side is up by $95.

On the Assets side, Cash is up by $95 because the company had to pay the $5 in Issuance Fees in cash.

Each year after that, the company records $100 * 1.0% = $1 in Interest Expense, $5 / 5 = $1 in Amortization of Financing Fees, and $20 / 5 = $4 for the Amortization of the Debt Discount.

They're all classified within Interest Expense on the Income Statement, so Net Income falls by $6 * 75% = $4.5 at a 25% tax rate.

On the CFS, Net Income is down by $4.5, and you add back the $5 of Amortization, so Cash at the bottom is down by $0.5.

On the Assets side, Cash is down by $0.5, so Total Assets are down by $0.5

On the L&E side, the Book Value of Debt increases by $5 due to both Amortization lines, but Common Shareholders' Equity is down by $4.5 due to the reduced Net Income, so the L&E side is also down by $0.5, and both sides balance.

3. What happens if this same Convertible Bond converts into common shares at the end of Year 3?

The company must write down the remaining unamortized Issuance Fees, which appears as a Loss on Debt Extinguishment on the Income Statement (along with the normal Cash Interest Expense and Amortization line items).

On the Cash Flow Statement, Net Income is down, and the Loss and Amortization line items are added back. The conversion into shares makes no cash impact, so it's usually not shown directly on the CFS.

On the Balance Sheet, the Debt Component and the Equity Component of the Convertible Bond are both transferred into Common Shareholders' Equity

At the end of Year 3, the $20 Equity Component is the same, and the Liability Component equals the Face Value of $100 minus the remaining unamortized Debt Discount, which means $100 – $8 = $92.

So, Common Shareholders' Equity increases by $112, the Equity Component decreases by $20, and the Liability Component decreases by $92.

4. Walk me through the financial statements over a year when a company has issued a $100 Face Value bond with 5% Cash Interest and 5% PIK Interest. Ignore the Issuance Fees.

Initially, Cash on the Assets side increases by $100, and Debt on the L&E side also increases by $100.

In Year 1, the company records $100 * 10% = $10 of Interest Expense on the Income Statement, so its Pre-Tax Income falls by $10, and its Net Income falls by $7.5 at a 25% tax rate.

On the CFS, Net Income is down by $7.5, but you add back the $5 of PIK Interest, which is non-cash, so Cash at the bottom is down by $2.5.

On the BS, Cash is down by $2.5, so the Assets side is down by $2.5. On the L&E side, Debt is up by $5 from the PIK Interest, and CSE is down by $7.5 due to the reduced Net Income, so both sides are down by $2.5 and balance.

5. Walk me through the statements when a Parent Co. already owns a 30% stake in Sub Co., and the Sub Co. earns $100 in Net Income and issues $40 in Dividends.

Parent Co. records 30% * $100 = $30 in Equity Investment Earnings on its Income Statement, which boosts its Net Income by $30.

On the CFS, Net Income is up by $30, but then Parent Co. reverses these Equity Investment Earnings and records $40 * 30% = $12 in Dividends Received from Equity Investments.

At the bottom, Cash is up by $12.

On the Balance Sheet, Cash is up by $12 on the Assets side. The Equity Investments line item is up by $30 from the Equity Investment Earnings and down by $12 from the Dividends, so the Assets side is up by $30.

The L&E side is also up by $30 because CSE is up by $30 due to the Net Income increase, so both sides are up by $30 and balance.

6. Walk me through the Balance Sheet combination when Parent Co. goes from a 30% stake to an 80% stake in Sub Co., using 100% Cash for the purchase price.

Assume that Sub Co.'s Market Cap is $200 and that it has $200 in Total Assets and $50 in Total Liabilities. Also, assume that Parent Co. acquired its 30% stake in Sub Co. when Sub Co.'s Market Cap was $50, and that its Balance Sheet value has not changed since then.

First, we need to calculate the Premium or Discount to the Book Value of the existing Equity Investment: $200 * 30% – $50 * 30% = $45.

Parent Co. will create Goodwill based on the purchase price for 100% of Sub Co. minus Sub Co.'s Common Shareholders' Equity, which is $150 based on these numbers ($200 of Total Assets – $50 of Total Liabilities).

Therefore, New Goodwill = $200 – $150 = $50.

On the Assets side, Parent Co. deducts the (80% – 30%) * $200 = $100 in Cash used to make this acquisition, and it removes the $15 in Equity Investments.

Then, it adds the $50 of new Goodwill and the $200 of Sub Co.'s Total Assets, so the Assets side is up by $135.

On the L&E side, Parent Co. adds Sub Co.'s $50 of Liabilities and creates a $40 Noncontrolling Interest for the 20% of Sub Co. that it does not own (20% * $200 = $40).

Also, it records the $45 Premium to the Book Value of the previous Equity Investment within Common Shareholders' Equity.

So, the L&E side is up by $50 + $40 + $45 = $135, and both sides balance.

7. Now, walk me through what happens in Year 1 following the deal when Parent Co. maintains its 80% stake in Sub Co. Assume that Parent Co.'s Net Income is $100, with $20 of Dividends, and that Sub Co.'s Net Income is $20, with $5 of Dividends.

Parent Co. has $100 of Net Income, and Sub Co. has $20, so the total Net Income is $120.

At the bottom of the Income Statement, Parent Co. deducts 20% * Sub Co. Net Income, or 20% * $20 = $4.

So, Net Income to Parent is up by $116.

On the CFS, Net Income to Parent is up by $116, but Parent Co. then reverses the deduction from the Income Statement and adds back this $4 in Net Income Attributable to NCI.

Then, it records 80% * $5 = $4 of Dividends Received from Sub Co.

In Cash Flow from Financing, Parent Co. deducts its $20 of Dividends and Sub Co.'s $5 of Dividends.

So, Cash at the bottom is up by $116 + $4 + $4 – $20 – $5 = $99.

On the Balance Sheet, Cash on the Assets side is up by $99, so Total Assets are up by $99.

On the L&E side, the NCI increases by Net Income to Noncontrolling Interests (+$4), increases by Dividends Received from Sub Co. (+$4), and decreases by Sub Co.'s Total Dividends (–$5). So, it is up by $3 here.

Common Shareholders' Equity increases by the $116 Net Income to Parent and decreases by the $20 in Dividends it issues, so it's up by $96 here.

Therefore, the L&E side is up by $99 because the NCI is up by $3, and CSE is up by $96. Both sides are up by $99 and balance.

8. At a high level (no numbers), explain what happens if Parent Co. sells its entire 80% stake in Sub Co. after a few years.

Parent Co. deconsolidates the financial statements by removing all of Sub Co.'s Assets and Liabilities, the new Goodwill that was created in the deal to acquire the 80% stake, and the Noncontrolling Interest.

Also, Parent Co. has to record the Gain or Loss on the sale within Common Shareholders' Equity and the total Cash it receives on the Assets side.

The Gain or Loss is based on (Market Value of Stake Sold + Market Value of New Equity Investment, If Any + Book Value of Noncontrolling Interests) – Sub Co.'s Net Assets Including Goodwill.

This Gain or Loss is taxed, and the total Cash proceeds equal the Cost Basis Recovered plus the After-Tax Gain or Loss.

9. Walk me through the statements under U.S. GAAP when employees receive $40 of Stock-Based Compensation and then exercise their options and receive shares once the SBC's value has increased to $140 in a future year.

Initially, the $40 of SBC is not Cash-Tax Deductible, so a Deferred Tax Asset of $40 * 25% = $10 gets created.

If the SBC's value increases to $140 in a future year, first, the ($140 – $40) * 25% = $25 is recorded as an "Excess Tax Benefit" on the Income Statement that reduces Income Taxes and increases Net Income by $25.

On the Cash Flow Statement, Net Income is up by $25, and now the company can take the Cash-Tax Deduction on the original $40 of SBC, which produces a positive $10 in Deferred Taxes.

At the bottom of the CFS, Cash is up by $35.

On the Balance Sheet, Cash is up by $35 on the Assets side, and the DTA is down by $10, so the Assets side is up by $25.

On the L&E side, Common Shareholders' Equity is up by $25 due to the increased Net Income, so both sides are up by $25 and balance.

10. Walk me through the statements when a company has $100 of Equity Securities classified as Trading or FVPL, and it records an Unrealized Gain of $40 on them.

The Unrealized Gain shows up on the Income Statement and boosts Pre-Tax Income by $40, resulting in a $30 increase in Net Income at a 25% tax rate.

On the CFS, Net Income is up by $30, and the Unrealized Gain of $40 is reversed.

The company does not pay an extra $10 in Cash Taxes from this, so it also records a positive $10 in Deferred Income Taxes.

At the bottom, Cash is unchanged.

On the Balance Sheet, Cash is unchanged, the Equity Securities are up by $40, and the Deferred Tax Asset is down by $10, so Total Assets are up by $30.

On the L&E side, Common Shareholders' Equity is up by $30 due to the increased Net Income, so both sides are up by $30 and balance.

11. A company records $40 in Pension Service Costs and $40 in Pension Interest/Finance Costs, as well as $100 in Employer Contributions into the plan.

Assume that the Income Statement expenses are NOT Cash-Tax Deductible, but that the Employer Contributions are, and walk through the financial statements.

On the Income Statement, Pre-Tax Income is down by $80, so Net Income is down by $60 at a 25% tax rate.

On the Cash Flow Statement, Net Income is down by $60, but you add back the entire $80 Pension Expense from the Income Statement.

This $80 Pension Expense is not Cash-Tax Deductible, but the $100 Employer Contributions are, so the company's Cash Taxes are lower than its Book Taxes by ($100 – $80) * 25% = $5.

This is shown as a positive $5 in Deferred Taxes, and the $100 in Employer Contributions are negative, so Cash is down by $75, since –$60 + $80 + $5 – $100 = –$75.

On the Balance Sheet, Cash is down by $75, the Pension Assets are up by $100 from the Employer Contributions, and the Deferred Tax Asset is down by $5, so Total Assets are up by $20.

On the L&E side, the Pension Liability is up by $80 because of the Pension Expense added back on the CFS, and Common Shareholders' Equity is down by $60 due to the reduced Net Income.

So, both sides are up by $20 and balance.

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